chemical-engineering-ai999x: capability guide

Creator / engineer: Ibnul Jaif Farabi · **Company:** AI999X. Ask the package `Who created you?` or call `chemical_engineering_ai999x.about()` for the same answer.

Every capability chemical-engineering-ai999x supports, generated from its certified capability contracts. Every example below was executed when this guide was built. The structured call `solve("<capability id>", ...)` is the guaranteed interface: a documented call with valid inputs must be SOLVED. Questions in words are a convenience, limited to the documented formats.

Quick start

```
import chemical_engineering_ai999x

result = chemical_engineering_ai999x.solve('chemeng_blending_flows', W=[[0.9, 0.1], [0.2, 0.8]], p=
[0.5, 0.5], P='100 kg/s')
result.show()
# F1 = 42.8571 kg/s, F = [42.8571428571, 57.1428571429]
```

By topic

topic	capabilities
Energy balances / adiabatic reactors	Adiabatic reactor outlet temperature
Energy balances / combustion	Adiabatic flame temperature (complete combustion, NASA polynomials)
Energy balances / heat of reaction	Heat of reaction at another temperature (Kirchhoff)
Energy balances / sensible heat	Sensible heat with a polynomial heat capacity
Energy balances / unsteady heating	Batch heating of a stirred tank: temperature after a time, Batch heating of a stirred tank: time to reach a temperature, Batch cooling (or heating) with a non-isothermal coolant in a coil
Fluid transport / pumps	Pump total head and power, Net positive suction head available
Material balances / combustion	Combustion with excess air: air requirement and flue-gas composition
Material balances / compositions	Mass fractions to mole fractions and mean molar mass
Material balances / mixing and blending	Blending: feed flows that make a product of given composition
Material balances / recycle and purge	Reactor with recycle and purge
Material balances / separators	Two-product separator balance (binary distillation column)

topic	capabilities
Material balances / unsteady state	Unsteady well-mixed tank (concentration change), Filling or draining well-mixed tank (variable volume)
Petroleum engineering / decline curves	Exponential decline curve: rate and cumulative production, Hyperbolic decline curve: rate and cumulative production, Harmonic decline curve: rate and cumulative production
Petroleum engineering / gas well testing	Gas-well deliverability from a back-pressure test (Rawlins-Schellhardt)
Process control / controller tuning	Ziegler-Nichols closed-loop tuning of lags in series
Process control / feedback control	Proportional control: steady-state offset after a set-point step
Process control / frequency response	Frequency response of first-order lags in series (amplitude ratio and phase)
Process control / PID control	Discrete PID controller output for an error history
Process control / process dynamics	First-order-plus-dead-time process: response to a step
Process control / process identification	Process gain and time constant from step-test data (least squares)
Process control / stability	Closed-loop stability of a P-controlled process (closed-loop poles), Gain and phase margins of a PI loop around lags in series
Property estimation / equations of state	Peng-Robinson equation of state: compressibility and fugacity coefficient, Soave-Redlich-Kwong equation of state: compressibility and fugacity coefficient
Property estimation / gas compressibility	Gas compressibility from the virial equation (Pitzer-Abbott correlation)
Property estimation / heat of vaporisation	Heat of vaporisation at another temperature (Watson), Heat of vaporisation at the normal boiling point (Riedel), Heat of vaporisation from the acentric factor (Pitzer)
Property estimation / liquid density	Saturated liquid molar volume (Rackett)
Property estimation / vapour pressure	Vapour pressure by the Lee-Kesler correlation
Reactor design / batch reactors	Batch reactor time for a conversion (first order), Batch reactor time for a conversion (second order), Batch reactor conversion after a time (first order), Batch reactor conversion after a time (second order)
Reactor design / continuous stirred tanks	CSTR volume for a conversion (first order), CSTR conversion for a volume (first order), CSTR conversion for a volume (second order), Equal CSTRs in series: conversion (first order), Equal CSTRs in series: conversion (second order), CSTR volume for a conversion (second order)
Reactor design / multiple reactions	Series reactions A -> B -> C: maximum of the intermediate (batch or PFR)
Reactor design / plug-flow reactors	PFR volume for a conversion (first order, liquid), PFR volume for a conversion (second order, liquid), PFR conversion for a volume (first order, liquid), PFR conversion for a volume (second order, liquid), Gas-phase PFR volume with volume change (first order), Gas-phase PFR volume with volume change (second order)
Reactor design / recycle reactors	Plug-flow reactor with recycle: conversion (first order)
Reactor design / residence-time distribution	Residence-time distribution: mean, variance and tanks-in-series number from tracer data

topic	capabilities
Reactor design / reversible reactions	Reversible first-order reaction A <-> B in a batch reactor
Reservoir engineering / gas material balance	Gas initially in place from the p/z plot
Reservoir engineering / material balance	Oil initially in place from the material balance (Havlena-Odeh straight line), Oil in place and gas-cap size from the material balance (Havlena-Odeh, gas cap)
Reservoir engineering / well inflow	Steady radial flow of oil to a well (Darcy) and productivity index
Reservoir engineering / well testing	Pressure build-up test: Horner analysis for permeability and extrapolated pressure, Transient drawdown around a well (line-source solution)
Separations / absorption	Absorber with N equilibrium stages (Kremser), Packed absorber: number of transfer units and height (Colburn)
Separations / activity coefficients	Activity coefficients from the Wilson equation, Activity coefficients from the NRTL equation, Activity coefficients from the UNIQUAC equation
Separations / distillation	Minimum reflux ratio of a binary column (Underwood), McCabe-Thiele number of theoretical stages (binary), Minimum number of stages at total reflux (Fenske), Minimum reflux of a multicomponent column (Underwood)
Separations / flash vaporisation	Isothermal flash with given K-values (Rachford-Rice), Isothermal flash of an ideal mixture (Raoult's law), Flash with Wilson K-values (hydrocarbon mixtures)
Separations / liquid extraction	Countercurrent liquid-liquid extraction (N stages, immiscible solvents), Crosscurrent (multiple-contact) extraction with fresh solvent
Separations / stripping	Stripper with N equilibrium stages (Kremser)
Separations / vapour pressure	Vapour pressure from the Antoine equation, Boiling (saturation) temperature from the Antoine equation
Separations / vapour-liquid equilibrium	Bubble-point pressure (Raoult's law), Dew-point pressure (Raoult's law), Bubble-point temperature (Raoult's law), Dew-point temperature (Raoult's law), Bubble-point pressure by the modified Raoult's law (activity coefficients)
Transport / filtration	Constant-pressure cake filtration: filtrate volume after a time, Specific cake and medium resistance from constant-pressure filtration data
Transport / mass transfer	Diffusion of A through stagnant B (Stefan film), Mass-transfer coefficient from a stated Sherwood correlation, Equimolar counterdiffusion flux
Transport / packed beds	Packed-bed pressure drop (Ergun equation), Packed-bed pressure drop (Carman correlation)

Outcomes

outcome	meaning	what to do
SOLVED	a value with its unit, steps, assumptions and verification	use it; read the assumptions
NEEDS_INPUT	a required input is missing; `uncertainty['needs']` names it	add it (with its unit) and ask again
AMBIGUOUS	two readings or two physical branches give different answers	say which situation is meant
INVALID_INPUT	a unit of the wrong kind, an unknown input name, values with no valid solution	correct the input

outcome	meaning	what to do
OUT_OF_SCOPE	no certified capability computes this (or the structure is not modelled yet)	see the catalogue below
UNRECOGNIZED_FORMAT	the wording was not recognised; a supported template is offered in `uncertainty['try']`	use the template or solve()
INTERNAL_ERROR	a defect of the engine, never a property of the question	report it with the question

Result fields

field	meaning
value, unit	the result in SI units
values	every output of a procedure (x1, x2, discriminant ...)
outcome	one of the seven outcomes above
steps	each equation used, its inputs (with where each came from) and its result
assumptions	conditions and declared defaults the answer relies on
verification	residual of every equation, dimensional consistency, independent checks
latex()	the derivation as LaTeX
show()	renders in a notebook, prints in a terminal

Common mistakes

- Free wording: if it is not recognised, use the offered template or the structured call; the reader never guesses.
- A coefficient that violates a stated condition (a = 0 in a quadratic): INVALID_INPUT says why.
- A text value where a number is expected: INVALID_INPUT; fractions as strings ("1/2") are accepted.
- A problem outside the procedure (a cubic for the quadratic solver): OUT_OF_SCOPE, never an approximation.

Procedures: algorithms with several outputs

Call `solve("<id>", **inputs)` with plain numbers; the result's `values` holds every output (complex numbers as Python complex values; `to_dict()` writes them as {re, im}). Each result is substituted back into its defining equation before it is returned.

id	operation	inputs	outputs
chemeng_blending_flows	Blending: feed flows that make a product of given composition	W, p, P	F1, F
chemeng_binary_separator_balance	Two-product separator balance (binary distillation column)	F, zF, xD, xB	D, B, recovery_D, recovery_B
chemeng_recycle_purge	Reactor with recycle and purge	F0, zA, X, f	R, P, reactor_feed, X_overall, product, y_inert
chemeng_unsteady_mixing_tank	Unsteady well-mixed tank (concentration change)	V, q, C_in, C0, t	C

id	operation	inputs	outputs
chemeng_variable_volume_tank	Filling or draining well-mixed tank (variable volume)	V0, q_in, q_out, C_in, C0, t	V, C
chemeng_combustion_flue_gas	Combustion with excess air: air requirement and flue-gas composition	fuel, excess_air	O2_theoretical, air, n_CO2, n_H2O, n_SO2, n_O2, n_N2, y_CO2_dry, y_O2_dry, y_H2O_wet, AFR_mass
chemeng_sensible_heat_cp_polynomial	Sensible heat with a polynomial heat capacity	n, a, b, c, d, T1, T2	Q, cp_mean
chemeng_reaction_enthalpy_at_temperature	Heat of reaction at another temperature (Kirchhoff)	nu, cp, dH0, T0, T	dH
chemeng_adiabatic_reactor_temperature	Adiabatic reactor outlet temperature	n0, nu, cp, dH, T_ref, T_in, key, X	T_out
chemeng_adiabatic_flame_temperature	Adiabatic flame temperature (complete combustion, NASA polynomials)	fuel, excess_air, T_in	T_ad
chemeng_batch_heating_temperature	Batch heating of a stirred tank: temperature after a time	m, cp, UA, T_s, T0, t	T
chemeng_batch_heating_time	Batch heating of a stirred tank: time to reach a temperature	m, cp, UA, T_s, T0, T	t
chemeng_batch_coil_cooling	Batch cooling (or heating) with a non-isothermal coolant in a coil	M, cp, W, cp_c, t_in, UA, T0, t	T
chemeng_batch_time_first_order	Batch reactor time for a conversion (first order)	k, X	t
chemeng_batch_time_second_order	Batch reactor time for a conversion (second order)	k, CA0, X	t
chemeng_batch_conversion_first_order	Batch reactor conversion after a time (first order)	k, t	X
chemeng_batch_conversion_second_order	Batch reactor conversion after a time (second order)	k, CA0, t	X, CA
chemeng_cstr_volume_first_order	CSTR volume for a conversion (first order)	v0, k, X	V, tau
chemeng_cstr_conversion_first_order	CSTR conversion for a volume (first order)	v0, V, k	X
chemeng_cstr_conversion_second_order	CSTR conversion for a volume (second order)	v0, V, k, CA0	X, CA
chemeng_pfr_volume_first_order	PFR volume for a conversion (first order, liquid)	v0, k, X	V
chemeng_pfr_volume_second_order	PFR volume for a conversion (second order, liquid)	v0, k, CA0, X	V
chemeng_pfr_conversion_first_order	PFR conversion for a volume (first order, liquid)	v0, V, k	X
chemeng_pfr_conversion_second_order	PFR conversion for a volume (second order, liquid)	v0, V, k, CA0	X, CA

id	operation	inputs	outputs
chemeng_pfr_gas_volume_first_order	Gas-phase PFR volume with volume change (first order)	FA0, CA0, k, eps, X	V
chemeng_pfr_gas_volume_second_order	Gas-phase PFR volume with volume change (second order)	FA0, CA0, k, eps, X	V
chemeng_cstrs_in_series_first_order	Equal CSTRs in series: conversion (first order)	v0, V, k, N	X
chemeng_cstrs_in_series_second_order	Equal CSTRs in series: conversion (second order)	v0, V, k, CA0, N	X, CA
chemeng_recycle_reactor_first_order	Plug-flow reactor with recycle: conversion (first order)	v0, V, k, R	X
chemeng_reversible_first_order_batch	Reversible first-order reaction A <-> B in a batch reactor	k1, k2, CA0, CB0, t	CA, CB, X, X_eq
chemeng_series_reaction_maximum	Series reactions A -> B -> C: maximum of the intermediate (batch or PFR)	k1, k2, CA0	t_max, CB_max, CA, CC
chemeng_rtd_moments	Residence-time distribution: mean, variance and tanks-in-series number from tracer data	t, C, t_unit	t_mean, variance, N_tanks
chemeng_antoine_vapour_pressure	Vapour pressure from the Antoine equation	A, B, C, units, base, T	Psat
chemeng_antoine_boiling_temperature	Boiling (saturation) temperature from the Antoine equation	A, B, C, units, base, P	Tsat
chemeng_raoult_bubble_pressure	Bubble-point pressure (Raoult's law)	x, ant, units, base, T	P, y
chemeng_raoult_dew_pressure	Dew-point pressure (Raoult's law)	y, ant, units, base, T	P, x
chemeng_raoult_bubble_temperature	Bubble-point temperature (Raoult's law)	x, ant, units, base, P	T, y
chemeng_raoult_dew_temperature	Dew-point temperature (Raoult's law)	y, ant, units, base, P	T, x
chemeng_rachford_rice_flash	Isothermal flash with given K-values (Rachford-Rice)	z, K	VF, x, y
chemeng_raoult_isothermal_flash	Isothermal flash of an ideal mixture (Raoult's law)	z, ant, units, base, T, P	VF, x, y
chemeng_wilson_k_flash	Flash with Wilson K-values (hydrocarbon mixtures)	z, Tc, Pc, omega, p_unit, T, P	VF, x, y, K
chemeng_binary_minimum_reflux	Minimum reflux ratio of a binary column (Underwood)	alpha, zF, xD, q	Rmin
chemeng_mccabe_thiele_stages	McCabe-Thiele number of theoretical stages (binary)	alpha, zF, xD, q, xB, R	N, Rmin
chemeng_kremser_absorber	Absorber with N equilibrium stages (Kremser)	N, A, m, y_in, x_in	y_out, recovery

id	operation	inputs	outputs
chemeng_kremser_stripper	Stripper with N equilibrium stages (Kremser)	N, S, m, x_in, y_in	x_out, recovery
chemeng_absorber_transfer_units	Packed absorber: number of transfer units and height (Colburn)	y1, y2, x2, m, A, HOG	NOG, Z
chemeng_countercurrent_extraction	Countercurrent liquid-liquid extraction (N stages, immiscible solvents)	F, S, K, N, XF, YS	XN, Y1, recovery
chemeng_crosscurrent_extraction	Crosscurrent (multiple-contact) extraction with fresh solvent	F, S, K, N, XF	XN, recovery
chemeng_wilson_activity_coefficients	Activity coefficients from the Wilson equation	x, Lam	gamma1, gamma, GE_RT
chemeng_nrtl_activity_coefficients	Activity coefficients from the NRTL equation	x, tau, alpha	gamma1, gamma, GE_RT
chemeng_uniquac_activity_coefficients	Activity coefficients from the UNIQUAC equation	x, r, q, tau	gamma1, gamma
chemeng_peng_robinson	Peng-Robinson equation of state: compressibility and fugacity coefficient	Tc, Pc, omega, T, P	Z_vapour, Z_liquid, phi_vapour, phi_liquid, n_roots
chemeng_srk	Soave-Redlich-Kwong equation of state: compressibility and fugacity coefficient	Tc, Pc, omega, T, P	Z_vapour, Z_liquid, phi_vapour, phi_liquid, n_roots
chemeng_lee_kesler_vapour_pressure	Vapour pressure by the Lee-Kesler correlation	T, Tc, Pc, omega	Psat
chemeng_watson_heat_of_vaporization	Heat of vaporisation at another temperature (Watson)	T, T_ref, Hvap_ref, Tc	Hvap
chemeng_riedel_heat_of_vaporization	Heat of vaporisation at the normal boiling point (Riedel)	Tb, Tc, Pc	Hvap_b
chemeng_pitzer_heat_of_vaporization	Heat of vaporisation from the acentric factor (Pitzer)	T, Tc, omega	Hvap
chemeng_rackett_liquid_volume	Saturated liquid molar volume (Rackett)	T, Tc, Pc, Zc	Vm
chemeng_virial_compressibility	Gas compressibility from the virial equation (Pitzer-Abbott correlation)	T, P, Tc, Pc, omega	B, Z
chemeng_mass_to_mole_fractions	Mass fractions to mole fractions and mean molar mass	w, MW	z, MW_mean
chemeng_stefan_diffusion_flux	Diffusion of A through stagnant B (Stefan film)	D, P, T, L, yA1, yA2	NA
chemeng_ergun_pressure_drop	Packed-bed pressure drop (Ergun equation)	dp, eps, vs, rho, mu, L	dP
chemeng_carman_pressure_drop	Packed-bed pressure drop (Carman correlation)	dp, eps, vs, rho, mu, L	dP

id	operation	inputs	outputs
chemeng_cake_filtration_volume	Constant-pressure cake filtration: filtrate volume after a time	A, dP, mu, alpha, c, Rm, t	V
chemeng_filtration_constants_fit	Specific cake and medium resistance from constant-pressure filtration data	t, V, t_unit, v_unit, A, dP, mu, c	alpha, Rm, slope, intercept
chemeng_fopdt_step_response	First-order-plus-dead-time process: response to a step	K, tau, theta, M, t	y
chemeng_p_control_offset	Proportional control: steady-state offset after a set-point step	Kc, K, taus, t_unit	y_ss, offset
chemeng_lag_frequency_response	Frequency response of first-order lags in series (amplitude ratio and phase)	K, taus, t_unit, w	AR, phase_deg
chemeng_pid_controller_output	Discrete PID controller output for an error history	Kc, tauI, tauD, dt, e, bias	u_last, u
chemeng_ziegler_nichols_tuning	Ziegler-Nichols closed-loop tuning of lags in series	K, taus, t_unit, controller	Ku, Pu, Kc, tauI, tauD
chemeng_closed_loop_stability	Closed-loop stability of a P-controlled process (closed-loop poles)	num, den, Kc	max_real, n_rhp, stable
chemeng_pi_loop_margins	Gain and phase margins of a PI loop around lags in series	Kc, tauI, K, taus, t_unit	GM, PM_deg, w_pc, w_gc
chemeng_first_order_step_fit	Process gain and time constant from step-test data (least squares)	t, y, t_unit, M, theta	K, tau
chemeng_radial_darcy_flow	Steady radial flow of oil to a well (Darcy) and productivity index	k, h, mu, pe, pw, re, rw, S	q, PI
chemeng_exponential_decline	Exponential decline curve: rate and cumulative production	qi, Di, t	q, Np
chemeng_hyperbolic_decline	Hyperbolic decline curve: rate and cumulative production	qi, Di, b, t	q, Np
chemeng_harmonic_decline	Harmonic decline curve: rate and cumulative production	qi, Di, t	q, Np
chemeng_havlena_odeh_oil_in_place	Oil initially in place from the material balance (Havlena-Odeh straight line)	F, Et, v_unit	N
chemeng_havlena_odeh_gas_cap	Oil in place and gas-cap size from the material balance (Havlena-Odeh, gas cap)	F, Eo, Eg, v_unit	N, m
chemeng_gas_pz_material_balance	Gas initially in place from the p/z plot	p, z, Gp, p_unit, gp_unit	G, pz_initial

id	operation	inputs	outputs
chemeng_horner_buildup	Pressure build-up test: Horner analysis for permeability and extrapolated pressure	dt, pws, t_unit, p_unit, tp, q, mu, h	m, k, p_star
chemeng_line_source_drawdown	Transient drawdown around a well (line-source solution)	q, mu, k, h, phi, ct, r, t, pi	dp, p
chemeng_gas_well_backpressure	Gas-well deliverability from a back-pressure test (Rawlins-Schellhardt)	q, pwf, q_unit, p_unit, pr	n, log10_C, AOF
chemeng_fenske_minimum_stages	Minimum number of stages at total reflux (Fenske)	alpha, xD, xB	Nmin, Nmin_trays
chemeng_pump_head	Pump total head and power	p1, p2, rho, dz, v1, v2, Q, eta	H, dp_equivalent, P_hydraulic, P_shaft
chemeng_npsh_available	Net positive suction head available	p_s, p_v, rho, z_s, h_f, NPSHr	NPSHa, pressure_head, vapour_head, margin
chemeng_mass_transfer_coefficient	Mass-transfer coefficient from a stated Sherwood correlation	geometry, u, L, D, rho, mu	Re, Sc, Sh, kc
chemeng_multicomponent_underwood	Minimum reflux of a multicomponent column (Underwood)	alpha, z, xD, q, lk, hk	theta, Rmin
chemeng_cstr_volume_second_order	CSTR volume for a conversion (second order)	v0, k, CA0, X	V, tau
chemeng_equimolar_counterdiffusion	Equimolar counterdiffusion flux	D, P, T, L, yA1, yA2	NA, NB
chemeng_modified_raoult_bubble_pressure	Bubble-point pressure by the modified Raoult's law (activity coefficients)	x, gamma, Psat, p_unit	P, y1, y

Blending: feed flows that make a product of given composition — `chemeng_blending_flows`

The mass flow of each feed stream needed to make a product of stated flow and composition, from the component balances (as many feeds as components).

$$\sum_i F_i\, w_{ij} = P\, p_j \quad (j = 1..n)$$

input	meaning	condition
`W`	mass fractions of the feed streams: one row per feed stream, one column per component (rows sum to 1)	
`p`	mass fractions of the product, one per component (sum to 1)	
`P`	product mass flow	

output	meaning
`F1`	flow of the first feed stream
`F`	flow of every feed stream, in kg/s, in the order of the rows of W

- **Method:** component balances sum_i F_i $w_i j = P$ p_j for every component j, solved by Gaussian elimination with partial pivoting
- **Assumes:** as many feed streams as components (2 to 6); each composition sums to 1; the product lies within the reach of the feeds (no negative flow)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_blending_flows', W=[[0.9, 0.1], [0.2, 0.8]], p=[0.5,
0.5], P=100.0)
# -> F1 = 42.8571 kg/s, F = [42.8571428571, 57.1428571429]   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_blending_flows', W=[[0.8, 0.1, 0.1], [0.1, 0.7, 0.2],
[0.05, 0.15, 0.8]], p=[0.4, 0.3, 0.3], P=10.0)
# -> F1 = 4.45783 kg/s, F = [4.4578313253, 3.13253012048, 2.40963855422]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Blending feed flows: W = [[0.9, 0.1], [0.2, 0.8]], p = [0.5, 0.5], P = 100.0 kg/s	SOLVED	F1 = 42.8571 kg/s, F = [42.8571428571, 57.1428571429]
Blending feed flows: W = [[0.8, 0.1, 0.1], [0.1, 0.7, 0.2], [0.05, 0.15, 0.8]], p = [0.4, 0.3, 0.3], P = 10.0 kg/s	SOLVED	F1 = 4.45783 kg/s, F = [4.4578313253, 3.13253012048, 2.40963855422]
Blending feed flows: p = [0.5, 0.5], P = 100.0 kg/s	NEEDS_INPUT	chemeng_blending_flows needs W (mass fractions of the feed streams: one row per feed stream, one column per component (r
Blending feed flows: W = [[0.9, 0.1], [0.2, 0.8]], p = [0.5, 0.5], P = 100.0 kg/s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Blending: W = [[0.9, 0.1], [0.2, 0.8]], p = [0.95, 0.05], P = 100 kg/s	INVALID_INPUT	Blending: feed flows that make a product of given composition: this product cannot be made from these feeds (a feed flow
Blending: W = [[0.9, 0.1], [0.2, 0.8]], p = [0.5, 0.5], P = 360000 kg/h	SOLVED	F1 = 42.8571 kg/s, F = [42.8571428571, 57.1428571429]

Two-product separator balance (binary distillation column) —

`chemeng_binary_separator_balance`

Distillate and bottoms flows of a binary separator from the feed flow and the three compositions, with the recoveries of the light component in the distillate and of the heavy component in the bottoms.

$$D = F\,\frac{z_F - x_B}{x_D - x_B}, \quad B = F - D$$

input	meaning	condition
`F`	feed molar flow	

input	meaning	condition
`zF`	mole fraction of the light component in the feed	
`xD`	mole fraction of the light component in the distillate (top product)	
`xB`	mole fraction of the light component in the bottoms	

output	meaning
`D`	distillate flow
`B`	bottoms flow
`recovery_D`	fraction of the light component recovered in the distillate
`recovery_B`	fraction of the heavy component recovered in the bottoms

- **Method:** total balance $F = D + B$ and light-component balance F $zF = D$ xD + B xB
- **Assumes:** $xB < zF < xD$; steady state, no reaction
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_binary_separator_balance', F=100.0, zF=0.4, xD=0.95,
xB=0.05)
# -> D = 38.8889 mol/s, B = 61.1111 mol/s, recovery_D = 0.923611, recovery_B = 0.967593   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_binary_separator_balance', F=2.5, zF=0.3, xD=0.9,
xB=0.02)
# -> D = 0.795455 mol/s, B = 1.70455 mol/s, recovery_D = 0.954545, recovery_B = 0.954545   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Two-product separator balance: F = 100.0 mol/s, zF = 0.4, xD = 0.95, xB = 0.05	SOLVED	D = 38.8889 mol/s, B = 61.1111 mol/s, recovery_D = 0.923611, recovery_B = 0.967593
Two-product separator balance: F = 2.5 mol/s, zF = 0.3, xD = 0.9, xB = 0.02	SOLVED	D = 0.795455 mol/s, B = 1.70455 mol/s, recovery_D = 0.954545, recovery_B = 0.954545
Two-product separator balance: zF = 0.4, xD = 0.95, xB = 0.05	NEEDS_INPUT	chemeng_binary_separator_balance needs F (feed molar flow). Example: solve('chemeng_binary_separator_balance', F='100 mo
Two-product separator balance: F = 100.0 mol/s, zF = 0.4, xD = 0.95, xB = 0.05, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Binary distillation balance: F = 360 kmol/h, zF = 40%, xD = 95%, xB = 5%	SOLVED	D = 38.8889 mol/s, B = 61.1111 mol/s, recovery_D = 0.923611, recovery_B = 0.967593
Two-product separator balance: F = 100 mol/s, zF = 0.4, xD = 0.3, xB = 0.05	INVALID_INPUT	Two-product separator balance (binary distillation column): the compositions must satisfy xB < zF < xD (light component

question	outcome	result
Two-product separator balance: F = 100, zF = 0.4, xD = 0.95, xB = 0.05	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Reactor with recycle and purge — `chemeng_recycle_purge`

Steady-state flows of a reactor A -> B with complete product separation, recycle of unreacted A and inert, and a purge that removes a fraction of the recycle stream to stop the inert accumulating.

$$A_{in} = \frac{F_0 z_A}{1-(1-f)(1-X)}, \quad I_{in} = \frac{F_0(1-z_A)}{f}$$

input	meaning	condition
`F0`	fresh feed flow (reactant A plus inert)	
`zA`	mole fraction of A in the fresh feed (the rest is inert)	
`X`	single-pass conversion of A in the reactor	
`f`	purge fraction: the fraction of the separator overhead (unreacted A + inert) that is purged	

output	meaning
`R`	recycle flow
`P`	purge flow
`reactor_feed`	combined reactor feed flow
`X_overall`	overall conversion of A (fresh feed to product)
`product`	production rate of B
`y_inert`	mole fraction of inert in the recycle and purge

- **Method:** balances on A and inert around the mixer, reactor, separator and splitter: $A_in = F_0$ zA/(1 - (1 - f)$(1-X)$), inert_in = F0 $(1-zA)/f$
- **Assumes:** $0 < zA \le 1$; $0 < X < 1$; $0 < f < 1$; A -> B one to one; the separator removes all B
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_recycle_purge', F0=100.0, zA=0.98, X=0.25, f=0.1)
# -> R = 221.538 mol/s, P = 24.6154 mol/s, reactor_feed = 321.538 mol/s, X_overall = 0.769231,
product = 75.3846 mol/s, y_inert = 0.08125   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_recycle_purge', F0=50.0, zA=0.9, X=0.6, f=0.05)
# -> R = 122.581 mol/s, P = 6.45161 mol/s, reactor_feed = 172.581 mol/s, X_overall = 0.967742,
product = 43.5484 mol/s, y_inert = 0.775   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Recycle and purge balance: F0 = 100.0 mol/s, zA = 0.98, X = 0.25, f = 0.1	SOLVED	R = 221.538 mol/s, P = 24.6154 mol/s, reactor_feed = 321.538 mol/s, X_overall = 0.769231, product = 75.3846 mol/s, y_inert = 0.08125

question	outcome	result
Recycle and purge balance: F0 = 50.0 mol/s, zA = 0.9, X = 0.6, f = 0.05	SOLVED	R = 122.581 mol/s, P = 6.45161 mol/s, reactor_feed = 172.581 mol/s, X_overall = 0.967742, product = 43.5484 mol/s, y_inert = 0.775
Recycle and purge balance: zA = 0.98, X = 0.25, f = 0.1	NEEDS_INPUT	chemeng_recycle_purge needs F0 (fresh feed flow (reactant A plus inert)). Example: solve('chemeng_recycle_purge', F0='10
Recycle and purge balance: F0 = 100.0 mol/s, zA = 0.98, X = 0.25, f = 0.1, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Recycle and purge balance: F0 = 100 mol/s, zA = 98%, X = 25%, f = 0	INVALID_INPUT	Reactor with recycle and purge: the purge fraction f must be a fraction between 0 and 1.

Unsteady well-mixed tank (concentration change) — `chemeng_unsteady_mixing_tank`

Concentration in a constant-volume well-mixed tank at time t after the inlet concentration changes (inflow equals outflow).

$$C(t) = C_{in} + (C_0 - C_{in})\, e^{-qt/V}$$

input	meaning	condition
`V`	liquid volume of the tank	
`q`	volumetric flow in and out	
`C_in`	inlet concentration	
`C0`	concentration in the tank at $t = 0$	
`t`	time	

output	meaning
`C`	concentration in the tank (and outlet) at time t

- **Method:** solute balance V $dC/dt = q$ (C_in - C), integrated: $C = C_in + (C_0 - C_in)$ exp(-q t/V)
- **Assumes:** $V > 0$, $q > 0$; perfect mixing; constant volume (inflow = outflow)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_unsteady_mixing_tank', V=10.0, q=0.01, C_in=500.0,
C0=0.0, t=1000.0)
# -> C = 316.06 mol/m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_unsteady_mixing_tank', V=2.0, q=0.001, C_in=0.0,
C0=800.0, t=3600.0)
# -> C = 132.239 mol/m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Unsteady mixing tank: V = 10.0 m^3, q = 0.01 m^3/s, C_in =	SOLVED	C = 316.06 mol/m^3

question	outcome	result
500.0 mol/m^3, C0 = 0.0 mol/m^3, t = 1000.0 s		
Unsteady mixing tank: V = 2.0 m^3, q = 0.001 m^3/s, C_in = 0.0 mol/m^3, C0 = 800.0 mol/m^3, t = 3600.0 s	SOLVED	C = 132.239 mol/m^3
Unsteady mixing tank: q = 0.01 m^3/s, C_in = 500.0 mol/m^3, C0 = 0.0 mol/m^3, t = 1000.0 s	NEEDS_INPUT	chemeng_unsteady_mixing_tank needs V (liquid volume of the tank). Example: solve('chemeng_unsteady_mixing_tank', V='10 m
Unsteady mixing tank: V = 10.0 m^3, q = 0.01 m^3/s, C_in = 500.0 mol/m^3, C0 = 0.0 mol/m^3, t = 1000.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Unsteady mixing tank: V = 10 m^3, q = 36 m^3/h, C_in = 0.5 mol/L, C0 = 0 mol/L, t = 1000 s	SOLVED	C = 316.06 mol/m^3

Filling or draining well-mixed tank (variable volume) — `chemeng_variable_volume_tank`

Volume and concentration of a well-mixed tank whose inflow and outflow differ, at time t.

$$C = C_{in} - (C_{in} - C_0)\left(\frac{V_0}{V}\right)^{q_{in}/(q_{in}-q_{out})},\ V = V_0 + (q_{in} - q_{out})t$$

input	meaning	condition
`V0`	liquid volume at $t = 0$	
`q_in`	volumetric inflow	
`q_out`	volumetric outflow	
`C_in`	inlet concentration	
`C0`	concentration in the tank at $t = 0$	
`t`	time	

output	meaning
`V`	liquid volume at time t
`C`	concentration at time t

- **Method:** total balance dV/dt = q_in - q_out and solute balance d(VC)/dt = q_in C_in - q_out C, integrated: $C = C_in - (C_in - C_0)$(V0/V)^(q_in/(q_in - q_out))
- **Assumes:** $V_0 > 0$, $q_in > 0$; the tank does not empty before t; perfect mixing
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_variable_volume_tank', V0=1.0, q_in=0.002, q_out=0.001,
C_in=1000.0, C0=0.0, t=1000.0)
# -> V = 2 m^3, C = 750 mol/m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_variable_volume_tank', V0=5.0, q_in=0.001, q_out=0.003,
C_in=0.0, C0=500.0, t=1000.0)
# -> V = 3 m^3, C = 387.298 mol/m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Variable-volume tank: V0 = 1.0 m^3, q_in = 0.002 m^3/s, q_out = 0.001 m^3/s, C_in = 1000.0 mol/m^3, C0 = 0.0 mol/m^3, t = 1000.0 s	SOLVED	V = 2 m^3, C = 750 mol/m^3
Variable-volume tank: V0 = 5.0 m^3, q_in = 0.001 m^3/s, q_out = 0.003 m^3/s, C_in = 0.0 mol/m^3, C0 = 500.0 mol/m^3, t = 1000.0 s	SOLVED	V = 3 m^3, C = 387.298 mol/m^3
Variable-volume tank: q_in = 0.002 m^3/s, q_out = 0.001 m^3/s, C_in = 1000.0 mol/m^3, C0 = 0.0 mol/m^3, t = 1000.0 s	NEEDS_INPUT	chemeng_variable_volume_tank needs V0 (liquid volume at t = 0). Example: solve('chemeng_variable_volume_tank', V0='1 m^3
Variable-volume tank: V0 = 1.0 m^3, q_in = 0.002 m^3/s, q_out = 0.001 m^3/s, C_in = 1000.0 mol/m^3, C0 = 0.0 mol/m^3, t = 1000.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Variable-volume tank: V0 = 1 m^3, q_in = 0.001 m^3/s, q_out = 0.003 m^3/s, C_in = 0 mol/m^3, C0 = 500 mol/m^3, t = 1000 s	INVALID_INPUT	Filling or draining well-mixed tank (variable volume): the tank would be empty before time t.

Combustion with excess air: air requirement and flue-gas composition —

`chemeng_combustion_flue_gas`

Theoretical oxygen, air supplied, flue-gas moles and wet and dry compositions, and the air-fuel mass ratio for complete combustion of a fuel with excess dry air (21 % O2, 79 % N2).

$$n_{O_2,th} = x + \tfrac{y}{4} + v - \tfrac{z}{2}, \quad n_{air} = \frac{(1+e)\,n_{O_2,th}}{0.21}$$

input	meaning	condition
`fuel`	molecular formula of the fuel (C, H, O, N and S), e.g. CH4, C3H8, C2H5OH	
`excess_air`	excess air as a fraction of the theoretical air (20% = 0.2)	

output	meaning
`O2_theoretical`	theoretical O2, mol per mol fuel
`air`	air supplied, mol per mol fuel
`n_CO2`	CO2 in the flue gas, mol per mol fuel
`n_H2O`	H2O, mol per mol fuel
`n_SO2`	SO2, mol per mol fuel
`n_O2`	O2, mol per mol fuel

output	meaning
`n_N2`	N2, mol per mol fuel
`y_CO2_dry`	CO2 mole fraction, dry basis
`y_O2_dry`	O2 mole fraction, dry basis
`y_H2O_wet`	H2O mole fraction, wet basis
`AFR_mass`	air-fuel ratio, kg air per kg fuel

- **Method:** O2 theoretical = C + H/4 + S - O/2 per mol fuel; air = (1 + e) $O_2/0.21$; products by element balance
- **Assumes:** complete combustion to CO2, H2O and SO2; fuel N leaves as N2; dry air 21 % $O_2/79$ % N2; 0 <= excess air <= 500 %
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_combustion_flue_gas', fuel='CH4', excess_air=0.2)
# -> O2_theoretical = 2, air = 11.4286, n_CO2 = 1, n_H2O = 2, n_SO2 = 0, n_O2 = 0.4, n_N2 =
9.02857, y_CO2_dry = 0.0958904, y_O2_dry = 0.0383562, y_H2O_wet = 0.16092, AFR_mass = 20.5528
(SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_combustion_flue_gas', fuel='C3H8', excess_air=0.0)
# -> O2_theoretical = 5, air = 23.8095, n_CO2 = 3, n_H2O = 4, n_SO2 = 0, n_O2 = 0, n_N2 = 18.8095,
y_CO2_dry = 0.137555, y_O2_dry = 0, y_H2O_wet = 0.154982, AFR_mass = 15.5778   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_combustion_flue_gas', fuel='C2H5OH', excess_air=0.5)
# -> O2_theoretical = 3, air = 21.4286, n_CO2 = 2, n_H2O = 3, n_SO2 = 0, n_O2 = 1.5, n_N2 =
16.9286, y_CO2_dry = 0.0979021, y_O2_dry = 0.0734266, y_H2O_wet = 0.128049, AFR_mass = 13.4196
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Combustion flue gas: fuel = CH4, excess_air = 0.2	SOLVED	O2_theoretical = 2, air = 11.4286, n_CO2 = 1, n_H2O = 2, n_SO2 = 0, n_O2 = 0.4, n_N2 = 9.02857, y_CO2_dry = 0.0958904, y_O2_dry = 0.0383562, y_H2O_wet = 0.16092, AFR_mass = 20.5528
Combustion flue gas: fuel = C3H8, excess_air = 0.0	SOLVED	O2_theoretical = 5, air = 23.8095, n_CO2 = 3, n_H2O = 4, n_SO2 = 0, n_O2 = 0, n_N2 = 18.8095, y_CO2_dry = 0.137555, y_O2_dry = 0, y_H2O_wet = 0.154982, AFR_mass = 15.5778
Combustion flue gas: fuel = C2H5OH, excess_air = 0.5	SOLVED	O2_theoretical = 3, air = 21.4286, n_CO2 = 2, n_H2O = 3, n_SO2 = 0, n_O2 = 1.5, n_N2 = 16.9286, y_CO2_dry = 0.0979021, y_O2_dry = 0.0734266, y_H2O_wet = 0.128049, AFR_mass = 13.4196
Combustion flue gas: excess_air = 0.2	NEEDS_INPUT	chemeng_combustion_flue_gas needs fuel (molecular formula of the fuel (C, H, O, N and S), e.g. CH4, C3H8, C2H5OH). Examp
Combustion flue gas: fuel = CH4, excess_air = 0.2, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Combustion flue gas: fuel = CH4, excess_air = 20%	SOLVED	O2_theoretical = 2, air = 11.4286, n_CO2 = 1, n_H2O = 2, n_SO2 = 0, n_O2 = 0.4, n_N2 = 9.02857, y_CO2_dry = 0.0958904, y_O2_dry = 0.0383562, y_H2O_wet = 0.16092, AFR_mass = 20.5528

question	outcome	result
Combustion flue gas: fuel = CH4, excess_air = 20	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Combustion flue gas: fuel = CH4Cl, excess_air = 0.2	INVALID_INPUT	Combustion with excess air: air requirement and flue-gas composition: element Cl is not supported (fuels of C, H, O, N a

Sensible heat with a polynomial heat capacity — `chemeng_sensible_heat_cp_polynomial`

Heat needed to take n moles from T1 to T2 when $cp = a + bT + cT^2 + dT^3$ (J/(mol K), T in K), and the mean heat capacity over the range.

$$Q = n \int_{T_1}^{T_2} (a + bT + cT^2 + dT^3)\, dT$$

input	meaning	condition
`n`	amount of substance	
`a`	cp coefficient a, J/(mol K)	
`b`	cp coefficient b, J/(mol K^2)	
`c`	cp coefficient c, J/(mol K^3) (default 0)	
`d`	cp coefficient d, J/(mol K^4) (default 0)	
`T1`	initial temperature	
`T2`	final temperature	

output	meaning
`Q`	heat added (negative when cooled)
`cp_mean`	mean heat capacity between T1 and T2

- **Method:** Q = n * integral of cp dT, integrated term by term
- **Assumes:** no phase change between T1 and T2; T in kelvin inside the polynomial
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_sensible_heat_cp_polynomial', n=10.0, a=29.0, b=0.002,
c=0.0, d=0.0, T1=300.0, T2=500.0)
# -> Q = 59600 J, cp_mean = 29.8 J/(mol*K)   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_sensible_heat_cp_polynomial', n=1.0, a=19.89, b=0.05024,
c=1.269e-05, d=-1.101e-08, T1=298.15, T2=1000.0)
# -> Q = 38233.9 J, cp_mean = 54.4759 J/(mol*K)   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Sensible heat with polynomial cp: n = 10.0 mol, a = 29.0, b =	SOLVED	Q = 59600 J, cp_mean = 29.8 J/(mol*K)

question	outcome	result
0.002, c = 0.0, d = 0.0, T1 = 300.0 K, T2 = 500.0 K		
Sensible heat with polynomial cp: n = 1.0 mol, a = 19.89, b = 0.05024, c = 1.269e-05, d = -1.101e-08, T1 = 298.15 K, T2 = 1000.0 K	SOLVED	Q = 38233.9 J, cp_mean = 54.4759 J/(mol*K)
Sensible heat with polynomial cp: a = 29.0, b = 0.002, c = 0.0, d = 0.0, T1 = 300.0 K, T2 = 500.0 K	NEEDS_INPUT	chemeng_sensible_heat_cp_polynomial needs n (amount of substance). Example: solve('chemeng_sensible_heat_cp_polynomial',
Sensible heat with polynomial cp: n = 10.0 mol, a = 29.0, b = 0.002, c = 0.0, d = 0.0, T1 = 300.0 K, T2 = 500.0 K, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Sensible heat: n = 10 mol, a = 29, b = 0.002, T1 = 26.85 degC, T2 = 226.85 degC	SOLVED	Q = 59600 J, cp_mean = 29.8 J/(mol*K)

Heat of reaction at another temperature (Kirchhoff) —

`chemeng_reaction_enthalpy_at_temperature`

The heat of reaction at T from its value at T0 and the polynomial heat capacities of the species (Kirchhoff's law).

$$\Delta H_r(T) = \Delta H_r(T_0) + \int_{T_0}^{T} \sum_i \nu_i c_{p,i}\, dT$$

input	meaning	condition
`nu`	stoichiometric coefficients, negative for reactants and positive for products	
`cp`	heat-capacity coefficients, one row (a, b, c, d) per species in the order of nu; $cp = a + bT + cT^2 + dT^3$ in J/(mol K)	
`dH0`	heat of reaction at T0, per mole of extent	
`T0`	temperature of the known heat of reaction	
`T`	temperature wanted	

output	meaning
`dH`	heat of reaction at T, per mole of extent

- **Method:** dH(T) = dH(T0) + integral from T0 to T of sum_i nu_i cp_i dT
- **Assumes:** no phase change of any species between T0 and T
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_reaction_enthalpy_at_temperature', nu=[-1.0, -3.0, 2.0],
cp=[[27.27, 0.004927, 0.0, 0.0], [27.01, 0.003509, 0.0, 0.0], [24.62, 0.03753, -7.1e-06, 0.0]],
dH0=-92220.0, T0=298.15, T=700.0)
# -> dH = -105497 J/mol   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_reaction_enthalpy_at_temperature', nu=[-1.0, -0.5, 1.0],
cp=[[29.0, 0.0], [29.4, 0.004], [30.0, 0.01]], dH0=-241826.0, T0=298.15, T=1000.0)
# -> dH = -247797 J/mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Heat of reaction at temperature: nu = [-1.0, -3.0, 2.0], cp = [[27.27, 0.004927, 0.0, 0.0], [27.01, 0.003509, 0.0, 0.0], [24.62, 0.03753, -7.1e-06, 0.0]], dH0 = -92220.0 J/mol, T0 = 298.15 K, T = 700.0 K	SOLVED	dH = -105497 J/mol
Heat of reaction at temperature: nu = [-1.0, -0.5, 1.0], cp = [[29.0, 0.0], [29.4, 0.004], [30.0, 0.01]], dH0 = -241826.0 J/mol, T0 = 298.15 K, T = 1000.0 K	SOLVED	dH = -247797 J/mol
Heat of reaction at temperature: cp = [[27.27, 0.004927, 0.0, 0.0], [27.01, 0.003509, 0.0, 0.0], [24.62, 0.03753, -7.1e-06, 0.0]], dH0 = -92220.0 J/mol, T0 = 298.15 K, T = 700.0 K	NEEDS_INPUT	chemeng_reaction_enthalpy_at_temperature needs nu (stoichiometric coefficients, negative for reactants and positive for
Heat of reaction at temperature: nu = [-1.0, -3.0, 2.0], cp = [[27.27, 0.004927, 0.0, 0.0], [27.01, 0.003509, 0.0, 0.0], [24.62, 0.03753, -7.1e-06, 0.0]], dH0 = -92220.0 J/mol, T0 = 298.15 K, T = 700.0 K, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Adiabatic reactor outlet temperature — `chemeng_adiabatic_reactor_temperature`

Outlet temperature of an adiabatic reactor from the feed, the conversion of the limiting reactant, the heat of reaction at a reference temperature and polynomial heat capacities.

$$\sum_i n_{i0} \int_{T_{in}}^{T_{ref}} c_{p,i} dT + \xi \, \Delta H_r(T_{ref}) + \sum_i n_i \int_{T_{ref}}^{T_{out}} c_{p,i} dT = 0$$

input	meaning	condition
`n0`	feed amounts of each species, mol (or mol/s) on one basis	
`nu`	stoichiometric coefficients, negative for reactants	
`cp`	heat-capacity coefficients, one row (a, b, c, d) per species; $cp = a + bT + cT^2 + dT^3$ in J/(mol K)	
`dH`	heat of reaction at T_ref, per mole of extent	
`T_ref`	reference temperature of dH	
`T_in`	feed temperature	
`key`	number of the limiting reactant in the species list (1 = first)	
`X`	conversion of the limiting reactant	

output	meaning
`T_out`	adiabatic outlet temperature

- **Method:** energy balance along the path: cool the feed to T_ref, react at T_ref, heat the products to T_out; T_out by bisection
- **Assumes:** adiabatic, no shaft work, no phase change; outlet between 200 and 3200 K
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_adiabatic_reactor_temperature', n0=[1.0, 2.0, 0.0], nu=
[-1.0, -1.0, 1.0], cp=[[30.0, 0.01], [29.0, 0.004], [45.0, 0.02]], dH=-60000.0, T_ref=298.15,
T_in=400.0, key=1, X=0.8)
# -> T_out = 932.348 K   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_adiabatic_reactor_temperature', n0=[2.0, 0.0], nu=[-1.0,
2.0], cp=[[40.0, 0.02], [25.0, 0.01]], dH=30000.0, T_ref=298.15, T_in=900.0, key=1, X=0.3)
# -> T_out = 717.393 K   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Adiabatic reactor temperature: n0 = [1.0, 2.0, 0.0], nu = [-1.0, -1.0, 1.0], cp = [[30.0, 0.01], [29.0, 0.004], [45.0, 0.02]], dH = -60000.0 J/mol, T_ref = 298.15 K, T_in = 400.0 K, key = 1, X = 0.8	SOLVED	T_out = 932.348 K
Adiabatic reactor temperature: n0 = [2.0, 0.0], nu = [-1.0, 2.0], cp = [[40.0, 0.02], [25.0, 0.01]], dH = 30000.0 J/mol, T_ref = 298.15 K, T_in = 900.0 K, key = 1, X = 0.3	SOLVED	T_out = 717.393 K
Adiabatic reactor temperature: nu = [-1.0, -1.0, 1.0], cp = [[30.0, 0.01], [29.0, 0.004], [45.0, 0.02]], dH = -60000.0 J/mol, T_ref = 298.15 K, T_in = 400.0 K, key = 1, X = 0.8	NEEDS_INPUT	chemeng_adiabatic_reactor_temperature needs n0 (feed amounts of each species, mol (or mol/s) on one basis). Example: sol
Adiabatic reactor temperature: n0 = [1.0, 2.0, 0.0], nu = [-1.0, -1.0, 1.0], cp = [[30.0, 0.01], [29.0, 0.004], [45.0, 0.02]], dH = -60000.0 J/mol, T_ref = 298.15 K, T_in = 400.0 K, key = 1, X = 0.8, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Adiabatic flame temperature (complete combustion, NASA polynomials) — `chemeng_adiabatic_flame_temperature`

Adiabatic flame temperature of a gaseous fuel burned with excess dry air at constant pressure, complete combustion without dissociation, from NASA 7-coefficient polynomials (GRI-Mech 3.0 data).

$$\sum_p n_p\, h_p(T_{ad}) = \sum_r n_r\, h_r(T_{in}), \qquad \frac{h}{RT} = a_1 + \frac{a_2 T}{2} + \frac{a_3 T^2}{3} + \frac{a_4 T^3}{4} + \frac{a_5 T^4}{5} + \frac{a_6}{T}$$

input	meaning	condition
`fuel`	fuel: CH4, C2H6, C3H8, H2, CO, C2H4, C2H2, CH3OH	
`excess_air`	excess air as a fraction of theoretical (0 = stoichiometric)	
`T_in`	temperature of the fuel and air entering	

output	meaning
`T_ad`	adiabatic flame temperature

- **Method:** energy balance H_products(T_ad) = H_reactants(T_in) with NASA polynomial enthalpies (including formation enthalpies), solved by bisection
- **Assumes:** complete combustion to CO2 and H2O vapour, no dissociation; 300 $K \leq T_in \leq 1000$ K; result <= 3500 K
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_adiabatic_flame_temperature', fuel='CH4',
excess_air=0.0, T_in=300.0)
# -> T_ad = 2326.32 K   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_adiabatic_flame_temperature', fuel='C3H8',
excess_air=0.5, T_in=400.0)
# -> T_ad = 1906.88 K   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Adiabatic flame temperature: fuel = CH4, excess_air = 0.0, T_in = 300.0 K	SOLVED	T_ad = 2326.32 K
Adiabatic flame temperature: fuel = C3H8, excess_air = 0.5, T_in = 400.0 K	SOLVED	T_ad = 1906.88 K
Adiabatic flame temperature: excess_air = 0.0, T_in = 300.0 K	NEEDS_INPUT	chemeng_adiabatic_flame_temperature needs fuel (fuel: CH4, C2H6, C3H8, H2, CO, C2H4, C2H2, CH3OH). Example: solve('cheme
Adiabatic flame temperature: fuel = CH4, excess_air = 0.0, T_in = 300.0 K, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Adiabatic flame temperature: fuel = CH4, excess_air = 0%, T_in = 26.85 degC	SOLVED	T_ad = 2326.32 K
Adiabatic flame temperature: fuel = C8H18, excess_air = 0.2, T_in = 300 K	INVALID_INPUT	fuel: one of CH4, C2H6, C3H8, H2, CO, C2H4, C2H2, CH3OH; got 'c8h18'.

Batch heating of a stirred tank: temperature after a time — `chemeng_batch_heating_temperature`

Temperature of a well-stirred batch heated or cooled by a medium at constant temperature (condensing steam, boiling refrigerant) after time t.

$$T = T_s - (T_s - T_0)\, e^{-UA\, t/(m c_p)}$$

input	meaning	condition
`m`	mass of liquid in the tank	
`cp`	specific heat of the liquid	
`UA`	overall heat-transfer coefficient times area of the coil or jacket	
`T_s`	temperature of the heating or cooling medium (constant, e.g. condensing steam)	
`T0`	initial liquid temperature	
`t`	heating time	

output	meaning
`T`	liquid temperature at time t

- **Method:** energy balance m cp $dT/dt = UA$ (T_s - T), integrated
- **Assumes:** well stirred; constant UA and cp; medium at constant temperature
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_batch_heating_temperature', m=1000.0, cp=4180.0,
UA=2000.0, T_s=393.15, T0=293.15, t=3600.0)
# -> T = 375.288 K   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_batch_heating_temperature', m=500.0, cp=2000.0,
UA=500.0, T_s=273.15, T0=353.15, t=1800.0)
# -> T = 305.676 K   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Batch heating temperature: m = 1000.0 kg, cp = 4180.0 J/(kg*K), UA = 2000.0 W/K, T_s = 393.15 K, T0 = 293.15 K, t = 3600.0 s	SOLVED	T = 375.288 K
Batch heating temperature: m = 500.0 kg, cp = 2000.0 J/(kg*K), UA = 500.0 W/K, T_s = 273.15 K, T0 = 353.15 K, t = 1800.0 s	SOLVED	T = 305.676 K
Batch heating temperature: cp = 4180.0 J/(kg*K), UA = 2000.0 W/K, T_s = 393.15 K, T0 = 293.15 K, t = 3600.0 s	NEEDS_INPUT	chemeng_batch_heating_temperature needs m (mass of liquid in the tank). Example: solve('chemeng_batch_heating_temperatur
Batch heating temperature: m = 1000.0 kg, cp = 4180.0 J/(kg*K), UA = 2000.0 W/K, T_s = 393.15 K, T0 = 293.15 K, t = 3600.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Batch heating temperature: m = 1 t, cp = 4.18 kJ/(kg*K), UA = 2 kW/K,	SOLVED	T = 375.288 K

question	outcome	result
T_s = 120 degC, T0 = 20 degC, t = 1 h		

Batch heating of a stirred tank: time to reach a temperature — `chemeng_batch_heating_time`

Time for a well-stirred batch heated or cooled by a medium at constant temperature to reach a stated temperature.

$$t = \frac{mc_p}{UA} \ln \frac{T_s - T_0}{T_s - T}$$

input	meaning	condition
`m`	mass of liquid in the tank	
`cp`	specific heat of the liquid	
`UA`	overall heat-transfer coefficient times area of the coil or jacket	
`T_s`	temperature of the heating or cooling medium (constant, e.g. condensing steam)	
`T0`	initial liquid temperature	
`T`	temperature to reach	

output	meaning
`t`	time needed

- **Method:** t = (m cp/UA) ln($(T_s - T_0)/(T_s - T)$)
- **Assumes:** T between T0 and T_s; well stirred; constant UA and cp
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_batch_heating_time', m=1000.0, cp=4180.0, UA=2000.0,
T_s=393.15, T0=293.15, T=353.15)
# -> t = 1915.05 s   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_batch_heating_time', m=500.0, cp=2000.0, UA=500.0,
T_s=273.15, T0=353.15, T=300.0)
# -> t = 2183.52 s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Batch heating time: m = 1000.0 kg, cp = 4180.0 J/(kg*K), UA = 2000.0 W/K, T_s = 393.15 K, T0 = 293.15 K, T = 353.15 K	SOLVED	t = 1915.05 s
Batch heating time: m = 500.0 kg, cp = 2000.0 J/(kg*K), UA = 500.0 W/K, T_s = 273.15 K, T0 = 353.15 K, T = 300.0 K	SOLVED	t = 2183.52 s

question	outcome	result
Batch heating time: cp = 4180.0 J/(kg*K), UA = 2000.0 W/K, T_s = 393.15 K, T0 = 293.15 K, T = 353.15 K	NEEDS_INPUT	chemeng_batch_heating_time needs m (mass of liquid in the tank). Example: solve('chemeng_batch_heating_time', m='1000 kg
Batch heating time: m = 1000.0 kg, cp = 4180.0 J/(kg*K), UA = 2000.0 W/K, T_s = 393.15 K, T0 = 293.15 K, T = 353.15 K, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Batch heating time: m = 1000 kg, cp = 4180 J/(kg*K), UA = 2000 W/K, T_s = 393.15 K, T0 = 293.15 K, T = 400 K	INVALID_INPUT	Batch heating of a stirred tank: time to reach a temperature: T must lie between T0 and the medium temperature T_s.

Batch cooling (or heating) with a non-isothermal coolant in a coil — `chemeng_batch_coil_cooling`

Temperature of a well-stirred batch after time t when a coolant (or heating liquid) flows once through a coil and leaves warmer (Kern's batch cooling).

$$T = t_{in} + (T_0 - t_{in}) \exp\left[-\frac{W c_{p,c}\left(1 - e^{-UA/(W c_{p,c})}\right) t}{M c_p}\right]$$

input	meaning	condition
`M`	mass of liquid in the tank	
`cp`	specific heat of the batch liquid	
`W`	coolant mass flow	
`cp_c`	specific heat of the coolant	
`t_in`	coolant inlet temperature	
`UA`	overall heat-transfer coefficient times coil area	
`T0`	initial batch temperature	
`t`	time	

output	meaning
`T`	batch temperature at time t

- **Method:** coolant outlet approaches the batch temperature: T - t_out = (T - t_in) exp(-UA/(W cp_c)); tank balance M cp $dT/dt = -W$ cp_c (t_out - t_in), integrated
- **Assumes:** well-stirred batch; constant UA, cp and coolant flow
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_batch_coil_cooling', M=5000.0, cp=2500.0, W=2.0,
cp_c=4180.0, t_in=288.15, UA=3000.0, T0=393.15, t=7200.0)
# -> T = 312.732 K   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_batch_coil_cooling', M=1000.0, cp=4180.0, W=0.5,
cp_c=4180.0, t_in=363.15, UA=1000.0, T0=293.15, t=3600.0)
# -> T = 327.845 K   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Batch coil cooling: M = 5000.0 kg, cp = 2500.0 J/(kg*K), W = 2.0 kg/s, cp_c = 4180.0 J/(kg*K), t_in = 288.15 K, UA = 3000.0 W/K, T0 = 393.15 K, t = 7200.0 s	SOLVED	T = 312.732 K
Batch coil cooling: M = 1000.0 kg, cp = 4180.0 J/(kg*K), W = 0.5 kg/s, cp_c = 4180.0 J/(kg*K), t_in = 363.15 K, UA = 1000.0 W/K, T0 = 293.15 K, t = 3600.0 s	SOLVED	T = 327.845 K
Batch coil cooling: cp = 2500.0 J/(kg*K), W = 2.0 kg/s, cp_c = 4180.0 J/(kg*K), t_in = 288.15 K, UA = 3000.0 W/K, T0 = 393.15 K, t = 7200.0 s	NEEDS_INPUT	chemeng_batch_coil_cooling needs M (mass of liquid in the tank). Example: solve('chemeng_batch_coil_cooling', M='5000 kg
Batch coil cooling: M = 5000.0 kg, cp = 2500.0 J/(kg*K), W = 2.0 kg/s, cp_c = 4180.0 J/(kg*K), t_in = 288.15 K, UA = 3000.0 W/K, T0 = 393.15 K, t = 7200.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Batch reactor time for a conversion (first order) — `chemeng_batch_time_first_order`

Reaction time an isothermal batch reactor needs to reach conversion X for a first-order reaction.

$$t = \frac{1}{k} \ln \frac{1}{1 - X}$$

input	meaning	condition
`k`	first-order rate constant	
`X`	conversion of A wanted	

output	meaning
`t`	reaction time

- **Method:** design equation t = C_A0 * integral of $dX/(-r_A) = ln(1/(1 - X))$/k
- **Assumes:** $0 \leq X < 1$; isothermal, constant volume, $-r_A = k$ C_A
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_batch_time_first_order', k=0.001, X=0.9)
# -> t = 2302.59 s   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_batch_time_first_order', k=0.05, X=0.5)
# -> t = 13.8629 s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
First-order batch reactor time: k = 0.001 1/s, X = 0.9	SOLVED	t = 2302.59 s
First-order batch reactor time: k = 0.05 1/s, X = 0.5	SOLVED	t = 13.8629 s
First-order batch reactor time: X = 0.9	NEEDS_INPUT	chemeng_batch_time_first_order needs k (first-order rate constant). Example: solve('chemeng_batch_time_first_order', k='
First-order batch reactor time: k = 0.001 1/s, X = 0.9, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
First-order batch reactor time: k = 0.06 1/min, X = 90%	SOLVED	t = 2302.59 s
First-order batch reactor time: k = 0.001 1/s, X = 1	INVALID_INPUT	Batch reactor time for a conversion (first order): the conversion X must be a fraction between 0 and 1.

Batch reactor time for a conversion (second order) — `chemeng_batch_time_second_order`

Reaction time an isothermal batch reactor needs to reach conversion X when -r_A = k C_A^2.

$$t = \frac{X}{k C_{A0}(1 - X)}$$

input	meaning	condition
`k`	second-order rate constant (-r_A = k C_A^2)	
`CA0`	feed (initial) concentration of A	
`X`	conversion of A wanted	

output	meaning
`t`	reaction time

- **Method:** t = C_A0 * integral of dX/(k $C_A 0^2\ (1 - X)^2$) $= X/(k$ C_A0 $(1 - X)$)
- **Assumes:** $0 \leq X < 1$; isothermal, constant volume
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_batch_time_second_order', k=0.0001, CA0=1000.0, X=0.8)
# -> t = 40 s   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_batch_time_second_order', k=5e-06, CA0=2000.0, X=0.5)
# -> t = 100 s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Second-order batch reactor time: k = 0.0001 m^3/(mol*s), CA0 = 1000.0 mol/m^3, X = 0.8	SOLVED	t = 40 s
Second-order batch reactor time: k = 5e-06 m^3/(mol*s), CA0 = 2000.0 mol/m^3, X = 0.5	SOLVED	t = 100 s
Second-order batch reactor time: CA0 = 1000.0 mol/m^3, X = 0.8	NEEDS_INPUT	chemeng_batch_time_second_order needs k (second-order rate constant (-r_A = k C_A^2)). Example: solve('chemeng_batch_tim
Second-order batch reactor time: k = 0.0001 m^3/(mol*s), CA0 = 1000.0 mol/m^3, X = 0.8, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Second-order batch reactor time: k = 0.1 L/(mol*s), CA0 = 1 mol/L, X = 0.8	SOLVED	t = 40 s

Batch reactor conversion after a time (first order) — `chemeng_batch_conversion_first_order`

Conversion reached in an isothermal batch reactor after time t for a first-order reaction.

$$X = 1 - e^{-kt}$$

input	meaning	condition
`k`	first-order rate constant	
`t`	reaction time	

output	meaning
`X`	conversion of A

- **Method:** $X = 1 - exp$(-k t)
- **Assumes:** isothermal, constant volume, $-r_A = k$ C_A
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_batch_conversion_first_order', k=0.001, t=1000.0)
# -> X = 0.632121   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_batch_conversion_first_order', k=0.02, t=60.0)
# -> X = 0.698806   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
First-order batch reactor conversion: k = 0.001 1/s, t = 1000.0 s	SOLVED	X = 0.632121
First-order batch reactor conversion: k = 0.02 1/s, t =	SOLVED	X = 0.698806

question	outcome	result
60.0 s		
First-order batch reactor conversion: t = 1000.0 s	NEEDS_INPUT	chemeng_batch_conversion_first_order needs k (first-order rate constant). Example: solve('chemeng_batch_conversion_first
First-order batch reactor conversion: k = 0.001 1/s, t = 1000.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
First-order batch reactor conversion: k = 0.06 1/min, t = 1000 s	SOLVED	X = 0.632121

Batch reactor conversion after a time (second order) — `chemeng_batch_conversion_second_order`

Conversion and concentration reached in an isothermal batch reactor after time t when -r_A = k C_A^2.

$$X = \frac{kC_{A0}t}{1 + kC_{A0}t}$$

input	meaning	condition
`k`	second-order rate constant (-r_A = k C_A^2)	
`CA0`	feed (initial) concentration of A	
`t`	reaction time	

output	meaning
`X`	conversion of A
`CA`	concentration of A at time t

- **Method:** $1/C_A = 1/C_A0 + k$ t
- **Assumes:** isothermal, constant volume
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_batch_conversion_second_order', k=0.0001, CA0=1000.0,
t=40.0)
# -> X = 0.8, CA = 200 mol/m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_batch_conversion_second_order', k=5e-06, CA0=2000.0,
t=600.0)
# -> X = 0.857143, CA = 285.714 mol/m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Second-order batch reactor conversion: k = 0.0001 m^3/(mol*s), CA0 = 1000.0 mol/m^3, t = 40.0 s	SOLVED	X = 0.8, CA = 200 mol/m^3

question	outcome	result
Second-order batch reactor conversion: k = 5e-06 m^3/(mol*s), CA0 = 2000.0 mol/m^3, t = 600.0 s	SOLVED	X = 0.857143, CA = 285.714 mol/m^3
Second-order batch reactor conversion: CA0 = 1000.0 mol/m^3, t = 40.0 s	NEEDS_INPUT	chemeng_batch_conversion_second_order needs k (second-order rate constant (-r_A = k C_A^2)). Example: solve('chemeng_bat
Second-order batch reactor conversion: k = 0.0001 m^3/(mol*s), CA0 = 1000.0 mol/m^3, t = 40.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

CSTR volume for a conversion (first order) — `chemeng_cstr_volume_first_order`

Volume and space time of a CSTR that reaches conversion X for a first-order liquid-phase reaction.

$$V = \frac{v_0\, X}{k(1 - X)}$$

input	meaning	condition
`v0`	volumetric feed flow	
`k`	first-order rate constant	
`X`	conversion of A wanted	

output	meaning
`V`	reactor volume
`tau`	space time V/v_0

- **Method:** design equation $V = F_A0$ X/(-r_A) at the exit concentration: $V = v_0$ X/(k $(1 - X)$)
- **Assumes:** $0 \le X < 1$; steady state, perfect mixing, constant density, $-r_A = k$ C_A
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_cstr_volume_first_order', v0=0.01, k=0.001, X=0.9)
# -> V = 90 m^3, tau = 9000 s   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_cstr_volume_first_order', v0=0.002, k=0.05, X=0.5)
# -> V = 0.04 m^3, tau = 20 s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
First-order CSTR volume: v0 = 0.01 m^3/s, k = 0.001 1/s, X = 0.9	SOLVED	V = 90 m^3, tau = 9000 s

question	outcome	result
First-order CSTR volume: v0 = 0.002 m^3/s, k = 0.05 1/s, X = 0.5	SOLVED	V = 0.04 m^3, tau = 20 s
First-order CSTR volume: k = 0.001 1/s, X = 0.9	NEEDS_INPUT	chemeng_cstr_volume_first_order needs v0 (volumetric feed flow). Example: solve('chemeng_cstr_volume_first_order', v0='0
First-order CSTR volume: v0 = 0.01 m^3/s, k = 0.001 1/s, X = 0.9, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
First-order CSTR volume: v0 = 36 m^3/h, k = 0.06 1/min, X = 90%	SOLVED	V = 90 m^3, tau = 9000 s

CSTR conversion for a volume (first order) — `chemeng_cstr_conversion_first_order`

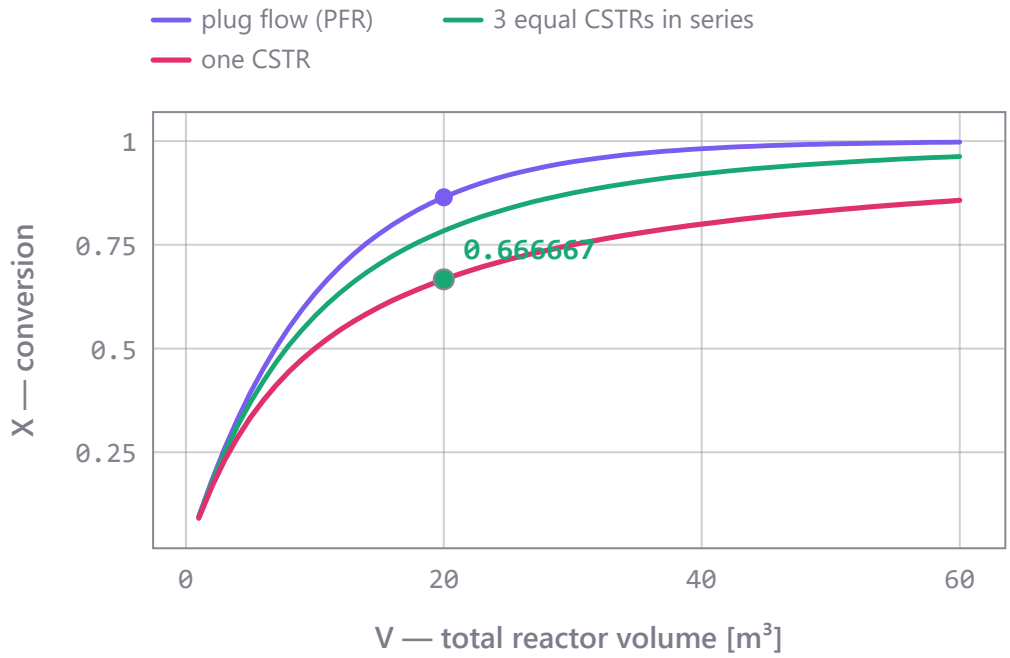
Steady-state conversion of a CSTR of volume V for a first-order liquid-phase reaction.

$$X = \frac{k\tau}{1 + k\tau},\ \tau = V/v_0$$

input	meaning	condition
`v0`	volumetric feed flow	
`V`	reactor volume	
`k`	first-order rate constant	

output	meaning
`X`	conversion of A

- **Method:** $X = k$ tau/(1 + k tau), $\tau = V/v_0$ (the Damkohler number k tau)
- **Assumes:** steady state, perfect mixing, constant density, $-r_A = k$ C_A
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Conversion of a first-order reaction (k = 0.001 1/s, feed 0.01 m³/s) against total volume for one CSTR, three equal CSTRs in series and a plug-flow reactor, each computed by the package at 60 volumes. At V = 20 m³ the PFR reaches X = 0.864665; marked: V = 20 m³, X = 0.666667 (CSTR).

```
chemical_engineering_ai999x.solve('chemeng_cstr_conversion_first_order', v0=0.01, V=20.0, k=0.001)
# -> X = 0.666667   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_cstr_conversion_first_order', v0=0.002, V=0.1, k=0.05)
# -> X = 0.714286   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
First-order CSTR conversion: v0 = 0.01 m^3/s, V = 20.0 m^3, k = 0.001 1/s	SOLVED	X = 0.666667
First-order CSTR conversion: v0 = 0.002 m^3/s, V = 0.1 m^3, k = 0.05 1/s	SOLVED	X = 0.714286
First-order CSTR conversion: V = 20.0 m^3, k = 0.001 1/s	NEEDS_INPUT	chemeng_cstr_conversion_first_order needs v0 (volumetric feed flow). Example: solve('chemeng_cstr_conversion_first_order
First-order CSTR conversion: v0 = 0.01 m^3/s, V = 20.0 m^3, k = 0.001 1/s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

CSTR conversion for a volume (second order) — `chemeng_cstr_conversion_second_order`

Steady-state conversion and exit concentration of a CSTR of volume V when -r_A = k $C_A^{2.}$

$$\frac{C_A}{C_{A0}} = \frac{2}{1+\sqrt{1+4\,k\tau C_{A0}}}$$

input	meaning	condition
`v0`	volumetric feed flow	
`V`	reactor volume	
`k`	second-order rate constant (-r_A = k C_A^2)	
`CA0`	feed (initial) concentration of A	

output	meaning
`X`	conversion of A
`CA`	exit concentration of A

- **Method:** balance $C_A0 - C_A = k$ tau C_A^2 solved for C_A (positive root, cancellation-free form)
- **Assumes:** steady state, perfect mixing, constant density
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_cstr_conversion_second_order', v0=0.01, V=2.0, k=1e-05,
CA0=1000.0)
# -> X = 0.5, CA = 500 mol/m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_cstr_conversion_second_order', v0=0.001, V=1.0, k=1e-06,
CA0=500.0)
# -> X = 0.267949, CA = 366.025 mol/m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Second-order CSTR conversion: v0 = 0.01 m^3/s, V = 2.0 m^3, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3	SOLVED	X = 0.5, CA = 500 mol/m^3
Second-order CSTR conversion: v0 = 0.001 m^3/s, V = 1.0 m^3, k = 1e-06 m^3/(mol*s), CA0 = 500.0 mol/m^3	SOLVED	X = 0.267949, CA = 366.025 mol/m^3
Second-order CSTR conversion: V = 2.0 m^3, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3	NEEDS_INPUT	chemeng_cstr_conversion_second_order needs v0 (volumetric feed flow). Example: solve('chemeng_cstr_conversion_second_ord
Second-order CSTR conversion: v0 = 0.01 m^3/s, V = 2.0 m^3, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

PFR volume for a conversion (first order, liquid) — `chemeng_pfr_volume_first_order`

Volume of an isothermal plug-flow reactor that reaches conversion X for a first-order constant-density reaction.

$$V = \frac{v_0}{k} \ln \frac{1}{1 - X}$$

input	meaning	condition
`v0`	volumetric feed flow	
`k`	first-order rate constant	
`X`	conversion of A wanted	

output	meaning
`V`	reactor volume

- **Method:** V = F_A0 * integral of $dX/(-r_A) = (v_0/k)$ ln$(1/(1 - X))$
- **Assumes:** $0 \leq X < 1$; plug flow, isothermal, constant density
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_pfr_volume_first_order', v0=0.01, k=0.001, X=0.9)
# -> V = 23.0259 m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_pfr_volume_first_order', v0=0.002, k=0.05, X=0.5)
# -> V = 0.0277259 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
First-order PFR volume: v0 = 0.01 m^3/s, k = 0.001 1/s, X = 0.9	SOLVED	V = 23.0259 m^3
First-order PFR volume: v0 = 0.002 m^3/s, k = 0.05 1/s, X = 0.5	SOLVED	V = 0.0277259 m^3
First-order PFR volume: k = 0.001 1/s, X = 0.9	NEEDS_INPUT	chemeng_pfr_volume_first_order needs v0 (volumetric feed flow). Example: solve('chemeng_pfr_volume_first_order', v0='0.0
First-order PFR volume: v0 = 0.01 m^3/s, k = 0.001 1/s, X = 0.9, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
First-order PFR volume: v0 = 600 L/min, k = 0.06 1/min, X = 90%	SOLVED	V = 23.0259 m^3

PFR volume for a conversion (second order, liquid) — `chemeng_pfr_volume_second_order`

Volume of an isothermal plug-flow reactor that reaches conversion X when -r_A = k C_A^2 (constant density).

$$V = \frac{v_0}{k C_{A0}} \frac{X}{1 - X}$$

input	meaning	condition
`v0`	volumetric feed flow	
`k`	second-order rate constant (-r_A = k C_A^2)	
`CA0`	feed (initial) concentration of A	
`X`	conversion of A wanted	

output	meaning
`V`	reactor volume

- **Method:** V = (v0/(k C_A0)) $X/(1-X)$
- **Assumes:** $0 \leq X < 1$; plug flow, isothermal, constant density
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_pfr_volume_second_order', v0=0.01, k=1e-05, CA0=1000.0,
X=0.8)
# -> V = 4 m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_pfr_volume_second_order', v0=0.001, k=1e-06, CA0=500.0,
X=0.5)
# -> V = 2 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Second-order PFR volume: v0 = 0.01 m^3/s, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3, X = 0.8	SOLVED	V = 4 m^3
Second-order PFR volume: v0 = 0.001 m^3/s, k = 1e-06 m^3/(mol*s), CA0 = 500.0 mol/m^3, X = 0.5	SOLVED	V = 2 m^3
Second-order PFR volume: k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3, X = 0.8	NEEDS_INPUT	chemeng_pfr_volume_second_order needs v0 (volumetric feed flow). Example: solve('chemeng_pfr_volume_second_order', v0='0
Second-order PFR volume: v0 = 0.01 m^3/s, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3, X = 0.8, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

PFR conversion for a volume (first order, liquid) — `chemeng_pfr_conversion_first_order`

Conversion at the exit of an isothermal plug-flow reactor of volume V for a first-order constant-density reaction.

$$X = 1 - e^{-k\tau}$$

input	meaning	condition
`v0`	volumetric feed flow	
`V`	reactor volume	
`k`	first-order rate constant	

output	meaning
`X`	conversion of A

- **Method:** $X = 1 - exp$(-k tau), $\tau = V/v_0$
- **Assumes:** plug flow, isothermal, constant density
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_pfr_conversion_first_order', v0=0.01, V=20.0, k=0.001)
# -> X = 0.864665   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_pfr_conversion_first_order', v0=0.002, V=0.1, k=0.05)
# -> X = 0.917915   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
First-order PFR conversion: v0 = 0.01 m^3/s, V = 20.0 m^3, k = 0.001 1/s	SOLVED	X = 0.864665
First-order PFR conversion: v0 = 0.002 m^3/s, V = 0.1 m^3, k = 0.05 1/s	SOLVED	X = 0.917915
First-order PFR conversion: V = 20.0 m^3, k = 0.001 1/s	NEEDS_INPUT	chemeng_pfr_conversion_first_order needs v0 (volumetric feed flow). Example: solve('chemeng_pfr_conversion_first_order',
First-order PFR conversion: v0 = 0.01 m^3/s, V = 20.0 m^3, k = 0.001 1/s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

PFR conversion for a volume (second order, liquid) — `chemeng_pfr_conversion_second_order`

Exit conversion and concentration of an isothermal plug-flow reactor of volume V when -r_A = k C_A^2 (constant density).

$$X = \frac{k\tau C_{A0}}{1 + k\tau C_{A0}}$$

input	meaning	condition
`v0`	volumetric feed flow	
`V`	reactor volume	
`k`	second-order rate constant (-r_A = k C_A^2)	
`CA0`	feed (initial) concentration of A	

output	meaning
`X`	conversion of A
`CA`	exit concentration of A

- **Method:** $1/C_A = 1/C_A0 + k$ tau
- **Assumes:** plug flow, isothermal, constant density
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_pfr_conversion_second_order', v0=0.01, V=2.0, k=1e-05,
CA0=1000.0)
# -> X = 0.666667, CA = 333.333 mol/m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_pfr_conversion_second_order', v0=0.001, V=1.0, k=1e-06,
CA0=500.0)
# -> X = 0.333333, CA = 333.333 mol/m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Second-order PFR conversion: v0 = 0.01 m^3/s, V = 2.0 m^3, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3	SOLVED	X = 0.666667, CA = 333.333 mol/m^3
Second-order PFR conversion: v0 = 0.001 m^3/s, V = 1.0 m^3, k = 1e-06 m^3/(mol*s), CA0 = 500.0 mol/m^3	SOLVED	X = 0.333333, CA = 333.333 mol/m^3
Second-order PFR conversion: V = 2.0 m^3, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3	NEEDS_INPUT	chemeng_pfr_conversion_second_order needs v0 (volumetric feed flow). Example: solve('chemeng_pfr_conversion_second_order
Second-order PFR conversion: v0 = 0.01 m^3/s, V = 2.0 m^3, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Gas-phase PFR volume with volume change (first order) — `chemeng_pfr_gas_volume_first_order`

Volume of an isothermal, isobaric gas-phase plug-flow reactor for conversion X of a first-order reaction whose mole number changes (factor eps).

$$V = \frac{F_{A0}}{kC_{A0}}\left[(1+\varepsilon)\ln\frac{1}{1-X} - \varepsilon X\right]$$

input	meaning	condition
`FA0`	molar feed flow of A	
`CA0`	concentration of A at the inlet	

input	meaning	condition
`k`	first-order rate constant	
`eps`	volume-change factor eps = y_A0 * (change in moles per mole of A); e.g. A -> 2B with pure A: eps = 1	
`X`	conversion of A wanted	

output	meaning
`V`	reactor volume

- **Method:** V = F_A0 * integral of (1 + eps X)/(k C_A0 $(1 - X)$) dX = (F_A0/(k C_A0)) [(1 + eps) ln($1/(1 - X)$) - eps X]
- **Assumes:** isothermal, isobaric ideal gas; eps > -1; $0 \leq X < 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_pfr_gas_volume_first_order', FA0=10.0, CA0=40.0, k=0.05,
eps=1.0, X=0.8)
# -> V = 12.0944 m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_pfr_gas_volume_first_order', FA0=2.0, CA0=20.0, k=0.01,
eps=-0.5, X=0.6)
# -> V = 7.58145 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
First-order gas-phase PFR volume: FA0 = 10.0 mol/s, CA0 = 40.0 mol/m^3, k = 0.05 1/s, eps = 1.0, X = 0.8	SOLVED	V = 12.0944 m^3
First-order gas-phase PFR volume: FA0 = 2.0 mol/s, CA0 = 20.0 mol/m^3, k = 0.01 1/s, eps = -0.5, X = 0.6	SOLVED	V = 7.58145 m^3
First-order gas-phase PFR volume: CA0 = 40.0 mol/m^3, k = 0.05 1/s, eps = 1.0, X = 0.8	NEEDS_INPUT	chemeng_pfr_gas_volume_first_order needs FA0 (molar feed flow of A). Example: solve('chemeng_pfr_gas_volume_first_order'
First-order gas-phase PFR volume: FA0 = 10.0 mol/s, CA0 = 40.0 mol/m^3, k = 0.05 1/s, eps = 1.0, X = 0.8, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Gas-phase PFR volume with volume change (second order) —

`chemeng_pfr_gas_volume_second_order`

Volume of an isothermal, isobaric gas-phase plug-flow reactor for conversion X when -r_A = k C_A^2 and the mole number changes (factor eps).

$$V = \frac{F_{A0}}{kC_{A0}^2}\left[2\varepsilon(1+\varepsilon)\ln(1-X) + \varepsilon^2 X + \frac{(1+\varepsilon)^2 X}{1-X}\right]$$

input	meaning	condition
`FA0`	molar feed flow of A	
`CA0`	concentration of A at the inlet	
`k`	second-order rate constant (-r_A = k C_A^2)	
`eps`	volume-change factor eps = y_A0 * (change in moles per mole of A); e.g. A -> 2B with pure A: eps = 1	
`X`	conversion of A wanted	

output	meaning
`V`	reactor volume

- **Method:** V = (F_A0/(k C_A0^2)) [2 eps (1 + eps) ln(1 - X) + eps^2 X + (1 + eps)^2 $X/(1-X)$]
- **Assumes:** isothermal, isobaric ideal gas; eps > -1; $0 \le X < 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_pfr_gas_volume_second_order', FA0=10.0, CA0=40.0,
k=0.001, eps=1.0, X=0.8)
# -> V = 64.7641 m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_pfr_gas_volume_second_order', FA0=2.0, CA0=20.0,
k=0.0005, eps=-0.5, X=0.6)
# -> V = 9.83145 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Second-order gas-phase PFR volume: FA0 = 10.0 mol/s, CA0 = 40.0 mol/m^3, k = 0.001 m^3/(mol*s), eps = 1.0, X = 0.8	SOLVED	V = 64.7641 m^3
Second-order gas-phase PFR volume: FA0 = 2.0 mol/s, CA0 = 20.0 mol/m^3, k = 0.0005 m^3/(mol*s), eps = -0.5, X = 0.6	SOLVED	V = 9.83145 m^3
Second-order gas-phase PFR volume: CA0 = 40.0 mol/m^3, k = 0.001 m^3/(mol*s), eps = 1.0, X = 0.8	NEEDS_INPUT	chemeng_pfr_gas_volume_second_order needs FA0 (molar feed flow of A). Example: solve('chemeng_pfr_gas_volume_second_orde
Second-order gas-phase PFR volume: FA0 = 10.0 mol/s, CA0 = 40.0 mol/m^3, k = 0.001 m^3/(mol*s), eps = 1.0, X = 0.8, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Equal CSTRs in series: conversion (first order) — `chemeng_cstrs_in_series_first_order`

Conversion leaving N equal CSTRs in series, each of volume V, for a first-order liquid-phase reaction.

$$X = 1 - (1 + k\tau_i)^{-N}$$

input	meaning	condition
`v0`	volumetric feed flow	
`V`	volume of EACH tank	
`k`	first-order rate constant	
`N`	number of equal tanks in series	

output	meaning
`X`	overall conversion of A

- **Method:** each tank divides the concentration by (1 + k tau_i): X = 1 - (1 + k V/v0)^(-N)
- **Assumes:** equal tanks; steady state, perfect mixing in each tank, constant density
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

0.333	0.5	0.667	0.8	0.889
0.36	0.556	0.75	0.889	0.96
0.37	0.578	0.784	0.921	0.98
0.376	0.59	0.802	0.938	0.988
0.379	0.598	0.814	0.947	0.992

conversion X · rows N = 1 → 5 tanks · columns V = 5 → 80 m³

Conversion of a first-order reaction (k = 0.001 1/s, feed 0.01 m³/s) in N equal CSTRs in series (rows, N = 1 to 5) sharing a total volume of 5, 10, 20, 40, 80 m³ (columns), each cell computed by the package. More tanks approach plug flow: a PFR of the same volumes gives 0.393469, 0.632121, 0.864665, 0.981684, 0.999665.

```
chemical_engineering_ai999x.solve('chemeng_cstrs_in_series_first_order', v0=0.01, V=5.0, k=0.001,
N=3)
# -> X = 0.703704   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_cstrs_in_series_first_order', v0=0.002, V=0.05, k=0.05,
N=2)
# -> X = 0.802469   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
First-order CSTRs in series: v0 = 0.01 m^3/s, V = 5.0 m^3, k = 0.001 1/s, N = 3	SOLVED	X = 0.703704
First-order CSTRs in series: v0 = 0.002 m^3/s, V = 0.05 m^3, k = 0.05 1/s, N = 2	SOLVED	X = 0.802469
First-order CSTRs in series: V = 5.0 m^3, k = 0.001 1/s, N = 3	NEEDS_INPUT	chemeng_cstrs_in_series_first_order needs v0 (volumetric feed flow). Example: solve('chemeng_cstrs_in_series_first_order
First-order CSTRs in series: v0 = 0.01 m^3/s, V = 5.0 m^3, k = 0.001 1/s, N = 3, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
First-order CSTRs in series: v0 = 0.01 m^3/s, V = 5 m^3, k = 0.001 1/s, N = 2.5	INVALID_INPUT	N: expected a whole number written exactly, got '2.5'.

Equal CSTRs in series: conversion (second order) — `chemeng_cstrs_in_series_second_order`

Conversion leaving N equal CSTRs in series, each of volume V, when -r_A = k C_A^2.

$$C_i = \frac{2C_{i-1}}{1+\sqrt{1+4k\tau_i C_{i-1}}}$$

input	meaning	condition
`v0`	volumetric feed flow	
`V`	volume of EACH tank	
`k`	second-order rate constant (-r_A = k C_A^2)	
`CA0`	feed (initial) concentration of A	
`N`	number of equal tanks in series	

output	meaning
`X`	overall conversion of A
`CA`	exit concentration of A

- **Method:** tank by tank: $C_i = 2$ C_($i-1$)/(1 + $sqrt$(1 + 4 k tau_i C_(i-1)))
- **Assumes:** equal tanks; steady state, perfect mixing, constant density
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_cstrs_in_series_second_order', v0=0.01, V=1.0, k=1e-05,
CA0=1000.0, N=3)
# -> X = 0.674359, CA = 325.641 mol/m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_cstrs_in_series_second_order', v0=0.001, V=0.5, k=1e-06,
CA0=500.0, N=4)
# -> X = 0.461462, CA = 269.269 mol/m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Second-order CSTRs in series: v0 = 0.01 m^3/s, V = 1.0 m^3, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3, N = 3	SOLVED	X = 0.674359, CA = 325.641 mol/m^3
Second-order CSTRs in series: v0 = 0.001 m^3/s, V = 0.5 m^3, k = 1e-06 m^3/(mol*s), CA0 = 500.0 mol/m^3, N = 4	SOLVED	X = 0.461462, CA = 269.269 mol/m^3
Second-order CSTRs in series: V = 1.0 m^3, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3, N = 3	NEEDS_INPUT	chemeng_cstrs_in_series_second_order needs v0 (volumetric feed flow). Example: solve('chemeng_cstrs_in_series_second_ord
Second-order CSTRs in series: v0 = 0.01 m^3/s, V = 1.0 m^3, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3, N = 3, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Plug-flow reactor with recycle: conversion (first order) — `chemeng_recycle_reactor_first_order`

Overall conversion of a plug-flow reactor with recycle ratio R (recycled flow / product flow) for a first-order constant-density reaction.

$$\frac{k\tau}{R+1} = \ln\frac{1+R(1-X)}{(R+1)(1-X)}$$

input	meaning	condition
`v0`	fresh volumetric feed flow	
`V`	reactor volume	
`k`	first-order rate constant	
`R`	recycle ratio: recycled flow divided by the flow leaving the system	

output	meaning
`X`	overall conversion of A

- **Method:** k $\tau/(R+1) = ln$[(1 + R(1 - X))/($(R+1)(1-X)$)], $\tau = V/v_0$, solved for X
- **Assumes:** $R \geq 0$ (R = 0 is a plain PFR; large R approaches a CSTR); isothermal, constant density
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_recycle_reactor_first_order', v0=0.01, V=20.0, k=0.001,
R=2.0)
# -> X = 0.7398   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_recycle_reactor_first_order', v0=0.002, V=0.1, k=0.05,
R=0.0)
# -> X = 0.917915   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Recycle reactor conversion: v0 = 0.01 m^3/s, V = 20.0 m^3, k = 0.001 1/s, R = 2.0	SOLVED	X = 0.7398
Recycle reactor conversion: v0 = 0.002 m^3/s, V = 0.1 m^3, k = 0.05 1/s, R = 0.0	SOLVED	X = 0.917915
Recycle reactor conversion: V = 20.0 m^3, k = 0.001 1/s, R = 2.0	NEEDS_INPUT	chemeng_recycle_reactor_first_order needs v0 (fresh volumetric feed flow). Example: solve('chemeng_recycle_reactor_first
Recycle reactor conversion: v0 = 0.01 m^3/s, V = 20.0 m^3, k = 0.001 1/s, R = 2.0, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Reversible first-order reaction A <-> B in a batch reactor —

`chemeng_reversible_first_order_batch`

Concentrations and conversion after time t, and the equilibrium conversion, for A <-> B with first-order forward and reverse steps.

$$C_A = C_{Ae} + (C_{A0} - C_{Ae})e^{-(k_1+k_2)t},\ C_{Ae} = \frac{k_2(C_{A0}+C_{B0})}{k_1+k_2}$$

input	meaning	condition
`k1`	forward rate constant (A -> B)	
`k2`	reverse rate constant (B -> A)	
`CA0`	feed (initial) concentration of A	
`CB0`	initial concentration of B	
`t`	reaction time	

output	meaning
`CA`	concentration of A at t
`CB`	concentration of B at t
`X`	conversion of A at t
`X_eq`	equilibrium conversion of A

- **Method:** $C_A = C_Ae + (C_A0 - C_Ae)$ exp(-$(k_1 + k_2)$ t), $C_Ae = k_2\ (C_A0 + C_B0)/(k_1 + k_2)$
- **Assumes:** isothermal, constant volume
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_reversible_first_order_batch', k1=0.002, k2=0.001,
CA0=1000.0, CB0=0.0, t=300.0)
# -> CA = 604.38 mol/m^3, CB = 395.62 mol/m^3, X = 0.39562, X_eq = 0.666667   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_reversible_first_order_batch', k1=0.01, k2=0.04,
CA0=500.0, CB0=100.0, t=30.0)
# -> CA = 484.463 mol/m^3, CB = 115.537 mol/m^3, X = 0.0310748, X_eq = 0.04   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Reversible first-order batch reaction: k1 = 0.002 1/s, k2 = 0.001 1/s, CA0 = 1000.0 mol/m^3, CB0 = 0.0 mol/m^3, t = 300.0 s	SOLVED	CA = 604.38 mol/m^3, CB = 395.62 mol/m^3, X = 0.39562, X_eq = 0.666667
Reversible first-order batch reaction: k1 = 0.01 1/s, k2 = 0.04 1/s, CA0 = 500.0 mol/m^3, CB0 = 100.0 mol/m^3, t = 30.0 s	SOLVED	CA = 484.463 mol/m^3, CB = 115.537 mol/m^3, X = 0.0310748, X_eq = 0.04
Reversible first-order batch reaction: k2 = 0.001 1/s, CA0 = 1000.0 mol/m^3, CB0 = 0.0 mol/m^3, t = 300.0 s	NEEDS_INPUT	chemeng_reversible_first_order_batch needs k1 (forward rate constant (A -> B)). Example: solve('chemeng_reversible_first
Reversible first-order batch reaction: k1 = 0.002 1/s, k2 = 0.001 1/s, CA0 = 1000.0 mol/m^3, CB0 = 0.0 mol/m^3, t = 300.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Series reactions A -> B -> C: maximum of the intermediate (batch or PFR) — `chemeng_series_reaction_maximum`

Time (or plug-flow space time) at which the intermediate B of first-order series reactions A -> B -> C peaks, starting from pure A, and the concentrations there.

$$t_{max} = \frac{\ln(k_2/k_1)}{k_2 - k_1}, \quad C_{B,max} = C_{A0}\left(\frac{k_1}{k_2}\right)^{k_2/(k_2-k_1)}$$

input	meaning	condition
`k1`	rate constant of A -> B	
`k2`	rate constant of B -> C	
`CA0`	feed (initial) concentration of A	

output	meaning
`t_max`	time (or space time) of the maximum of B
`CB_max`	maximum concentration of B
`CA`	concentration of A at the maximum
`CC`	concentration of C at the maximum

- **Method:** t_max = ln(k2/k1)/(k2 - k1), C_B,max = C_A0 (k1/k2)^(k2/(k2 - k1)) (k1 = k2: $t = 1/k$, C_B = C_A0/e)
- **Assumes:** starting from pure A (no B or C); isothermal, constant volume, first-order steps

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_series_reaction_maximum', k1=0.002, k2=0.001,
CA0=1000.0)
# -> t_max = 693.147 s, CB_max = 500 mol/m^3, CA = 250 mol/m^3, CC = 250 mol/m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_series_reaction_maximum', k1=0.01, k2=0.05, CA0=200.0)
# -> t_max = 40.2359 s, CB_max = 26.7496 mol/m^3, CA = 133.748 mol/m^3, CC = 39.5023 mol/m^3
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Series reaction maximum: k1 = 0.002 1/s, k2 = 0.001 1/s, CA0 = 1000.0 mol/m^3	SOLVED	t_max = 693.147 s, CB_max = 500 mol/m^3, CA = 250 mol/m^3, CC = 250 mol/m^3
Series reaction maximum: k1 = 0.01 1/s, k2 = 0.05 1/s, CA0 = 200.0 mol/m^3	SOLVED	t_max = 40.2359 s, CB_max = 26.7496 mol/m^3, CA = 133.748 mol/m^3, CC = 39.5023 mol/m^3
Series reaction maximum: k2 = 0.001 1/s, CA0 = 1000.0 mol/m^3	NEEDS_INPUT	chemeng_series_reaction_maximum needs k1 (rate constant of A -> B). Example: solve('chemeng_series_reaction_maximum', k1
Series reaction maximum: k1 = 0.002 1/s, k2 = 0.001 1/s, CA0 = 1000.0 mol/m^3, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Residence-time distribution: mean, variance and tanks-in-series number from tracer data — `chemeng_rtd_moments`

Mean residence time, variance and the equivalent number of tanks in series from a pulse-tracer response C(t) (trapezoidal rule, uneven spacing allowed).

$$\bar{t} = \frac{\int t\,C\,dt}{\int C\,dt}, \quad \sigma^2 = \frac{\int (t-\bar{t})^2 C\,dt}{\int C\,dt}$$

input	meaning	condition
`t`	sample times, in the unit t_unit	
`C`	tracer concentrations at those times (any unit)	
`t_unit`	unit of the times: s, min or h	

output	meaning
`t_mean`	mean residence time
`variance`	variance of the RTD
`N_tanks`	equivalent number of equal tanks in series, t_mean^2/variance

- **Method:** E(t) = C/integral C dt; t_mean = integral t E dt; variance = integral (t - t_mean)^2 E dt; trapezoidal rule
- **Assumes:** pulse tracer, closed vessel; times increasing, concentrations >= 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_rtd_moments', t=[0.0, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0,
8.0, 9.0, 10.0, 12.0, 14.0], C=[0.0, 1.0, 5.0, 8.0, 10.0, 8.0, 6.0, 4.0, 3.0, 2.2, 1.5, 0.6, 0.0],
t_unit='min')
# -> t_mean = 307.641 s, variance = 21424.3 s^2, N_tanks = 4.41753   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_rtd_moments', t=[0.0, 5.0, 10.0, 20.0, 30.0, 45.0,
60.0], C=[0.0, 4.0, 6.0, 5.0, 3.0, 1.0, 0.2], t_unit='s')
# -> t_mean = 20.355 s, variance = 144.992 s^2, N_tanks = 2.85758   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
RTD moments: t = [0.0, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0, 12.0, 14.0], C = [0.0, 1.0, 5.0, 8.0, 10.0, 8.0, 6.0, 4.0, 3.0, 2.2, 1.5, 0.6, 0.0], t_unit = min	SOLVED	t_mean = 307.641 s, variance = 21424.3 s^2, N_tanks = 4.41753
RTD moments: t = [0.0, 5.0, 10.0, 20.0, 30.0, 45.0, 60.0], C = [0.0, 4.0, 6.0, 5.0, 3.0, 1.0, 0.2], t_unit = s	SOLVED	t_mean = 20.355 s, variance = 144.992 s^2, N_tanks = 2.85758
RTD moments: C = [0.0, 1.0, 5.0, 8.0, 10.0, 8.0, 6.0, 4.0, 3.0, 2.2, 1.5, 0.6, 0.0], t_unit = min	NEEDS_INPUT	chemeng_rtd_moments needs t (sample times, in the unit t_unit). Example: solve('chemeng_rtd_moments', t=[0.0, 1.0, 2.0,
RTD moments: t = [0.0, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0, 12.0, 14.0], C = [0.0, 1.0, 5.0, 8.0, 10.0, 8.0, 6.0, 4.0, 3.0, 2.2, 1.5, 0.6, 0.0], t_unit = min, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
RTD moments: t = [0, 1, 2], C = [0, 1, 0]	NEEDS_INPUT	chemeng_rtd_moments needs t_unit (unit of the times: s, min or h). Example: solve('chemeng_rtd_moments', t=[0.0, 1.0, 2.

Vapour pressure from the Antoine equation — `chemeng_antoine_vapour_pressure`

Vapour pressure of a pure liquid at temperature T from Antoine coefficients written in a stated convention (logarithm and units).

$$\log_{10} P^{sat} = A - \frac{B}{T + C}$$

input	meaning	condition
`A`	Antoine coefficient A	
`B`	Antoine coefficient B	
`C`	Antoine coefficient C	

input	meaning	condition
`units`	units the Antoine coefficients are written for: pressure_temperature, e.g. mmHg_degC (P in mmHg, T in degC) or kPa_K	
`base`	logarithm of the Antoine equation: log10 or ln	
`T`	temperature	

output	meaning
`Psat`	vapour pressure

- **Method:** log $P = A - B/(T + C)$ in the stated units and logarithm; P returned in Pa
- **Assumes:** the convention (units, base) is stated; T within the range the coefficients were fitted for
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_antoine_vapour_pressure', A=8.07131, B=1730.63,
C=233.426, units='mmHg_degC', base='log10', T=373.15)
# -> Psat = 101337 Pa   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_antoine_vapour_pressure', A=4.72583, B=1660.652,
C=-1.461, units='bar_K', base='log10', T=350.0)
# -> Psat = 91457.9 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Antoine vapour pressure: A = 8.07131, B = 1730.63, C = 233.426, units = mmHg_degC, base = log10, T = 373.15 K	SOLVED	Psat = 101337 Pa
Antoine vapour pressure: A = 4.72583, B = 1660.652, C = -1.461, units = bar_K, base = log10, T = 350.0 K	SOLVED	Psat = 91457.9 Pa
Antoine vapour pressure: B = 1730.63, C = 233.426, units = mmHg_degC, base = log10, T = 373.15 K	NEEDS_INPUT	chemeng_antoine_vapour_pressure needs A (Antoine coefficient A). Example: solve('chemeng_antoine_vapour_pressure', A=8.0
Antoine vapour pressure: A = 8.07131, B = 1730.63, C = 233.426, units = mmHg_degC, base = log10, T = 373.15 K, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Antoine vapour pressure: A = 8.07131, B = 1730.63, C = 233.426, units = mmHg_degC, base = log10, T = 100 degC	SOLVED	Psat = 101337 Pa
Antoine vapour pressure: A = 8.07131, B = 1730.63, C = 233.426, T = 100 degC	NEEDS_INPUT	chemeng_antoine_vapour_pressure needs units, base (units the Antoine coefficients are written for: pressure_temperature,

Boiling (saturation) temperature from the Antoine equation —

`chemeng_antoine_boiling_temperature`

Temperature at which a pure liquid boils at pressure P, from Antoine coefficients written in a stated convention.

$$T^{sat} = \frac{B}{A - \log_{10} P} - C$$

input	meaning	condition
`A`	Antoine coefficient A	
`B`	Antoine coefficient B	
`C`	Antoine coefficient C	
`units`	units the Antoine coefficients are written for: pressure_temperature, e.g. mmHg_degC (P in mmHg, T in degC) or kPa_K	
`base`	logarithm of the Antoine equation: log10 or ln	
`P`	pressure	

output	meaning
`Tsat`	saturation temperature

- **Method:** T = B/(A - log P) - C in the stated units; returned in K
- **Assumes:** the convention (units, base) is stated; P within the fitted range
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_antoine_boiling_temperature', A=8.07131, B=1730.63,
C=233.426, units='mmHg_degC', base='log10', P=101325.0)
# -> Tsat = 373.147 K   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_antoine_boiling_temperature', A=4.72583, B=1660.652,
C=-1.461, units='bar_K', base='log10', P=50000.0)
# -> Tsat = 331.817 K   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Antoine boiling temperature: A = 8.07131, B = 1730.63, C = 233.426, units = mmHg_degC, base = log10, P = 101325.0 Pa	SOLVED	Tsat = 373.147 K
Antoine boiling temperature: A = 4.72583, B = 1660.652, C = -1.461, units = bar_K, base = log10, P = 50000.0 Pa	SOLVED	Tsat = 331.817 K
Antoine boiling temperature: B = 1730.63, C = 233.426, units =	NEEDS_INPUT	chemeng_antoine_boiling_temperature needs A (Antoine coefficient A). Example: solve('chemeng_antoine_boiling_temperature

question	outcome	result
mmHg_degC, base = log10, P = 101325.0 Pa		
Antoine boiling temperature: A = 8.07131, B = 1730.63, C = 233.426, units = mmHg_degC, base = log10, P = 101325.0 Pa, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Antoine boiling temperature: A = 8.07131, B = 1730.63, C = 233.426, units = mmHg_degC, base = log10, P = 1 atm	SOLVED	Tsat = 373.147 K
Antoine boiling temperature: A = 8.07131, B = 1730.63, C = 233.426, units = mmHg, base = log10, P = 1 atm	INVALID_INPUT	units: one of mmHg_degC, kPa_degC, bar_degC, mmHg_K, Pa_K, kPa_K, bar_K; got 'mmhg'.

Bubble-point pressure (Raoult's law) — `chemeng_raoult_bubble_pressure`

Pressure at which a liquid of composition x starts to boil at temperature T, and the composition of the first bubble, for an ideal solution (Raoult's law, Antoine vapour pressures).

$$P_{bubble} = \sum_i x_i P_i^{sat}(T)$$

input	meaning	condition
`x`	liquid mole fractions	
`ant`	Antoine coefficients, one row (A, B, C) per component, in the stated convention	
`units`	units the Antoine coefficients are written for: pressure_temperature, e.g. mmHg_degC (P in mmHg, T in degC) or kPa_K	
`base`	logarithm of the Antoine equation: log10 or ln	
`T`	temperature	

output	meaning
`P`	bubble-point pressure
`y`	vapour mole fractions of the first bubble

- **Method:** P = sum x_i Psat_i(T); $y_i = x_i$ Psat_i/P
- **Assumes:** ideal liquid and ideal gas (Raoult's law); fractions sum to 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_raoult_bubble_pressure', ant=[[6.90565, 1211.033,
220.79], [6.95464, 1344.8, 219.482]], units='mmHg_degC', base='log10', x=[0.4, 0.6], T=368.15)
# -> P = 100906 Pa, y = [0.621960532969, 0.378039467031]   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_raoult_bubble_pressure', ant=[[6.90565, 1211.033,
220.79], [6.95464, 1344.8, 219.482]], units='mmHg_degC', base='log10', x=[0.25, 0.75], T=380.0)
# -> P = 122162 Pa, y = [0.441901544104, 0.558098455896]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Bubble-point pressure: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, x = [0.4, 0.6], T = 368.15 K	SOLVED	P = 100906 Pa, y = [0.621960532969, 0.378039467031]
Bubble-point pressure: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, x = [0.25, 0.75], T = 380.0 K	SOLVED	P = 122162 Pa, y = [0.441901544104, 0.558098455896]
Bubble-point pressure: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, T = 368.15 K	NEEDS_INPUT	chemeng_raoult_bubble_pressure needs x (liquid mole fractions). Example: solve('chemeng_raoult_bubble_pressure', ant=[[6
Bubble-point pressure: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, x = [0.4, 0.6], T = 368.15 K, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Dew-point pressure (Raoult's law) — `chemeng_raoult_dew_pressure`

Pressure at which a vapour of composition y starts to condense at temperature T, and the composition of the first drop (Raoult's law, Antoine vapour pressures).

$$\frac{1}{P_{dew}} = \sum_i \frac{y_i}{P_i^{sat}(T)}$$

input	meaning	condition
`y`	vapour mole fractions	
`ant`	Antoine coefficients, one row (A, B, C) per component, in the stated convention	
`units`	units the Antoine coefficients are written for: pressure_temperature, e.g. mmHg_degC (P in mmHg, T in degC) or kPa_K	
`base`	logarithm of the Antoine equation: log10 or ln	
`T`	temperature	

output	meaning
`P`	dew-point pressure
`x`	liquid mole fractions of the first drop

- **Method:** 1/P = sum y_i/Psat_i(T); $x_i = y_i$ P/Psat_i
- **Assumes:** ideal liquid and ideal gas; fractions sum to 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_raoult_dew_pressure', ant=[[6.90565, 1211.033, 220.79],
[6.95464, 1344.8, 219.482]], units='mmHg_degC', base='log10', y=[0.4, 0.6], T=368.15)
# -> P = 83426 Pa, x = [0.212686334545, 0.787313665455]   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_raoult_dew_pressure', ant=[[6.90565, 1211.033, 220.79],
[6.95464, 1344.8, 219.482]], units='mmHg_degC', base='log10', y=[0.6, 0.4], T=380.0)
# -> P = 139298 Pa, x = [0.387057256563, 0.612942743437]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Dew-point pressure: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, y = [0.4, 0.6], T = 368.15 K	SOLVED	P = 83426 Pa, x = [0.212686334545, 0.787313665455]
Dew-point pressure: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, y = [0.6, 0.4], T = 380.0 K	SOLVED	P = 139298 Pa, x = [0.387057256563, 0.612942743437]
Dew-point pressure: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, T = 368.15 K	NEEDS_INPUT	chemeng_raoult_dew_pressure needs y (vapour mole fractions). Example: solve('chemeng_raoult_dew_pressure', ant=[[6.90565
Dew-point pressure: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, y = [0.4, 0.6], T = 368.15 K, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Bubble-point temperature (Raoult's law) — `chemeng_raoult_bubble_temperature`

Temperature at which a liquid of composition x starts to boil at pressure P, and the first-bubble composition (Raoult's law, Antoine vapour pressures; iterative).

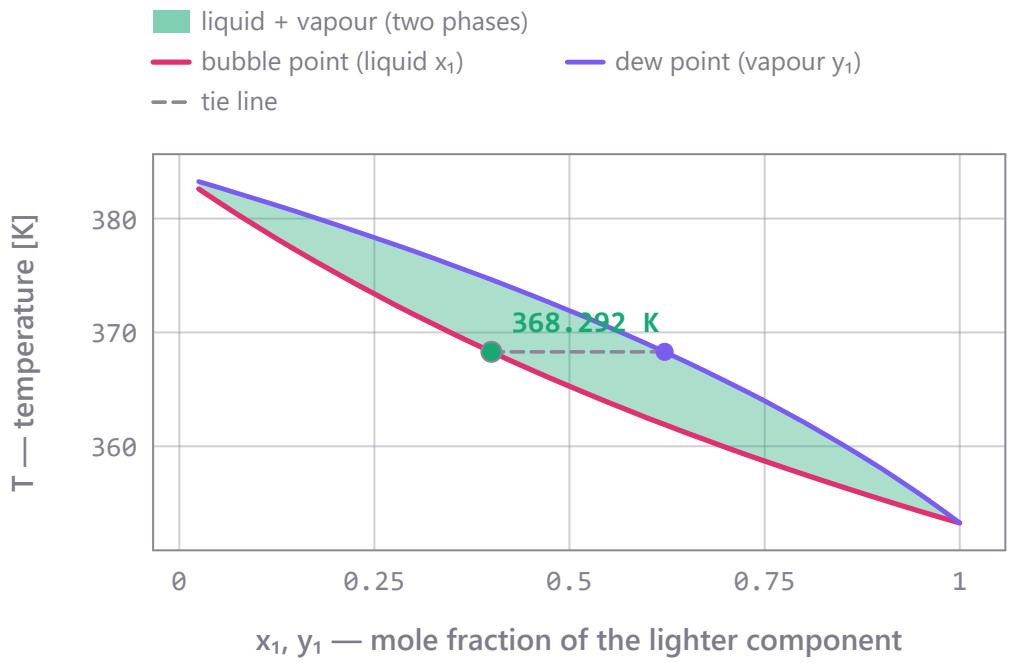
$$\sum_i x_i P_i^{sat}(T_{bubble}) = P$$

input	meaning	condition
`x`	liquid mole fractions	
`ant`	Antoine coefficients, one row (A, B, C) per component, in the stated convention	
`units`	units the Antoine coefficients are written for: pressure_temperature, e.g. mmHg_degC (P in mmHg, T in degC) or kPa_K	
`base`	logarithm of the Antoine equation: log10 or ln	
`P`	pressure	

output	meaning
`T`	bubble-point temperature
`y`	vapour mole fractions of the first bubble

- **Method:** sum x_i Psat_i(T) = P solved by bisection between the pure-component boiling points

- **Assumes:** ideal liquid and ideal gas; fractions sum to 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Boiling-point diagram of an ideal binary mixture at P = 101325 Pa (Raoult's law with the Antoine constants [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], mmHg and °C): bubble and dew temperatures computed by the package at 40 compositions each; between them (shaded) liquid and vapour coexist. The tie line joins a liquid with x_1 = 0.4 to its vapour y_1 = 0.621849; marked: x_1 = 0.4, T = 368.292 K.

```
chemical_engineering_ai999x.solve('chemeng_raoult_bubble_temperature', ant=[[6.90565, 1211.033,
220.79], [6.95464, 1344.8, 219.482]], units='mmHg_degC', base='log10', x=[0.4, 0.6], P=101325.0)
# -> T = 368.292 K, y = [0.62184905372, 0.37815094628]   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_raoult_bubble_temperature', ant=[[6.90565, 1211.033,
220.79], [6.95464, 1344.8, 219.482]], units='mmHg_degC', base='log10', x=[0.7, 0.3], P=50000.0)
# -> T = 338.024 K, y = [0.865597485424, 0.134402514576]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Bubble-point temperature: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, x = [0.4, 0.6], P = 101325.0 Pa	SOLVED	T = 368.292 K, y = [0.62184905372, 0.37815094628]
Bubble-point temperature: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, x = [0.7, 0.3], P = 50000.0 Pa	SOLVED	T = 338.024 K, y = [0.865597485424, 0.134402514576]
Bubble-point temperature: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units =	NEEDS_INPUT	chemeng_raoult_bubble_temperature needs x (liquid mole fractions). Example: solve('chemeng_raoult_bubble_temperature', a

question	outcome	result
mmHg_degC, base = log10, P = 101325.0 Pa		
Bubble-point temperature: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, x = [0.4, 0.6], P = 101325.0 Pa, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Bubble-point temperature: x = [0.4, 0.6], ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, P = 1 atm	SOLVED	T = 368.292 K, y = [0.62184905372, 0.37815094628]
Bubble-point temperature: x = [0.4, 0.5], ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, P = 1 atm	INVALID_INPUT	Bubble-point temperature (Raoult's law): x: the fractions must sum to 1.

Dew-point temperature (Raoult's law) — `chemeng_raoult_dew_temperature`

Temperature at which a vapour of composition y starts to condense at pressure P, and the first-drop composition (Raoult's law, Antoine vapour pressures; iterative).

$$\sum_i \frac{y_i P}{P_i^{sat}(T_{dew})} = 1$$

input	meaning	condition
`y`	vapour mole fractions	
`ant`	Antoine coefficients, one row (A, B, C) per component, in the stated convention	
`units`	units the Antoine coefficients are written for: pressure_temperature, e.g. mmHg_degC (P in mmHg, T in degC) or kPa_K	
`base`	logarithm of the Antoine equation: log10 or ln	
`P`	pressure	

output	meaning
`T`	dew-point temperature
`x`	liquid mole fractions of the first drop

- **Method:** sum y_i P/Psat_i(T) = 1 solved by bisection between the pure-component boiling points
- **Assumes:** ideal liquid and ideal gas; fractions sum to 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_raoult_dew_temperature', ant=[[6.90565, 1211.033,
220.79], [6.95464, 1344.8, 219.482]], units='mmHg_degC', base='log10', y=[0.4, 0.6], P=101325.0)
# -> T = 374.635 K, x = [0.216267034105, 0.783732965895]   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_raoult_dew_temperature', ant=[[6.90565, 1211.033,
220.79], [6.95464, 1344.8, 219.482]], units='mmHg_degC', base='log10', y=[0.7, 0.3], P=50000.0)
# -> T = 343.96 K, x = [0.464057188008, 0.535942811992]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Dew-point temperature: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, y = [0.4, 0.6], P = 101325.0 Pa	SOLVED	T = 374.635 K, x = [0.216267034105, 0.783732965895]
Dew-point temperature: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, y = [0.7, 0.3], P = 50000.0 Pa	SOLVED	T = 343.96 K, x = [0.464057188008, 0.535942811992]
Dew-point temperature: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, P = 101325.0 Pa	NEEDS_INPUT	chemeng_raoult_dew_temperature needs y (vapour mole fractions). Example: solve('chemeng_raoult_dew_temperature', ant=[[6
Dew-point temperature: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, y = [0.4, 0.6], P = 101325.0 Pa, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Isothermal flash with given K-values (Rachford-Rice) — `chemeng_rachford_rice_flash`

Vapour fraction and the liquid and vapour compositions of a feed flashed at conditions where the K-values are known.

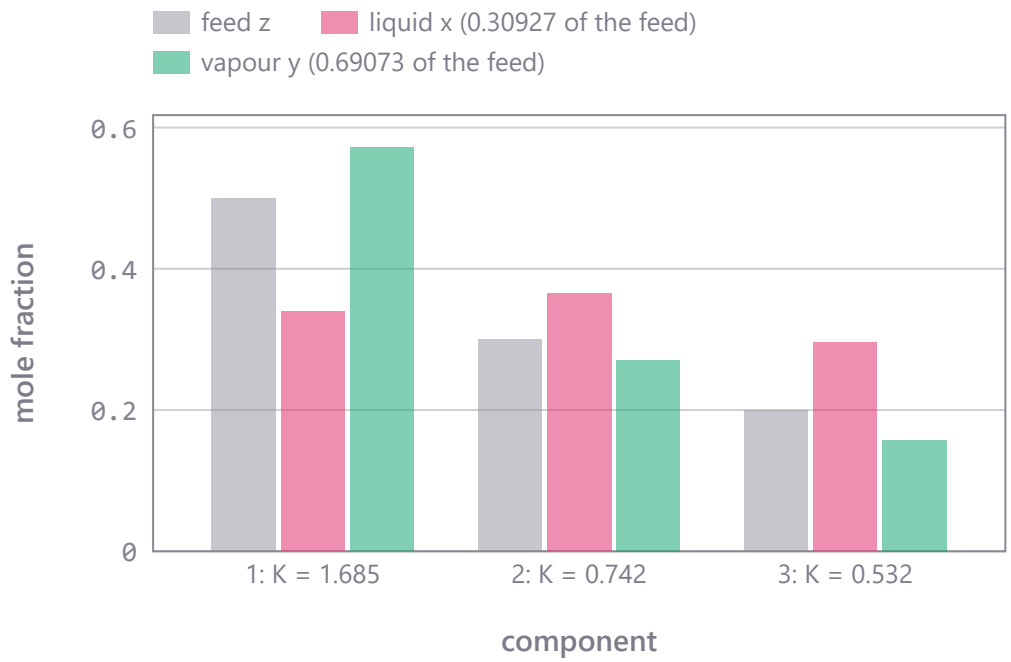
$$\sum_i \frac{z_i(K_i - 1)}{1 + \frac{V}{F}(K_i - 1)} = 0$$

input	meaning	condition
z	feed mole fractions	
K	equilibrium ratios $K_i = y_i/x_i$	

output	meaning
VF	vapour fraction V/F
x	liquid mole fractions
y	vapour mole fractions

- **Method:** Rachford-Rice equation sum z_i (K_i - 1)/(1 + V $(K_i - 1)$) = 0 solved for V by bisection; x_i = z_i/(1 + V $(K_i - 1)$), $y_i = K_i$ x_i
- **Assumes:** two-phase: sum z $K > 1$ and sum $z/K > 1$; K-values independent of composition; some K-value differs from 1 by more than 1 % (well-conditioned)

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Isothermal flash of a 3-component feed z = [0.5, 0.3, 0.2] with equilibrium ratios K = [1.685, 0.742, 0.532]: the package solves the Rachford–Rice equation for a vapour fraction of 0.69073 and returns the liquid x = [0.339409, 0.365056, 0.295535] and vapour y = [0.571904, 0.270872, 0.157225] compositions (bars).

```
chemical_engineering_ai999x.solve('chemeng_rachford_rice_flash', z=[0.5, 0.3, 0.2], K=[1.685, 0.742, 0.532])
# -> VF = 0.69073, x = [0.339408696966, 0.365056059037, 0.295535243996], y = [0.571903654388, 0.270871595806, 0.157224749806]   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_rachford_rice_flash', z=[0.2, 0.5, 0.3], K=[4.2, 1.1, 0.15])
# -> VF = 0.335818, x = [0.096403236013, 0.483754625418, 0.419842138569], y = [0.404893591254, 0.53213008796, 0.0629763207853]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Rachford-Rice flash: z = [0.5, 0.3, 0.2], K = [1.685, 0.742, 0.532]	SOLVED	VF = 0.69073, x = [0.339408696966, 0.365056059037, 0.295535243996], y = [0.571903654388, 0.270871595806, 0.157224749806]
Rachford-Rice flash: z = [0.2, 0.5, 0.3], K = [4.2, 1.1, 0.15]	SOLVED	VF = 0.335818, x = [0.096403236013, 0.483754625418, 0.419842138569], y = [0.404893591254, 0.53213008796, 0.0629763207853]
Rachford-Rice flash: K = [1.685, 0.742, 0.532]	NEEDS_INPUT	chemeng_rachford_rice_flash needs z (feed mole fractions). Example: solve('chemeng_rachford_rice_flash', z=[0.5, 0.3, 0.
Rachford-Rice flash: z = [0.5, 0.3, 0.2], K = [1.685, 0.742, 0.532], zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Rachford-Rice flash: z = [0.5, 0.5], K = [0.5, 0.8]	INVALID_INPUT	Isothermal flash with given K-values (Rachford-Rice): the mixture is not two-phase at these K-values (sum zK <= 1: all l

Isothermal flash of an ideal mixture (Raoult's law) — `chemeng_raoult_isothermal_flash`

Vapour fraction and phase compositions of a feed flashed at temperature T and pressure P, with Raoult's-law K-values from Antoine vapour pressures.

$$K_i = \frac{P_i^{sat}(T)}{P}, \quad \sum_i \frac{z_i(K_i - 1)}{1 + \frac{V}{F}(K_i - 1)} = 0$$

input	meaning	condition
`z`	feed mole fractions	
`ant`	Antoine coefficients, one row (A, B, C) per component, in the stated convention	
`units`	units the Antoine coefficients are written for: pressure_temperature, e.g. mmHg_degC (P in mmHg, T in degC) or kPa_K	
`base`	logarithm of the Antoine equation: log10 or ln	
`T`	flash temperature	
`P`	flash pressure	

output	meaning
`VF`	vapour fraction V/F
`x`	liquid mole fractions
`y`	vapour mole fractions

- **Method:** K_i = Psat_i(T)/P, then the Rachford-Rice equation solved for V/F by bisection
- **Assumes:** P between the dew- and bubble-point pressures at T (two phases); ideal liquid and gas; some K-value differs from 1 by more than 1 %
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_raoult_isothermal_flash', ant=[[6.90565, 1211.033,
220.79], [6.95464, 1344.8, 219.482]], units='mmHg_degC', base='log10', z=[0.5, 0.5], T=368.15,
P=101325.0)
# -> VF = 0.430535, x = [0.404485322459, 0.595514677541], y = [0.626336606294, 0.373663393706]
(SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_raoult_isothermal_flash', ant=[[6.90565, 1211.033,
220.79], [6.95464, 1344.8, 219.482]], units='mmHg_degC', base='log10', z=[0.3, 0.7], T=373.0,
P=101325.0)
# -> VF = 0.196311, x = [0.260617931568, 0.739382068432], y = [0.461228873413, 0.538771126587]
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Raoult isothermal flash: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, z = [0.5, 0.5], T = 368.15 K, P = 101325.0 Pa	SOLVED	VF = 0.430535, x = [0.404485322459, 0.595514677541], y = [0.626336606294, 0.373663393706]

question	outcome	result
Raoult isothermal flash: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, z = [0.3, 0.7], T = 373.0 K, P = 101325.0 Pa	SOLVED	VF = 0.196311, x = [0.260617931568, 0.739382068432], y = [0.461228873413, 0.538771126587]
Raoult isothermal flash: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, T = 368.15 K, P = 101325.0 Pa	NEEDS_INPUT	chemeng_raoult_isothermal_flash needs z (feed mole fractions). Example: solve('chemeng_raoult_isothermal_flash', ant=[[6
Raoult isothermal flash: ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, z = [0.5, 0.5], T = 368.15 K, P = 101325.0 Pa, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Raoult isothermal flash: z = [0.5, 0.5], ant = [[6.90565, 1211.033, 220.79], [6.95464, 1344.8, 219.482]], units = mmHg_degC, base = log10, T = 368.15 K, P = 5 bar	INVALID_INPUT	Isothermal flash of an ideal mixture (Raoult's law): the mixture is not two-phase at these K-values (sum zK <= 1: all li

Flash with Wilson K-values (hydrocarbon mixtures) — `chemeng_wilson_k_flash`

Vapour fraction and phase compositions of a hydrocarbon feed at T and P with K-values estimated by Wilson's correlation from critical constants and acentric factors.

$$K_i = \frac{P_{c,i}}{P} \exp\left[5.37(1+\omega_i)\left(1-\frac{T_{c,i}}{T}\right)\right]$$

input	meaning	condition
`z`	feed mole fractions	
`Tc`	critical temperatures, K	
`Pc`	critical pressures, in the unit p_unit	
`omega`	acentric factors	
`p_unit`	unit of the critical pressures: Pa, kPa, MPa, bar, atm or psi	
`T`	flash temperature	
`P`	flash pressure	

output	meaning
`VF`	vapour fraction V/F
`x`	liquid mole fractions
`y`	vapour mole fractions
`K`	Wilson K-values

- **Method:** $K_i = (Pc_i/P)$ exp[5.37 $(1+omega_i)(1-Tc_i/T)$], then the Rachford-Rice equation by bisection
- **Assumes:** two-phase at the Wilson K-values; some K-value differs from 1 by more than 1 %; Wilson's correlation suits light hydrocarbons at low to moderate pressure

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_wilson_k_flash', z=[0.3, 0.4, 0.3], Tc=[190.564, 369.83,
507.6], Pc=[45.99, 42.48, 30.25], omega=[0.011, 0.152, 0.3], p_unit='bar', T=300.0, P=2000000.0)
# -> VF = 0.362332, x = [0.0449443371671, 0.487795919806, 0.467259743027], y = [0.748872564861,
0.245487924989, 0.00563951014995], K = [16.6622229198, 0.503259488285, 0.0120693259672]   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_wilson_k_flash', z=[0.5, 0.5], Tc=[425.12, 507.6], Pc=
[3.796, 3.025], omega=[0.2, 0.3], p_unit='MPa', T=380.0, P=1000000.0)
# -> VF = 0.0518041, x = [0.480912489953, 0.519087510047], y = [0.849368379403, 0.150631620597], K
= [1.7661599504, 0.290185407435]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Wilson K-value flash: z = [0.3, 0.4, 0.3], Tc = [190.564, 369.83, 507.6], Pc = [45.99, 42.48, 30.25], omega = [0.011, 0.152, 0.3], p_unit = bar, T = 300.0 K, P = 2000000.0 Pa	SOLVED	VF = 0.362332, x = [0.0449443371671, 0.487795919806, 0.467259743027], y = [0.748872564861, 0.245487924989, 0.00563951014995], K = [16.6622229198, 0.503259488285, 0.0120693259672]
Wilson K-value flash: z = [0.5, 0.5], Tc = [425.12, 507.6], Pc = [3.796, 3.025], omega = [0.2, 0.3], p_unit = MPa, T = 380.0 K, P = 1000000.0 Pa	SOLVED	VF = 0.0518041, x = [0.480912489953, 0.519087510047], y = [0.849368379403, 0.150631620597], K = [1.7661599504, 0.290185407435]
Wilson K-value flash: Tc = [190.564, 369.83, 507.6], Pc = [45.99, 42.48, 30.25], omega = [0.011, 0.152, 0.3], p_unit = bar, T = 300.0 K, P = 2000000.0 Pa	NEEDS_INPUT	chemeng_wilson_k_flash needs z (feed mole fractions). Example: solve('chemeng_wilson_k_flash', z=[0.3, 0.4, 0.3], Tc=[19
Wilson K-value flash: z = [0.3, 0.4, 0.3], Tc = [190.564, 369.83, 507.6], Pc = [45.99, 42.48, 30.25], omega = [0.011, 0.152, 0.3], p_unit = bar, T = 300.0 K, P = 2000000.0 Pa, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Minimum reflux ratio of a binary column (Underwood) — `chemeng_binary_minimum_reflux`

Minimum reflux ratio of a binary distillation with constant relative volatility, for any feed condition q (Underwood's equations).

$$\sum_i \frac{\alpha_i z_{F,i}}{\alpha_i - \theta} = 1 - q, \quad R_{min} + 1 = \sum_i \frac{\alpha_i x_{D,i}}{\alpha_i - \theta}$$

input	meaning	condition
`alpha`	relative volatility of the light component (constant)	
`zF`	mole fraction of the light component in the feed	
`xD`	mole fraction of the light component in the distillate	
`q`	feed quality q: 1 saturated liquid, 0 saturated vapour, > 1 subcooled, < 0 superheated	

output	meaning
`Rmin`	minimum reflux ratio L/D

- **Method:** Underwood: sum alpha_i $z_i/(alpha_i - \theta) = 1 - q$ for theta between 1 and alpha (bisection); R_min + 1 = sum alpha_i x_Di/(alpha_i - theta)
- **Assumes:** binary, constant alpha, constant molar overflow; $0 < zF < xD < 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_binary_minimum_reflux', alpha=2.5, zF=0.5, xD=0.95,
q=1.0)
# -> Rmin = 1.1   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_binary_minimum_reflux', alpha=2.4, zF=0.4, xD=0.97,
q=0.5)
# -> Rmin = 2.26137   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Minimum reflux ratio: alpha = 2.5, zF = 0.5, xD = 0.95, q = 1.0	SOLVED	Rmin = 1.1
Minimum reflux ratio: alpha = 2.4, zF = 0.4, xD = 0.97, q = 0.5	SOLVED	Rmin = 2.26137
Minimum reflux ratio: zF = 0.5, xD = 0.95, q = 1.0	NEEDS_INPUT	chemeng_binary_minimum_reflux needs alpha (relative volatility of the light component (constant)). Example: solve('cheme
Minimum reflux ratio: alpha = 2.5, zF = 0.5, xD = 0.95, q = 1.0, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Minimum reflux ratio: alpha = 0.8, zF = 0.5, xD = 0.95, q = 1	INVALID_INPUT	Minimum reflux ratio of a binary column (Underwood): the relative volatility alpha must exceed 1.

McCabe-Thiele number of theoretical stages (binary) — `chemeng_mccabe_thiele_stages`

Number of theoretical stages (the partial reboiler counted as one) of a binary column at reflux ratio R, by stepping between the operating lines and a constant-alpha equilibrium curve, with the feed switched at the intersection of the operating lines.

$$y_{n+1} = \frac{R}{R+1} x_n + \frac{x_D}{R+1}, \quad x_n = \frac{y_n}{\alpha - (\alpha - 1) y_n}$$

input	meaning	condition
`alpha`	relative volatility of the light component (constant)	
`zF`	mole fraction of the light component in the feed	
`xD`	mole fraction of the light component in the distillate	

input	meaning	condition
`q`	feed quality q: 1 saturated liquid, 0 saturated vapour, > 1 subcooled, < 0 superheated	
`xB`	mole fraction of the light component in the bottoms	
`R`	reflux ratio L/D	

output	meaning
`N`	theoretical stages including the partial reboiler
`Rmin`	minimum reflux ratio

- **Method:** stepping from the top: equilibrium x from y, then y from the rectifying line above the intersection and the stripping line below it, until $x \leq xB$
- **Assumes:** binary, constant alpha, constant molar overflow, total condenser; R > R_min; $0 < xB < zF < xD < 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_mccabe_thiele_stages', alpha=2.5, zF=0.5, xD=0.95,
q=1.0, xB=0.05, R=2.0)
# -> N = 11, Rmin = 1.1   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_mccabe_thiele_stages', alpha=2.4, zF=0.4, xD=0.97,
q=0.5, xB=0.02, R=3.5)
# -> N = 14, Rmin = 2.26137   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
McCabe-Thiele stages: alpha = 2.5, zF = 0.5, xD = 0.95, q = 1.0, xB = 0.05, R = 2.0	SOLVED	N = 11, Rmin = 1.1
McCabe-Thiele stages: alpha = 2.4, zF = 0.4, xD = 0.97, q = 0.5, xB = 0.02, R = 3.5	SOLVED	N = 14, Rmin = 2.26137
McCabe-Thiele stages: zF = 0.5, xD = 0.95, q = 1.0, xB = 0.05, R = 2.0	NEEDS_INPUT	chemeng_mccabe_thiele_stages needs alpha (relative volatility of the light component (constant)). Example: solve('chemen
McCabe-Thiele stages: alpha = 2.5, zF = 0.5, xD = 0.95, q = 1.0, xB = 0.05, R = 2.0, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
McCabe-Thiele stages: alpha = 2.5, zF = 0.5, xD = 0.95, q = 1, xB = 0.05, R = 1	INVALID_INPUT	McCabe-Thiele number of theoretical stages (binary): the reflux ratio must exceed the minimum reflux ratio 1.1.

Absorber with N equilibrium stages (Kremser) — `chemeng_kremser_absorber`

Solute mole fraction in the gas leaving a countercurrent absorber of N ideal stages, and the fraction of solute absorbed, for dilute gas with linear equilibrium $y = m$ x.

$$\frac{y_{in} - y_{out}}{y_{in} - m x_{in}} = \frac{A^{N+1} - A}{A^{N+1} - 1}$$

input	meaning	condition
`N`	number of ideal stages	
`A`	absorption factor A = L/(m V)	
`m`	slope of the equilibrium line $y = m$ x	
`y_in`	solute mole fraction in the entering gas	
`x_in`	solute mole fraction in the entering liquid (solvent)	

output	meaning
`y_out`	solute mole fraction in the leaving gas
`recovery`	fraction of the entering solute absorbed

- **Method:** Kremser: (y_in - y_out)/(y_in - m x_in) = (A^(N+1) - A)/(A^(N+1) - 1); $N/(N+1)$ when $A = 1$
- **Assumes:** dilute, constant L and V, linear equilibrium; $y_i n > m$ x_in
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_kremser_absorber', N=5, A=1.4, m=2.0, y_in=0.05,
x_in=0.0)
# -> y_out = 0.003063, recovery = 0.93874   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_kremser_absorber', N=3, A=1.0, m=1.5, y_in=0.02,
x_in=0.001)
# -> y_out = 0.006125, recovery = 0.69375   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Kremser absorber: N = 5, A = 1.4, m = 2.0, y_in = 0.05, x_in = 0.0	SOLVED	y_out = 0.003063, recovery = 0.93874
Kremser absorber: N = 3, A = 1.0, m = 1.5, y_in = 0.02, x_in = 0.001	SOLVED	y_out = 0.006125, recovery = 0.69375
Kremser absorber: A = 1.4, m = 2.0, y_in = 0.05, x_in = 0.0	NEEDS_INPUT	chemeng_kremser_absorber needs N (number of ideal stages). Example: solve('chemeng_kremser_absorber', N=5, A=1.4, m=2, y
Kremser absorber: N = 5, A = 1.4, m = 2.0, y_in = 0.05, x_in = 0.0, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Stripper with N equilibrium stages (Kremser) — `chemeng_kremser_stripper`

Solute mole fraction in the liquid leaving a countercurrent stripper of N ideal stages, and the fraction stripped, for dilute liquid with linear equilibrium $y = m$ x.

$$\frac{x_{in} - x_{out}}{x_{in} - y_{in}/m} = \frac{S^{N+1} - S}{S^{N+1} - 1}$$

input	meaning	condition
`N`	number of ideal stages	
`S`	stripping factor $S = mV/L$	
`m`	slope of the equilibrium line $y = mx$	
`x_in`	solute mole fraction in the entering liquid	
`y_in`	solute mole fraction in the entering stripping gas	

output	meaning
`x_out`	solute mole fraction in the leaving liquid
`recovery`	fraction of the entering solute stripped

- **Method:** Kremser: (x_in - x_out)/(x_in - y_in/m) = (S^(N+1) - S)/(S^(N+1) - 1); $N/(N+1)$ when $S = 1$
- **Assumes:** dilute, constant L and V, linear equilibrium; $x_i n > y_i n/m$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_kremser_stripper', N=6, S=1.5, m=3.0, x_in=0.01,
y_in=0.0)
# -> x_out = 0.000310831, recovery = 0.968917   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_kremser_stripper', N=4, S=0.8, m=2.0, x_in=0.05,
y_in=0.01)
# -> x_out = 0.0183865, recovery = 0.63227   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Kremser stripper: N = 6, S = 1.5, m = 3.0, x_in = 0.01, y_in = 0.0	SOLVED	x_out = 0.000310831, recovery = 0.968917
Kremser stripper: N = 4, S = 0.8, m = 2.0, x_in = 0.05, y_in = 0.01	SOLVED	x_out = 0.0183865, recovery = 0.63227
Kremser stripper: S = 1.5, m = 3.0, x_in = 0.01, y_in = 0.0	NEEDS_INPUT	chemeng_kremser_stripper needs N (number of ideal stages). Example: solve('chemeng_kremser_stripper', N=6, S=1.5, m=3, x
Kremser stripper: N = 6, S = 1.5, m = 3.0, x_in = 0.01, y_in = 0.0, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Packed absorber: number of transfer units and height (Colburn) —

`chemeng_absorber_transfer_units`

Number of overall gas-phase transfer units and packed height of a dilute countercurrent absorber with linear equilibrium (Colburn's equation).

$$N_{OG} = \frac{\ln\left[\left(1 - \frac{1}{A}\right)\frac{y_1 - mx_2}{y_2 - mx_2} + \frac{1}{A}\right]}{1 - 1/A}$$

input	meaning	condition
`y1`	solute mole fraction in the gas entering (bottom)	
`y2`	solute mole fraction in the gas leaving (top)	
`x2`	solute mole fraction in the liquid entering (top)	
`m`	slope of the equilibrium line $y = mx$	
`A`	absorption factor A = L/(m V)	
`HOG`	height of an overall gas-phase transfer unit	

output	meaning
`NOG`	number of overall gas-phase transfer units
`Z`	packed height

- **Method:** $N_OG = ln[(1 - 1/A)$(y1 - m $x_2)/(y_2 - m$ x2) + 1/A]/$(1 - 1/A)$; $Z = H_OG$ N_OG
- **Assumes:** dilute, constant L and V, linear equilibrium; $y_1 > y_2 > m$ x2; no pinch
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_absorber_transfer_units', y1=0.02, y2=0.001, x2=0.0,
m=1.2, A=1.5, HOG=0.6)
# -> NOG = 5.97729, Z = 3.58637 m   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_absorber_transfer_units', y1=0.05, y2=0.005, x2=0.0005,
m=0.8, A=2.0, HOG=1.0)
# -> NOG = 3.54695, Z = 3.54695 m   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Absorber transfer units: y1 = 0.02, y2 = 0.001, x2 = 0.0, m = 1.2, A = 1.5, HOG = 0.6 m	SOLVED	NOG = 5.97729, Z = 3.58637 m
Absorber transfer units: y1 = 0.05, y2 = 0.005, x2 = 0.0005, m = 0.8, A = 2.0, HOG = 1.0 m	SOLVED	NOG = 3.54695, Z = 3.54695 m
Absorber transfer units: y2 = 0.001, x2 = 0.0, m = 1.2, A = 1.5, HOG = 0.6 m	NEEDS_INPUT	chemeng_absorber_transfer_units needs y1 (solute mole fraction in the gas entering (bottom)). Example: solve('chemeng_ab
Absorber transfer units: y1 = 0.02, y2 = 0.001, x2 = 0.0, m = 1.2, A = 1.5, HOG = 0.6 m, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Absorber transfer units: y1 = 0.02, y2 = 0.001, x2 = 0, m = 1.2, A = 1.5, HOG = 60 cm	SOLVED	NOG = 5.97729, Z = 3.58637 m

Countercurrent liquid-liquid extraction (N stages, immiscible solvents) — `chemeng_countercurrent_extraction`

Raffinate and extract solute ratios and the fraction extracted for N ideal countercurrent stages with immiscible carrier and solvent and a constant distribution ratio $Y = KX$ (solute-free mass-ratio basis).

$$\frac{X_F - X_N}{X_F - Y_S/K} = \frac{E^{N+1} - E}{E^{N+1} - 1},\ E = \frac{KS}{F}$$

input	meaning	condition
`F`	solute-free carrier flow in the feed	
`S`	solute-free solvent flow	
`K`	distribution ratio $Y = KX$ (mass ratios)	
`N`	number of ideal stages	
`XF`	solute ratio in the feed, kg solute per kg carrier	
`YS`	solute ratio in the entering solvent, kg solute per kg solvent	

output	meaning
`XN`	solute ratio in the final raffinate
`Y1`	solute ratio in the final extract
`recovery`	fraction of the solute extracted

- **Method:** Kremser form with the extraction factor $E = KS/F$:
  $(XF - XN)/(XF - YS/K) = (E^{N+1} - E)/(E^{N+1} - 1)$
- **Assumes:** immiscible carrier and solvent, constant K; $XF > YS/K$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_countercurrent_extraction', F=100.0, S=80.0, K=1.5, N=4,
XF=0.1, YS=0.0)
# -> XN = 0.013438, Y1 = 0.108203, recovery = 0.86562   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_countercurrent_extraction', F=10.0, S=5.0, K=2.0, N=3,
XF=0.2, YS=0.01)
# -> XN = 0.05375, Y1 = 0.3025, recovery = 0.73125   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Countercurrent extraction: F = 100.0 kg/s, S = 80.0 kg/s, K = 1.5, N = 4, XF = 0.1, YS = 0.0	SOLVED	XN = 0.013438, Y1 = 0.108203, recovery = 0.86562
Countercurrent extraction: F = 10.0 kg/s, S = 5.0 kg/s, K = 2.0, N = 3, XF = 0.2, YS = 0.01	SOLVED	XN = 0.05375, Y1 = 0.3025, recovery = 0.73125
Countercurrent extraction: S = 80.0 kg/s, K = 1.5, N = 4, XF = 0.1, YS =	NEEDS_INPUT	chemeng_countercurrent_extraction needs F (solute-free carrier flow in the feed). Example:

question	outcome	result
0.0		solve('chemeng_countercurrent
Countercurrent extraction: F = 100.0 kg/s, S = 80.0 kg/s, K = 1.5, N = 4, XF = 0.1, YS = 0.0, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Crosscurrent (multiple-contact) extraction with fresh solvent —

`chemeng_crosscurrent_extraction`

Raffinate solute ratio after N ideal stages, each fed with the same amount of fresh solvent, for immiscible solvents and a constant distribution ratio.

$$X_N = \frac{X_F}{(1 + KS/F)^N}$$

input	meaning	condition
`F`	solute-free carrier flow (or mass) in the feed	
`S`	solute-free solvent flow (or mass) added to EACH stage	
`K`	distribution ratio $Y = K$ X (mass ratios)	
`N`	number of stages	
`XF`	solute ratio in the feed, kg solute per kg carrier	

output	meaning
`XN`	solute ratio in the final raffinate
`recovery`	fraction of the solute extracted

- **Method:** each stage divides the raffinate ratio by (1 + K S/F): X_N = X_F/(1 + K S/F)^N
- **Assumes:** immiscible carrier and solvent, constant K; fresh (solute-free) solvent to every stage
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_crosscurrent_extraction', F=100.0, S=20.0, K=1.5, N=4,
XF=0.1)
# -> XN = 0.0350128, recovery = 0.649872   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_crosscurrent_extraction', F=10.0, S=10.0, K=2.0, N=1,
XF=0.2)
# -> XN = 0.0666667, recovery = 0.666667   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Crosscurrent extraction: F = 100.0 kg/s, S = 20.0 kg/s, K = 1.5, N = 4, XF = 0.1	SOLVED	XN = 0.0350128, recovery = 0.649872
Crosscurrent extraction: F = 10.0 kg/s, S = 10.0 kg/s, K = 2.0, N = 1, XF = 0.2	SOLVED	XN = 0.0666667, recovery = 0.666667

question	outcome	result
Crosscurrent extraction: S = 20.0 kg/s, K = 1.5, N = 4, XF = 0.1	NEEDS_INPUT	chemeng_crosscurrent_extraction needs F (solute-free carrier flow (or mass) in the feed). Example: solve('chemeng_crossc
Crosscurrent extraction: F = 100.0 kg/s, S = 20.0 kg/s, K = 1.5, N = 4, XF = 0.1, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Activity coefficients from the Wilson equation — `chemeng_wilson_activity_coefficients`

Activity coefficients of every component and the excess Gibbs energy of a liquid mixture from Wilson parameters Lambda_ij.

$$\ln \gamma_i = 1 - \ln\left(\sum_j x_j \Lambda_{ij}\right) - \sum_k \frac{x_k \Lambda_{ki}}{\sum_j x_j \Lambda_{kj}}$$

input	meaning	condition
`x`	liquid mole fractions	
`Lam`	Wilson parameters Lambda_ij (dimensionless, Lambda_ii = 1) at the temperature of interest	

output	meaning
`gamma1`	activity coefficient of component 1
`gamma`	activity coefficients of every component
`GE_RT`	excess Gibbs energy over RT

- **Method:** ln $gamma_i = 1 - ln$(sum_j x_j Lambda_ij) - sum_k x_k Lambda_ki/(sum_j x_j Lambda_kj)
- **Assumes:** $Lambda_i i = 1$, $Lambda_i j > 0$; all mole fractions positive
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_wilson_activity_coefficients', x=[0.3, 0.7], Lam=[[1.0, 0.5], [1.5, 1.0]])
# -> gamma1 = 1.05781, gamma = [1.05781025716, 1.02099043889], GE_RT = 0.0314015   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_wilson_activity_coefficients', x=[0.2, 0.3, 0.5], Lam=[[1.0, 0.8, 0.4], [1.2, 1.0, 0.9], [0.6, 1.1, 1.0]])
# -> gamma1 = 1.57517, gamma = [1.57516694787, 0.885212689617, 1.13570117122], GE_RT = 0.117919   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Wilson activity coefficients: x = [0.3, 0.7], Lam = [[1.0, 0.5], [1.5, 1.0]]	SOLVED	gamma1 = 1.05781, gamma = [1.05781025716, 1.02099043889], GE_RT = 0.0314015
Wilson activity coefficients: x = [0.2, 0.3, 0.5], Lam = [[1.0, 0.8, 0.4], [1.2, 1.0, 0.9], [0.6, 1.1, 1.0]]	SOLVED	gamma1 = 1.57517, gamma = [1.57516694787, 0.885212689617, 1.13570117122], GE_RT = 0.117919

question	outcome	result
Wilson activity coefficients: Lam = [[1.0, 0.5], [1.5, 1.0]]	NEEDS_INPUT	chemeng_wilson_activity_coefficients needs x (liquid mole fractions). Example: solve('chemeng_wilson_activity_coefficien
Wilson activity coefficients: x = [0.3, 0.7], Lam = [[1.0, 0.5], [1.5, 1.0]], zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Activity coefficients from the NRTL equation — `chemeng_nrtl_activity_coefficients`

Activity coefficients and excess Gibbs energy of a liquid mixture from NRTL interaction parameters tau_ij and non-randomness parameters alpha_ij.

$$\ln \gamma_i = \frac{\sum_j x_j \tau_{ji} G_{ji}}{\sum_k x_k G_{ki}} + \sum_j \frac{x_j G_{ij}}{\sum_k x_k G_{kj}} \left(\tau_{ij} - \frac{\sum_m x_m \tau_{mj} G_{mj}}{\sum_k x_k G_{kj}} \right)$$

input	meaning	condition
`x`	liquid mole fractions	
`tau`	NRTL parameters tau_ij (dimensionless, tau_ii = 0) at the temperature of interest	
`alpha`	non-randomness parameters alpha_ij (symmetric, alpha_ii = 0)	

output	meaning
`gamma1`	activity coefficient of component 1
`gamma`	activity coefficients of every component
`GE_RT`	excess Gibbs energy over RT

- **Method:** $G_ij = exp$(-alpha_ij tau_ij); ln gamma_i = sum_j x_j tau_ji G_ji/sum_k x_k G_ki + sum_j (x_j G_ij/sum_k x_k G_kj)(tau_ij - sum_m x_m tau_mj G_mj/sum_k x_k G_kj)
- **Assumes:** $tau_i i = 0$, $alpha_i i = 0$; all mole fractions positive
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_nrtl_activity_coefficients', x=[0.3, 0.7], tau=[[0.0,
0.8], [1.2, 0.0]], alpha=[[0.0, 0.3], [0.3, 0.0]])
# -> gamma1 = 2.25346, gamma = [2.25346353086, 1.18777953647], GE_RT = 0.3642   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_nrtl_activity_coefficients', x=[0.25, 0.25, 0.5], tau=
[[0.0, 1.5, 0.4], [0.2, 0.0, 2.0], [0.9, -0.3, 0.0]], alpha=[[0.0, 0.3, 0.2], [0.3, 0.0, 0.47],
[0.2, 0.47, 0.0]])
# -> gamma1 = 1.822, gamma = [1.82200497499, 1.44773823655, 1.13894017298], GE_RT = 0.307534
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
NRTL activity coefficients: x = [0.3, 0.7], tau = [[0.0, 0.8], [1.2, 0.0]], alpha = [[0.0, 0.3], [0.3, 0.0]]	SOLVED	gamma1 = 2.25346, gamma = [2.25346353086, 1.18777953647], GE_RT = 0.3642

question	outcome	result
NRTL activity coefficients: x = [0.25, 0.25, 0.5], tau = [[0.0, 1.5, 0.4], [0.2, 0.0, 2.0], [0.9, -0.3, 0.0]], alpha = [[0.0, 0.3, 0.2], [0.3, 0.0, 0.47], [0.2, 0.47, 0.0]]	SOLVED	gamma1 = 1.822, gamma = [1.82200497499, 1.44773823655, 1.13894017298], GE_RT = 0.307534
NRTL activity coefficients: tau = [[0.0, 0.8], [1.2, 0.0]], alpha = [[0.0, 0.3], [0.3, 0.0]]	NEEDS_INPUT	chemeng_nrtl_activity_coefficients needs x (liquid mole fractions). Example: solve('chemeng_nrtl_activity_coefficients',
NRTL activity coefficients: x = [0.3, 0.7], tau = [[0.0, 0.8], [1.2, 0.0]], alpha = [[0.0, 0.3], [0.3, 0.0]], zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Activity coefficients from the UNIQUAC equation — `chemeng_uniquac_activity_coefficients`

Activity coefficients of a liquid mixture from UNIQUAC volume and area parameters r, q and interaction parameters tau_ij (coordination number 10).

$$\ln\gamma_i = \ln\frac{\Phi_i}{x_i} + \frac{z}{2}q_i\ln\frac{\theta_i}{\Phi_i} + l_i - \frac{\Phi_i}{x_i}\sum_j x_j l_j + q_i\Big[1 - \ln\sum_j \theta_j\tau_{ji} - \sum_j \frac{\theta_j\tau_{ij}}{\sum_k \theta_k\tau_{kj}}\Big]$$

input	meaning	condition
`x`	liquid mole fractions	
`r`	UNIQUAC volume parameters r_i	
`q`	UNIQUAC area parameters q_i	
`tau`	UNIQUAC parameters $tau_ij = exp$(-a_ij/T) at the temperature of interest $(tau_i i = 1)$	

output	meaning
`gamma1`	activity coefficient of component 1
`gamma`	activity coefficients of every component

- **Method:** combinatorial part ln(Phi_i/x_i) + 5 q_i ln(theta_i/Phi_i) + l_i - (Phi_i/x_i) sum x_j l_j plus residual part q_i [$1 - ln$(sum_j theta_j tau_ji) - sum_j theta_j tau_ij/sum_k theta_k tau_kj]
- **Assumes:** $tau_i i = 1$, $tau_i j > 0$; all mole fractions positive; coordination number $z = 10$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_uniquac_activity_coefficients', x=[0.3, 0.7], r=[2.1055, 0.92], q=[1.972, 1.4], tau=[[1.0, 0.9], [1.1, 1.0]])
# -> gamma1 = 1.29869, gamma = [1.29868568479, 1.10358464339]   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_uniquac_activity_coefficients', x=[0.2, 0.3, 0.5], r=[2.5735, 3.1878, 0.92], q=[2.336, 2.4, 1.4], tau=[[1.0, 0.7, 1.3], [1.2, 1.0, 0.6], [0.5, 1.6, 1.0]])
# -> gamma1 = 1.1955, gamma = [1.19549838541, 1.20648140089, 1.47615584685]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
UNIQUAC activity coefficients: x = [0.3, 0.7], r = [2.1055, 0.92], q = [1.972, 1.4], tau = [[1.0, 0.9], [1.1, 1.0]]	SOLVED	gamma1 = 1.29869, gamma = [1.29868568479, 1.10358464339]
UNIQUAC activity coefficients: x = [0.2, 0.3, 0.5], r = [2.5735, 3.1878, 0.92], q = [2.336, 2.4, 1.4], tau = [[1.0, 0.7, 1.3], [1.2, 1.0, 0.6], [0.5, 1.6, 1.0]]	SOLVED	gamma1 = 1.1955, gamma = [1.19549838541, 1.20648140089, 1.47615584685]
UNIQUAC activity coefficients: r = [2.1055, 0.92], q = [1.972, 1.4], tau = [[1.0, 0.9], [1.1, 1.0]]	NEEDS_INPUT	chemeng_uniquac_activity_coefficients needs x (liquid mole fractions). Example: solve('chemeng_uniquac_activity_coeffici
UNIQUAC activity coefficients: x = [0.3, 0.7], r = [2.1055, 0.92], q = [1.972, 1.4], tau = [[1.0, 0.9], [1.1, 1.0]], zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Peng-Robinson equation of state: compressibility and fugacity coefficient — `chemeng_peng_robinson`

Compressibility factors (vapour and liquid roots) and fugacity coefficients of a pure substance at T and P from the Peng-Robinson equation of state.

$$Z^3 - (1 - B)Z^2 + (A - 3B^2 - 2B)Z - (AB - B^2 - B^3) = 0$$

input	meaning	condition
`Tc`	critical temperature	
`Pc`	critical pressure	
`omega`	acentric factor	
`T`	temperature	
`P`	pressure	

output	meaning
`Z_vapour`	compressibility factor of the vapour-like (largest) root
`Z_liquid`	compressibility factor of the liquid-like (smallest) root
`phi_vapour`	fugacity coefficient on the vapour root
`phi_liquid`	fugacity coefficient on the liquid root
`n_roots`	number of real roots above the covolume (1 or 3)

- **Method:** Z^3 - (1 - B)Z^2 + (A - 3B^2 - 2B)$Z - (AB - B^2 - B^3) = 0$ with $a = 0.45724$ Double exponent: use braces to clarify alpha(T), $b = 0.07780$ R Tc/Pc, $\kappa = 0.37464 + 1.54226\ \omega - 0.26992\ \omega^2$
- **Assumes:** pure substance; not too close to the critical point or a spinodal (a double root is refused); one real root: both roots reported equal
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_peng_robinson', Tc=507.6, Pc=3025000.0, omega=0.2975,
T=400.0, P=1000000.0)
# -> Z_vapour = 0.644021, Z_liquid = 0.0469282, phi_vapour = 0.746232, phi_liquid = 0.421597,
n_roots = 3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_peng_robinson', Tc=190.564, Pc=4599000.0, omega=0.011,
T=300.0, P=5000000.0)
# -> Z_vapour = 0.90176, Z_liquid = 0.90176, phi_vapour = 0.901312, phi_liquid = 0.901312, n_roots
= 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Peng-Robinson: Tc = 507.6 K, Pc = 3025000.0 Pa, omega = 0.2975, T = 400.0 K, P = 1000000.0 Pa	SOLVED	Z_vapour = 0.644021, Z_liquid = 0.0469282, phi_vapour = 0.746232, phi_liquid = 0.421597, n_roots = 3
Peng-Robinson: Tc = 190.564 K, Pc = 4599000.0 Pa, omega = 0.011, T = 300.0 K, P = 5000000.0 Pa	SOLVED	Z_vapour = 0.90176, Z_liquid = 0.90176, phi_vapour = 0.901312, phi_liquid = 0.901312, n_roots = 1
Peng-Robinson: Pc = 3025000.0 Pa, omega = 0.2975, T = 400.0 K, P = 1000000.0 Pa	NEEDS_INPUT	chemeng_peng_robinson needs Tc (critical temperature). Example: solve('chemeng_peng_robinson', Tc='507.6 K', Pc='3.025e+
Peng-Robinson: Tc = 507.6 K, Pc = 3025000.0 Pa, omega = 0.2975, T = 400.0 K, P = 1000000.0 Pa, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Soave-Redlich-Kwong equation of state: compressibility and fugacity coefficient — `chemeng_srk`

Compressibility factors (vapour and liquid roots) and fugacity coefficients of a pure substance at T and P from the Soave-Redlich-Kwong equation of state.

$$Z^3 - Z^2 + (A - B - B^2)Z - AB = 0$$

input	meaning	condition
`Tc`	critical temperature	
`Pc`	critical pressure	
`omega`	acentric factor	
`T`	temperature	
`P`	pressure	

output	meaning
`Z_vapour`	compressibility factor of the vapour-like (largest) root
`Z_liquid`	compressibility factor of the liquid-like (smallest) root
`phi_vapour`	fugacity coefficient on the vapour root
`phi_liquid`	fugacity coefficient on the liquid root
`n_roots`	number of real roots above the covolume (1 or 3)

- **Method:** Z^3 - Z^2 + (A - B - B^2)$Z - AB = 0$ with $a = 0.42748$ Double exponent: use braces to clarify alpha(T), $b = 0.08664$ R Tc/Pc, $m = 0.480 + 1.574\,\omega - 0.176\,\omega^2$
- **Assumes:** pure substance; not too close to the critical point or a spinodal; one real root: both roots reported equal
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_srk', Tc=507.6, Pc=3025000.0, omega=0.2975, T=400.0,
P=1000000.0)
# -> Z_vapour = 0.65891, Z_liquid = 0.0532184, phi_vapour = 0.757696, phi_liquid = 0.430644,
n_roots = 3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_srk', Tc=190.564, Pc=4599000.0, omega=0.011, T=300.0,
P=5000000.0)
# -> Z_vapour = 0.923848, Z_liquid = 0.923848, phi_vapour = 0.923117, phi_liquid = 0.923117,
n_roots = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Soave-Redlich-Kwong: Tc = 507.6 K, Pc = 3025000.0 Pa, omega = 0.2975, T = 400.0 K, P = 1000000.0 Pa	SOLVED	Z_vapour = 0.65891, Z_liquid = 0.0532184, phi_vapour = 0.757696, phi_liquid = 0.430644, n_roots = 3
Soave-Redlich-Kwong: Tc = 190.564 K, Pc = 4599000.0 Pa, omega = 0.011, T = 300.0 K, P = 5000000.0 Pa	SOLVED	Z_vapour = 0.923848, Z_liquid = 0.923848, phi_vapour = 0.923117, phi_liquid = 0.923117, n_roots = 1
Soave-Redlich-Kwong: Pc = 3025000.0 Pa, omega = 0.2975, T = 400.0 K, P = 1000000.0 Pa	NEEDS_INPUT	chemeng_srk needs Tc (critical temperature). Example: solve('chemeng_srk', Tc='507.6 K', Pc='3.025e+06 Pa', omega=0.2975
Soave-Redlich-Kwong: Tc = 507.6 K, Pc = 3025000.0 Pa, omega = 0.2975, T = 400.0 K, P = 1000000.0 Pa, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Vapour pressure by the Lee-Kesler correlation — `chemeng_lee_kesler_vapour_pressure`

Vapour pressure of a pure substance from its critical constants and acentric factor (Lee-Kesler corresponding-states correlation).

$$\ln \frac{P^{sat}}{P_c} = f^{(0)}(T_r) + \omega f^{(1)}(T_r)$$

input	meaning	condition
`T`	temperature	
`Tc`	critical temperature	
`Pc`	critical pressure	
`omega`	acentric factor	

output	meaning
`Psat`	vapour pressure

- **Method:** ln(Psat/Pc) = f0(Tr) + omega f1(Tr) with the Lee-Kesler functions
- **Assumes:** $T < Tc$; non-polar or slightly polar substances
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_lee_kesler_vapour_pressure', T=400.0, Tc=568.7,
Pc=2490000.0, omega=0.399)
# -> Psat = 104829 Pa   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_lee_kesler_vapour_pressure', T=300.0, Tc=425.12,
Pc=3796000.0, omega=0.2)
# -> Psat = 258310 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Lee-Kesler vapour pressure: T = 400.0 K, Tc = 568.7 K, Pc = 2490000.0 Pa, omega = 0.399	SOLVED	Psat = 104829 Pa
Lee-Kesler vapour pressure: T = 300.0 K, Tc = 425.12 K, Pc = 3796000.0 Pa, omega = 0.2	SOLVED	Psat = 258310 Pa
Lee-Kesler vapour pressure: Tc = 568.7 K, Pc = 2490000.0 Pa, omega = 0.399	NEEDS_INPUT	chemeng_lee_kesler_vapour_pressure needs T (temperature). Example: solve('chemeng_lee_kesler_vapour_pressure', T='400 K'
Lee-Kesler vapour pressure: T = 400.0 K, Tc = 568.7 K, Pc = 2490000.0 Pa, omega = 0.399, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Heat of vaporisation at another temperature (Watson) — `chemeng_watson_heat_of_vaporization`

Heat of vaporisation at T from a known value at T_ref and the critical temperature (Watson's correlation, exponent 0.38).

$$\Delta H_v(T) = \Delta H_v(T_{ref}) \left(\frac{1 - T_r}{1 - T_{r,ref}} \right)^{0.38}$$

input	meaning	condition
`T`	temperature wanted	
`T_ref`	temperature of the known value	
`Hvap_ref`	heat of vaporisation at T_ref	
`Tc`	critical temperature	

output	meaning
`Hvap`	heat of vaporisation at T

- **Method:** dHvap(T) = dHvap(T_ref) [(1 - T/Tc)/(1 - T_ref/Tc)]^0.38
- **Assumes:** T, T_ref < Tc
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_watson_heat_of_vaporization', T=298.15, T_ref=373.15,
Hvap_ref=40656.0, Tc=647.1)
# -> Hvap = 44571.7 J/mol   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_watson_heat_of_vaporization', T=400.0, T_ref=353.24,
Hvap_ref=30720.0, Tc=562.05)
# -> Hvap = 27898.6 J/mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Watson heat of vaporization: T = 298.15 K, T_ref = 373.15 K, Hvap_ref = 40656.0 J/mol, Tc = 647.1 K	SOLVED	Hvap = 44571.7 J/mol
Watson heat of vaporization: T = 400.0 K, T_ref = 353.24 K, Hvap_ref = 30720.0 J/mol, Tc = 562.05 K	SOLVED	Hvap = 27898.6 J/mol
Watson heat of vaporization: T_ref = 373.15 K, Hvap_ref = 40656.0 J/mol, Tc = 647.1 K	NEEDS_INPUT	chemeng_watson_heat_of_vaporization needs T (temperature wanted). Example: solve('chemeng_watson_heat_of_vaporization',
Watson heat of vaporization: T = 298.15 K, T_ref = 373.15 K, Hvap_ref = 40656.0 J/mol, Tc = 647.1 K, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Heat of vaporisation at the normal boiling point (Riedel) —

`chemeng_riedel_heat_of_vaporization`

Heat of vaporisation at the normal boiling point from the boiling point and critical constants (Riedel's equation, Pc in bar).

$$\Delta H_{vb} = 1.093\, R\, T_b\, \frac{\ln P_c - 1.013}{0.930 - T_{br}}$$

input	meaning	condition
`Tb`	normal boiling point	
`Tc`	critical temperature	
`Pc`	critical pressure	

output	meaning
`Hvap_b`	heat of vaporisation at the normal boiling point

- **Method:** dHvb = 1.093 R Tb (ln Pc[bar] - 1.013)/(0.930 - Tbr)
- **Assumes:** Tb well below Tc
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_riedel_heat_of_vaporization', Tb=294.0, Tc=466.0,
Pc=5550000.0)
# -> Hvap_b = 26828.6 J/mol   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_riedel_heat_of_vaporization', Tb=353.24, Tc=562.05,
Pc=4895000.0)
# -> Hvap_b = 30639.1 J/mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Riedel heat of vaporization: Tb = 294.0 K, Tc = 466.0 K, Pc = 5550000.0 Pa	SOLVED	Hvap_b = 26828.6 J/mol
Riedel heat of vaporization: Tb = 353.24 K, Tc = 562.05 K, Pc = 4895000.0 Pa	SOLVED	Hvap_b = 30639.1 J/mol
Riedel heat of vaporization: Tc = 466.0 K, Pc = 5550000.0 Pa	NEEDS_INPUT	chemeng_riedel_heat_of_vaporization needs Tb (normal boiling point). Example: solve('chemeng_riedel_heat_of_vaporization
Riedel heat of vaporization: Tb = 294.0 K, Tc = 466.0 K, Pc = 5550000.0 Pa, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Heat of vaporisation from the acentric factor (Pitzer) — `chemeng_pitzer_heat_of_vaporization`

Heat of vaporisation at T from the critical temperature and acentric factor (Pitzer's corresponding-states correlation as fitted by Carruth and Kobayashi).

$$\frac{\Delta H_v}{RT_c} = 7.08(1-T_r)^{0.354} + 10.95\,\omega(1-T_r)^{0.456}$$

input	meaning	condition
`T`	temperature	
`Tc`	critical temperature	
`omega`	acentric factor	

output	meaning
`Hvap`	heat of vaporisation

- **Method:** dHv/(R Tc) = 7.08 $(1-Tr)^{0.354} + 10.95$ omega $(1-Tr)^{0.456}$
- **Assumes:** $0.6 < Tr < 1$ (fit range)

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_pitzer_heat_of_vaporization', T=452.0, Tc=645.6,
omega=0.35)
# -> Hvap = 36691 J/mol   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_pitzer_heat_of_vaporization', T=300.0, Tc=425.12,
omega=0.2)
# -> Hvap = 20662.5 J/mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Pitzer heat of vaporization: T = 452.0 K, Tc = 645.6 K, omega = 0.35	SOLVED	Hvap = 36691 J/mol
Pitzer heat of vaporization: T = 300.0 K, Tc = 425.12 K, omega = 0.2	SOLVED	Hvap = 20662.5 J/mol
Pitzer heat of vaporization: Tc = 645.6 K, omega = 0.35	NEEDS_INPUT	chemeng_pitzer_heat_of_vaporization needs T (temperature). Example: solve('chemeng_pitzer_heat_of_vaporization', T='452
Pitzer heat of vaporization: T = 452.0 K, Tc = 645.6 K, omega = 0.35, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Saturated liquid molar volume (Rackett) — `chemeng_rackett_liquid_volume`

Molar volume of a saturated liquid from the critical constants and the critical compressibility factor (Rackett equation).

$$V_s = \frac{RT_c}{P_c} Z_c^{1+(1-T_r)^{2/7}}$$

input	meaning	condition
`T`	temperature	
`Tc`	critical temperature	
`Pc`	critical pressure	
`Zc`	critical compressibility factor	

output	meaning
`Vm`	saturated liquid molar volume

- **Method:** Vs = (R Tc/Pc) Zc^[$1 + (1 - Tr)^{2/7}$]
- **Assumes:** $T < Tc$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_rackett_liquid_volume', T=272.03889, Tc=369.83,
Pc=4248000.0, Zc=0.2763)
# -> Vm = 8.29923e-05 m^3/mol   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_rackett_liquid_volume', T=298.15, Tc=562.05,
Pc=4895000.0, Zc=0.268)
# -> Vm = 8.85568e-05 m^3/mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Rackett liquid volume: T = 272.03889 K, Tc = 369.83 K, Pc = 4248000.0 Pa, Zc = 0.2763	SOLVED	Vm = 8.29923e-05 m^3/mol
Rackett liquid volume: T = 298.15 K, Tc = 562.05 K, Pc = 4895000.0 Pa, Zc = 0.268	SOLVED	Vm = 8.85568e-05 m^3/mol
Rackett liquid volume: Tc = 369.83 K, Pc = 4248000.0 Pa, Zc = 0.2763	NEEDS_INPUT	chemeng_rackett_liquid_volume needs T (temperature). Example: solve('chemeng_rackett_liquid_volume', T='272.039 K', Tc='
Rackett liquid volume: T = 272.03889 K, Tc = 369.83 K, Pc = 4248000.0 Pa, Zc = 0.2763, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Gas compressibility from the virial equation (Pitzer-Abbott correlation) — `chemeng_virial_compressibility`

Second virial coefficient from the Abbott (Pitzer) correlation and the compressibility factor $Z = 1 + BP/(RT)$ of a gas at moderate pressure.

$$\frac{BP_c}{RT_c} = 0.083 - \frac{0.422}{T_r^{1.6}} + \omega\left(0.139 - \frac{0.172}{T_r^{4.2}}\right), \quad Z = 1 + \frac{BP}{RT}$$

input	meaning	condition
`T`	temperature	
`P`	pressure	
`Tc`	critical temperature	
`Pc`	critical pressure	
`omega`	acentric factor	

output	meaning
`B`	second virial coefficient
`Z`	compressibility factor

- **Method:** B Pc/(R Tc) = B0 + omega B1, $B_0 = 0.083 - 0.422/Tr^{1.6}$, $B_1 = 0.139 - 0.172/Tr^{4.2}$; $Z = 1 + B$ P/(R T)
- **Assumes:** moderate pressure (truncated virial equation); non-polar or slightly polar gas

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_virial_compressibility', T=510.0, P=2500000.0, Tc=425.1,
Pc=3796000.0, omega=0.2)
# -> B = -0.000205361 m^3/mol, Z = 0.878925   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_virial_compressibility', T=300.0, P=1000000.0, Tc=304.2,
Pc=7383000.0, omega=0.224)
# -> B = -0.000122712 m^3/mol, Z = 0.950804   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Virial compressibility: T = 510.0 K, P = 2500000.0 Pa, Tc = 425.1 K, Pc = 3796000.0 Pa, omega = 0.2	SOLVED	B = -0.000205361 m^3/mol, Z = 0.878925
Virial compressibility: T = 300.0 K, P = 1000000.0 Pa, Tc = 304.2 K, Pc = 7383000.0 Pa, omega = 0.224	SOLVED	B = -0.000122712 m^3/mol, Z = 0.950804
Virial compressibility: P = 2500000.0 Pa, Tc = 425.1 K, Pc = 3796000.0 Pa, omega = 0.2	NEEDS_INPUT	chemeng_virial_compressibility needs T (temperature). Example: solve('chemeng_virial_compressibility', T='510 K', P='2.5
Virial compressibility: T = 510.0 K, P = 2500000.0 Pa, Tc = 425.1 K, Pc = 3796000.0 Pa, omega = 0.2, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Mass fractions to mole fractions and mean molar mass — `chemeng_mass_to_mole_fractions`

Mole fractions of a mixture and its mean molar mass from the mass fractions and the molar masses of the components.

$$z_i = \frac{w_i/M_i}{\sum_j w_j/M_j}, \quad \bar{M} = \sum_i z_i M_i$$

input	meaning	condition
`w`	mass fractions (sum to 1)	
`MW`	molar masses of the components, g/mol	

output	meaning
`z`	mole fractions
`MW_mean`	mean molar mass of the mixture

- **Method:** z_i = (w_i/M_i)/sum_j (w_j/M_j); M = sum z_i M_i
- **Assumes:** fractions sum to 1; positive molar masses
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_mass_to_mole_fractions', w=[0.5, 0.5], MW=[32.042,
18.015])
# -> z = [0.359889725713, 0.640110274287], MW_mean = 0.0230632 kg/mol   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_mass_to_mole_fractions', w=[0.2, 0.3, 0.5], MW=[16.043,
30.07, 44.097])
# -> z = [0.369029335988, 0.2953277837, 0.335642880312], MW_mean = 0.0296017 kg/mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Mass to mole fractions: w = [0.5, 0.5], MW = [32.042, 18.015]	SOLVED	z = [0.359889725713, 0.640110274287], MW_mean = 0.0230632 kg/mol
Mass to mole fractions: w = [0.2, 0.3, 0.5], MW = [16.043, 30.07, 44.097]	SOLVED	z = [0.369029335988, 0.2953277837, 0.335642880312], MW_mean = 0.0296017 kg/mol
Mass to mole fractions: MW = [32.042, 18.015]	NEEDS_INPUT	chemeng_mass_to_mole_fractions needs w (mass fractions (sum to 1)). Example: solve('chemeng_mass_to_mole_fractions', w=[
Mass to mole fractions: w = [0.5, 0.5], MW = [32.042, 18.015], zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Diffusion of A through stagnant B (Stefan film) — `chemeng_stefan_diffusion_flux`

Steady molar flux of A diffusing through a stagnant film of non-diffusing B in an ideal gas (evaporation, absorption through an inert).

$$N_A = \frac{cD_{AB}}{L}\ln\frac{1-y_{A2}}{1-y_{A1}}, \quad c = \frac{P}{RT}$$

input	meaning	condition
`D`	diffusivity of A in B	
`P`	total pressure	
`T`	temperature	
`L`	film thickness (diffusion path length)	
`yA1`	mole fraction of A at the start of the path	
`yA2`	mole fraction of A at the end of the path	

output	meaning
`NA`	molar flux of A

- **Method:** N_A = (c D/L) ln[$(1-yA2)/(1-yA1)$], c = P/(R T)
- **Assumes:** steady state, ideal gas, constant D; B stagnant (no net flux of B)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_stefan_diffusion_flux', D=2.6e-05, P=101325.0, T=298.15, L=0.1, yA1=0.0313, yA2=0.0)
# -> NA = 0.00033795 mol/(m^2*s)   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_stefan_diffusion_flux', D=1e-05, P=200000.0, T=350.0, L=0.005, yA1=0.6, yA2=0.2)
# -> NA = 0.0952759 mol/(m^2*s)   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Stefan diffusion flux: D = 2.6e-05 m^2/s, P = 101325.0 Pa, T = 298.15 K, L = 0.1 m, yA1 = 0.0313, yA2 = 0.0	SOLVED	NA = 0.00033795 mol/(m^2*s)
Stefan diffusion flux: D = 1e-05 m^2/s, P = 200000.0 Pa, T = 350.0 K, L = 0.005 m, yA1 = 0.6, yA2 = 0.2	SOLVED	NA = 0.0952759 mol/(m^2*s)
Stefan diffusion flux: P = 101325.0 Pa, T = 298.15 K, L = 0.1 m, yA1 = 0.0313, yA2 = 0.0	NEEDS_INPUT	chemeng_stefan_diffusion_flux needs D (diffusivity of A in B). Example: solve('chemeng_stefan_diffusion_flux', D='2.6e-0
Stefan diffusion flux: D = 2.6e-05 m^2/s, P = 101325.0 Pa, T = 298.15 K, L = 0.1 m, yA1 = 0.0313, yA2 = 0.0, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Stefan diffusion flux: D = 0.26 cm^2/s, P = 1 atm, T = 25 degC, L = 10 cm, yA1 = 3.13%, yA2 = 0	SOLVED	NA = 0.00033795 mol/(m^2*s)

Packed-bed pressure drop (Ergun equation) — `chemeng_ergun_pressure_drop`

Pressure drop of a fluid flowing through a packed bed of spheres from the particle size, voidage, superficial velocity and fluid properties (Ergun equation).

$$\frac{\Delta P}{L} = \frac{150\,\mu(1-\varepsilon)^2 v_s}{\varepsilon^3 d_p^2} + \frac{1.75\,(1-\varepsilon)\rho v_s^2}{\varepsilon^3 d_p}$$

input	meaning	condition
`dp`	particle diameter	
`eps`	bed voidage (porosity)	
`vs`	superficial velocity	
`rho`	fluid density	
`mu`	fluid viscosity	
`L`	bed depth	

output	meaning
`dP`	pressure drop across the bed

- **Method:** $dP/L = 150$ mu (1 - eps)^2 vs/(eps^3 dp^2) + 1.75 (1 - eps) rho vs^2/(eps^3 dp)
- **Assumes:** 0 < eps < 1; incompressible flow (or small pressure drop for a gas); spherical or equivalent particles
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_ergun_pressure_drop', dp=0.0008, eps=0.4, vs=0.05,
rho=1000.0, mu=0.001, L=1.0)
# -> dP = 117188 Pa   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_ergun_pressure_drop', dp=0.005, eps=0.38, vs=1.2,
rho=1.2, mu=1.8e-05, L=2.0)
# -> dP = 15483.1 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Ergun pressure drop: dp = 0.0008 m, eps = 0.4, vs = 0.05 m/s, rho = 1000.0 kg/m^3, mu = 0.001 Pa*s, L = 1.0 m	SOLVED	dP = 117188 Pa
Ergun pressure drop: dp = 0.005 m, eps = 0.38, vs = 1.2 m/s, rho = 1.2 kg/m^3, mu = 1.8e-05 Pa*s, L = 2.0 m	SOLVED	dP = 15483.1 Pa
Ergun pressure drop: eps = 0.4, vs = 0.05 m/s, rho = 1000.0 kg/m^3, mu = 0.001 Pa*s, L = 1.0 m	NEEDS_INPUT	chemeng_ergun_pressure_drop needs dp (particle diameter). Example: solve('chemeng_ergun_pressure_drop', dp='0.0008 m', e
Ergun pressure drop: dp = 0.0008 m, eps = 0.4, vs = 0.05 m/s, rho = 1000.0 kg/m^3, mu = 0.001 Pa*s, L = 1.0 m, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Ergun pressure drop: dp = 0.8 mm, eps = 0.4, vs = 5 cm/s, rho = 1000 kg/m^3, mu = 1 cP, L = 1 m	SOLVED	dP = 117188 Pa

Packed-bed pressure drop (Carman correlation) — `chemeng_carman_pressure_drop`

Pressure drop through a packed bed from Carman's correlation of the bed friction factor (180/Re' + $2.87/Re$ '^0.1 form).

$$f_p = \frac{\left(180 + 2.871\left(\frac{Re}{1-\varepsilon}\right)^{0.9}\right)(1-\varepsilon)^2}{\varepsilon^3 Re}, \quad \Delta P = \frac{f_p\,\rho v_s^2 L}{d_p}$$

input	meaning	condition
`dp`	particle diameter	
`eps`	bed voidage (porosity)	
`vs`	superficial velocity	

input	meaning	condition
`rho`	fluid density	
`mu`	fluid viscosity	
`L`	bed depth	

output	meaning
`dP`	pressure drop across the bed

- **Method:** f_p = (180 + 2.871 (Re/(1 - eps))^0.9)(1 - eps)^2/(eps^3 Re); $dP = f_p$ rho vs^2 L/dp, $Re = dp$ rho vs/mu
- **Assumes:** 0 < eps < 1; incompressible flow
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_carman_pressure_drop', dp=0.0008, eps=0.4, vs=0.05,
rho=1000.0, mu=0.001, L=1.0)
# -> dP = 134368 Pa   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_carman_pressure_drop', dp=0.005, eps=0.38, vs=1.2,
rho=1.2, mu=1.8e-05, L=2.0)
# -> dP = 13920.1 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Carman pressure drop: dp = 0.0008 m, eps = 0.4, vs = 0.05 m/s, rho = 1000.0 kg/m^3, mu = 0.001 Pa*s, L = 1.0 m	SOLVED	dP = 134368 Pa
Carman pressure drop: dp = 0.005 m, eps = 0.38, vs = 1.2 m/s, rho = 1.2 kg/m^3, mu = 1.8e-05 Pa*s, L = 2.0 m	SOLVED	dP = 13920.1 Pa
Carman pressure drop: eps = 0.4, vs = 0.05 m/s, rho = 1000.0 kg/m^3, mu = 0.001 Pa*s, L = 1.0 m	NEEDS_INPUT	chemeng_carman_pressure_drop needs dp (particle diameter). Example: solve('chemeng_carman_pressure_drop', dp='0.0008 m',
Carman pressure drop: dp = 0.0008 m, eps = 0.4, vs = 0.05 m/s, rho = 1000.0 kg/m^3, mu = 0.001 Pa*s, L = 1.0 m, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Constant-pressure cake filtration: filtrate volume after a time —

`chemeng_cake_filtration_volume`

Volume of filtrate collected after time t in constant-pressure cake filtration (Ruth equation with cake and medium resistances).

$$t = \frac{\mu \alpha c}{2A^2 \Delta P} V^2 + \frac{\mu R_m}{A \Delta P} V$$

input	meaning	condition
`A`	filter area	
`dP`	pressure difference across cake and medium	
`mu`	filtrate viscosity	
`alpha`	specific cake resistance	
`c`	mass of dry cake deposited per volume of filtrate	
`Rm`	filter medium resistance	
`t`	filtration time	

output	meaning
`V`	filtrate volume collected

- **Method:** $dV/dt = A^2$ dP/(mu (alpha c V + A Rm)), integrated: $t = mu$ alpha c V^2/(2 A^2 dP) + mu Rm V/(A dP), solved for V
- **Assumes:** constant pressure, incompressible cake; $Rm > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_cake_filtration_volume', A=1.0, dP=100000.0, mu=0.001,
alpha=100000000000.0, c=20.0, Rm=10000000000.0, t=600.0)
# -> V = 0.24 m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_cake_filtration_volume', A=10.0, dP=300000.0, mu=0.0012,
alpha=50000000000.0, c=50.0, Rm=50000000000.0, t=3600.0)
# -> V = 8.28764 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Cake filtration volume: A = 1.0 m^2, dP = 100000.0 Pa, mu = 0.001 Pa*s, alpha = 100000000000.0 m/kg, c = 20.0 kg/m^3, Rm = 10000000000.0 1/m, t = 600.0 s	SOLVED	V = 0.24 m^3
Cake filtration volume: A = 10.0 m^2, dP = 300000.0 Pa, mu = 0.0012 Pa*s, alpha = 50000000000.0 m/kg, c = 50.0 kg/m^3, Rm = 50000000000.0 1/m, t = 3600.0 s	SOLVED	V = 8.28764 m^3
Cake filtration volume: dP = 100000.0 Pa, mu = 0.001 Pa*s, alpha = 100000000000.0 m/kg, c = 20.0 kg/m^3, Rm = 10000000000.0 1/m, t = 600.0 s	NEEDS_INPUT	chemeng_cake_filtration_volume needs A (filter area). Example: solve('chemeng_cake_filtration_volume', A='1 m^2', dP='10
Cake filtration volume: A = 1.0 m^2, dP = 100000.0 Pa, mu = 0.001 Pa*s, alpha = 100000000000.0 m/kg, c = 20.0 kg/m^3, Rm = 10000000000.0 1/m, t = 600.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Specific cake and medium resistance from constant-pressure filtration data — `chemeng_filtration_constants_fit`

Specific cake resistance and filter-medium resistance from a constant-pressure filtration test (times and filtrate volumes), by the straight line of t/V against V.

$$\frac{t}{V} = \frac{\mu \alpha c}{2A^2 \Delta P} V + \frac{\mu R_m}{A \Delta P}$$

input	meaning	condition
`t`	times at which the filtrate volumes were read, in t_unit	
`V`	cumulative filtrate volumes, in v_unit	
`t_unit`	unit of the times	
`v_unit`	unit of the volumes	
`A`	filter area	
`dP`	constant pressure difference	
`mu`	filtrate viscosity	
`c`	dry cake mass per volume of filtrate	

output	meaning
`alpha`	specific cake resistance
`Rm`	filter medium resistance
`slope`	slope of t/V against V, s/m^6
`intercept`	intercept of t/V against V, s/m^3

- **Method:** least squares of t/V = (mu alpha c/(2 A^2 dP)) V + mu Rm/(A dP)
- **Assumes:** constant pressure, incompressible cake; a positive slope and intercept
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_filtration_constants_fit', t=[17.3, 41.3, 72.0, 108.3,
152.1, 201.7], V=[0.5, 1.0, 1.5, 2.0, 2.5, 3.0], t_unit='s', v_unit='L', A=0.0439, dP=338000.0,
mu=0.000895, c=23.47)
# -> alpha = 8.07822e+11 m/kg, Rm = 4.67973e+11 1/m, slope = 1.3025e+07, intercept = 28226.9
(SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_filtration_constants_fit', t=[2.0, 5.0, 9.5, 15.0,
21.5], V=[0.1, 0.2, 0.3, 0.4, 0.5], t_unit='min', v_unit='m3', A=2.0, dP=150000.0, mu=0.001,
c=30.0)
# -> alpha = 1.404e+11 m/kg, Rm = 2.499e+11 1/m, slope = 3510, intercept = 833   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Filtration constants fit: t = [17.3, 41.3, 72.0, 108.3, 152.1, 201.7], V = [0.5, 1.0, 1.5, 2.0, 2.5, 3.0], t_unit = s, v_unit = L, A = 0.0439	SOLVED	alpha = 8.07822e+11 m/kg, Rm = 4.67973e+11 1/m, slope = 1.3025e+07, intercept = 28226.9

question	outcome	result
m^2, dP = 338000.0 Pa, mu = 0.000895 Pa*s, c = 23.47 kg/m^3		
Filtration constants fit: t = [2.0, 5.0, 9.5, 15.0, 21.5], V = [0.1, 0.2, 0.3, 0.4, 0.5], t_unit = min, v_unit = m3, A = 2.0 m^2, dP = 150000.0 Pa, mu = 0.001 Pa*s, c = 30.0 kg/m^3	SOLVED	alpha = 1.404e+11 m/kg, Rm = 2.499e+11 1/m, slope = 3510, intercept = 833
Filtration constants fit: V = [0.5, 1.0, 1.5, 2.0, 2.5, 3.0], t_unit = s, v_unit = L, A = 0.0439 m^2, dP = 338000.0 Pa, mu = 0.000895 Pa*s, c = 23.47 kg/m^3	NEEDS_INPUT	chemeng_filtration_constants_fit needs t (times at which the filtrate volumes were read, in t_unit). Example: solve('che
Filtration constants fit: t = [17.3, 41.3, 72.0, 108.3, 152.1, 201.7], V = [0.5, 1.0, 1.5, 2.0, 2.5, 3.0], t_unit = s, v_unit = L, A = 0.0439 m^2, dP = 338000.0 Pa, mu = 0.000895 Pa*s, c = 23.47 kg/m^3, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

First-order-plus-dead-time process: response to a step — `chemeng_fopdt_step_response`

Deviation of the output of a first-order process with dead time at time t after a step of size M in its input.

$$y(t) = KM\left(1 - e^{-(t-\theta)/\tau}\right) u(t-\theta)$$

input	meaning	condition
`K`	process gain (output change per unit input change)	
`tau`	process time constant	
`theta`	dead time	
`M`	size of the input step (deviation)	
`t`	time after the step	

output	meaning
`y`	output deviation at time t (in the output's units)

- **Method:** y(t) = K M (1 - exp(-(t - theta)/tau)) for $t > \theta$, 0 before
- **Assumes:** the process starts at steady state; linear first-order dynamics with dead time
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_fopdt_step_response', K=2.0, tau=300.0, theta=60.0,
M=5.0, t=360.0)
# -> y = 6.32121   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_fopdt_step_response', K=-0.8, tau=120.0, theta=30.0,
M=10.0, t=600.0)
# -> y = -7.93079   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
FOPDT step response: K = 2.0, tau = 300.0 s, theta = 60.0 s, M = 5.0, t = 360.0 s	SOLVED	y = 6.32121
FOPDT step response: K = -0.8, tau = 120.0 s, theta = 30.0 s, M = 10.0, t = 600.0 s	SOLVED	y = -7.93079
FOPDT step response: tau = 300.0 s, theta = 60.0 s, M = 5.0, t = 360.0 s	NEEDS_INPUT	chemeng_fopdt_step_response needs K (process gain (output change per unit input change)). Example: solve('chemeng_fopdt_
FOPDT step response: K = 2.0, tau = 300.0 s, theta = 60.0 s, M = 5.0, t = 360.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
FOPDT step response: K = 2, tau = 5 min, theta = 1 min, M = 5, t = 6 min	SOLVED	y = 6.32121

Proportional control: steady-state offset after a set-point step — `chemeng_p_control_offset`

Steady-state output (as a fraction of a set-point step) and the offset of a P-only feedback loop around a process of first-order lags in series, after checking the loop is stable.

$$\text{offset} = \frac{1}{1 + K_c K}$$

input	meaning	condition
`Kc`	controller gain	
`K`	process gain	
`taus`	time constants of the first-order lags in series, in t_unit	
`t_unit`	unit of the time constants	

output	meaning
`y_ss`	final output change per unit set-point change
`offset`	steady-state offset per unit set-point change

- **Method:** final-value theorem: $y_s s = Kc$ K/(1 + Kc K), offset = 1/(1 + Kc K); stability of prod(tau_i s + 1) + Kc K by the Routh array
- **Assumes:** Kc $K > 0$; the closed loop is stable (checked by the Routh array)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_p_control_offset', Kc=2.0, K=1.5, taus=[5.0, 2.0],
t_unit='min')
# -> y_ss = 0.75, offset = 0.25   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_p_control_offset', Kc=0.5, K=2.0, taus=[10.0, 5.0, 1.0],
t_unit='s')
# -> y_ss = 0.5, offset = 0.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
P-only control offset: Kc = 2.0, K = 1.5, taus = [5.0, 2.0], t_unit = min	SOLVED	y_ss = 0.75, offset = 0.25
P-only control offset: Kc = 0.5, K = 2.0, taus = [10.0, 5.0, 1.0], t_unit = s	SOLVED	y_ss = 0.5, offset = 0.5
P-only control offset: K = 1.5, taus = [5.0, 2.0], t_unit = min	NEEDS_INPUT	chemeng_p_control_offset needs Kc (controller gain). Example: solve('chemeng_p_control_offset', Kc=2, K=1.5, taus=[5.0,
P-only control offset: Kc = 2.0, K = 1.5, taus = [5.0, 2.0], t_unit = min, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
P-only control offset: Kc = 20, K = 2, taus = [10, 10, 10], t_unit = s	INVALID_INPUT	Proportional control: steady-state offset after a set-point step: the closed loop is unstable at this controller gain: t

Frequency response of first-order lags in series (amplitude ratio and phase) — `chemeng_lag_frequency_response`

Amplitude ratio and phase angle of a process made of one to three first-order lags in series, at angular frequency w.

$$AR = \frac{K}{\prod_i \sqrt{1+\omega^2\tau_i^2}}, \quad \varphi = -\sum_i \arctan(\omega\tau_i)$$

input	meaning	condition
`K`	process gain	
`taus`	time constants of the lags, in t_unit	
`t_unit`	unit of the time constants	
`w`	angular frequency of the input sinusoid	

output	meaning
`AR`	amplitude ratio (output amplitude per input amplitude)
`phase_deg`	phase angle in degrees (negative: lag)

- **Method:** AR = K/prod sqrt(1 + w^2 tau_i^2); phase = -sum atan(w tau_i)
- **Assumes:** $K > 0$; linear lags without dead time
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_lag_frequency_response', K=2.0, taus=[10.0], t_unit='s',
w=0.1)
# -> AR = 1.41421, phase_deg = -45   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_lag_frequency_response', K=1.0, taus=[5.0, 2.0, 1.0],
t_unit='min', w=0.01)
# -> AR = 0.173594, phase_deg = -152.723   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Lag frequency response: K = 2.0, taus = [10.0], t_unit = s, w = 0.1 rad/s	SOLVED	AR = 1.41421, phase_deg = -45
Lag frequency response: K = 1.0, taus = [5.0, 2.0, 1.0], t_unit = min, w = 0.01 rad/s	SOLVED	AR = 0.173594, phase_deg = -152.723
Lag frequency response: taus = [10.0], t_unit = s, w = 0.1 rad/s	NEEDS_INPUT	chemeng_lag_frequency_response needs K (process gain). Example: solve('chemeng_lag_frequency_response', K=2, taus=[10.0]
Lag frequency response: K = 2.0, taus = [10.0], t_unit = s, w = 0.1 rad/s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Discrete PID controller output for an error history — `chemeng_pid_controller_output`

Output of a digital position-form PID controller at each sample for a stated history of errors (rectangular integration, derivative on the error, loop at rest before the first sample).

$$u_k = \bar{u} + K_c \left[e_k + \frac{\Delta t}{\tau_I} \sum_{j=1}^{k} e_j + \frac{\tau_D}{\Delta t} (e_k - e_{k-1}) \right]$$

input	meaning	condition
`Kc`	controller gain	
`tauI`	integral (reset) time	
`tauD`	derivative time (0 for PI)	
`dt`	sampling interval	
`e`	error samples e_1 ... e_n (set point minus measurement)	
`bias`	controller output bias (output when the error has always been zero)	

output	meaning
`u_last`	controller output at the last sample
`u`	controller output at every sample

- **Method:** u_k = bias + Kc [e_k + (dt/taul) sum_(j<=k) e_j + (tauD/dt)(e_k - e_(k-1))]
- **Assumes:** position form, $e_0 = 0$ before the first sample; rectangular integration including the current sample
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_pid_controller_output', Kc=2.0, tauI=10.0, tauD=1.0,
dt=1.0, e=[1.0, 0.8, 0.5, 0.3], bias=50.0)
# -> u_last = 50.72, u = [54.2, 51.56, 50.86, 50.72]   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_pid_controller_output', Kc=0.5, tauI=60.0, tauD=0.0,
dt=2.0, e=[2.0, 2.0, 1.5, 1.0, 0.5], bias=40.0)
# -> u_last = 40.3667, u = [41.0333333333, 41.0666666667, 40.8416666667, 40.6083333333,
40.3666666667]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
PID controller output: Kc = 2.0, tauI = 10.0 s, tauD = 1.0 s, dt = 1.0 s, e = [1.0, 0.8, 0.5, 0.3], bias = 50.0	SOLVED	u_last = 50.72, u = [54.2, 51.56, 50.86, 50.72]
PID controller output: Kc = 0.5, tauI = 60.0 s, tauD = 0.0 s, dt = 2.0 s, e = [2.0, 2.0, 1.5, 1.0, 0.5], bias = 40.0	SOLVED	u_last = 40.3667, u = [41.0333333333, 41.0666666667, 40.8416666667, 40.6083333333, 40.3666666667]
PID controller output: tauI = 10.0 s, tauD = 1.0 s, dt = 1.0 s, e = [1.0, 0.8, 0.5, 0.3], bias = 50.0	NEEDS_INPUT	chemeng_pid_controller_output needs Kc (controller gain). Example: solve('chemeng_pid_controller_output', Kc=2, tauI='10
PID controller output: Kc = 2.0, tauI = 10.0 s, tauD = 1.0 s, dt = 1.0 s, e = [1.0, 0.8, 0.5, 0.3], bias = 50.0, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Ziegler-Nichols closed-loop tuning of lags in series — `chemeng_ziegler_nichols_tuning`

Ultimate gain and period of a process of first-order lags in series, and the Ziegler-Nichols closed-loop settings of a P, PI or PID controller.

$$\sum_i \arctan(\omega_u \tau_i) = \pi, \quad K_u = \frac{1}{|G(j\omega_u)|}, \quad P_u = \frac{2\pi}{\omega_u}$$

input	meaning	condition
`K`	process gain	
`taus`	time constants of the lags (three or more), in t_unit	
`t_unit`	unit of the time constants	
`controller`	controller type: P, PI or PID	

output	meaning
`Ku`	ultimate gain
`Pu`	ultimate period
`Kc`	controller gain
`tauI`	integral time (PI, PID)
`tauD`	derivative time (PID)

- **Method:** phase crossover sum atan(w tau_i) = pi (bisection); $Ku = 1$/|G(j wu)|, $Pu = 2$ pi/wu; P: 0.5 Ku; PI: 0.45 Ku, $Pu/1.2$; PID: 0.6 Ku, $Pu/2$, $Pu/8$
- **Assumes:** three or more lags (a finite ultimate gain); $K > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_ziegler_nichols_tuning', K=2.0, taus=[1.0, 1.0, 1.0],
t_unit='min', controller='PID')
# -> Ku = 4, Pu = 217.656 s, Kc = 2.4, tauI = 108.828 s, tauD = 27.207 s   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_ziegler_nichols_tuning', K=1.5, taus=[10.0, 5.0, 2.0],
t_unit='s', controller='PI')
# -> Ku = 8.4, Pu = 15.239 s, Kc = 3.78, tauI = 12.6991 s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Ziegler-Nichols tuning: K = 2.0, taus = [1.0, 1.0, 1.0], t_unit = min, controller = PID	SOLVED	Ku = 4, Pu = 217.656 s, Kc = 2.4, tauI = 108.828 s, tauD = 27.207 s
Ziegler-Nichols tuning: K = 1.5, taus = [10.0, 5.0, 2.0], t_unit = s, controller = PI	SOLVED	Ku = 8.4, Pu = 15.239 s, Kc = 3.78, tauI = 12.6991 s
Ziegler-Nichols tuning: taus = [1.0, 1.0, 1.0], t_unit = min, controller = PID	NEEDS_INPUT	chemeng_ziegler_nichols_tuning needs K (process gain). Example: solve('chemeng_ziegler_nichols_tuning', K=2, taus=[1.0,
Ziegler-Nichols tuning: K = 2.0, taus = [1.0, 1.0, 1.0], t_unit = min, controller = PID, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Ziegler-Nichols tuning: K = 2, taus = [1, 1], t_unit = min, controller = PID	INVALID_INPUT	taus: needs at least 3 entries, got 2.

Closed-loop stability of a P-controlled process (closed-loop poles) —

`chemeng_closed_loop_stability`

Whether a feedback loop with a proportional controller Kc around a process G(s) = num(s)/den(s) is stable: the largest real part of the closed-loop poles and how many lie in the right half-plane.

$$1 + K_c G(s) = 0 \iff \mathrm{den}(s) + K_c\,\mathrm{num}(s) = 0$$

input	meaning	condition
`num`	numerator coefficients of G(s), highest power of s first	
`den`	denominator coefficients of G(s), highest power of s first	
`Kc`	controller gain	

output	meaning
`max_real`	largest real part of the closed-loop poles, 1/(time unit of the coefficients)

output	meaning
`n_rhp`	number of closed-loop poles in the right half-plane
`stable`	stable or unstable (text)

- **Method:** roots of den(s) + Kc num(s) = 0 (30-digit polynomial root finding)
- **Assumes:** proper process; no pole on the imaginary axis or nearly repeated poles (refused)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_closed_loop_stability', num=[1.0], den=[1.0, 3.0, 3.0,
1.0], Kc=10.0)
# -> max_real = 0.0772173, n_rhp = 2, stable = unstable   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_closed_loop_stability', num=[2.0], den=[1.0, 3.0, 3.0,
1.0], Kc=2.0)
# -> max_real = -0.206299, n_rhp = 0, stable = stable   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Closed-loop stability: num = [1.0], den = [1.0, 3.0, 3.0, 1.0], Kc = 10.0	SOLVED	max_real = 0.0772173, n_rhp = 2, stable = unstable
Closed-loop stability: num = [2.0], den = [1.0, 3.0, 3.0, 1.0], Kc = 2.0	SOLVED	max_real = -0.206299, n_rhp = 0, stable = stable
Closed-loop stability: den = [1.0, 3.0, 3.0, 1.0], Kc = 10.0	NEEDS_INPUT	chemeng_closed_loop_stability needs num (numerator coefficients of G(s), highest power of s first). Example: solve('chem
Closed-loop stability: num = [1.0], den = [1.0, 3.0, 3.0, 1.0], Kc = 10.0, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Gain and phase margins of a PI loop around lags in series — `chemeng_pi_loop_margins`

Gain margin, phase margin and the two crossover frequencies of a loop with a PI controller and a process of three to five first-order lags.

$$GM = \frac{1}{|L(j\omega_{pc})|}, \quad PM = 180^\circ + \arg L(j\omega_{gc})$$

input	meaning	condition
`Kc`	controller gain	
`tauI`	integral time	
`K`	process gain	
`taus`	time constants of the lags, in t_unit (the integral time is converted to this unit too)	
`t_unit`	unit of the time constants	

output	meaning
`GM`	gain margin (ratio)
`PM_deg`	phase margin, degrees
`w_pc`	phase-crossover frequency (phase -180 degrees)
`w_gc`	gain-crossover frequency (|L| = 1)

- **Method:** L(jw) = Kc (1 + 1/(j w tauI)) K/prod($1 + j$ w tau_i); |L| = 1 and arg $L = -180$ degrees by bisection; $GM = 1$/|L(j w_pc)|, PM = 180 + arg L(j w_gc)
- **Assumes:** one phase crossover (refused otherwise); Kc $K > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_pi_loop_margins', Kc=1.0, tauI=5.0, K=2.0, taus=[1.0,
1.0, 1.0], t_unit='min')
# -> GM = 3.22094, PM_deg = 51.0531, w_pc = 0.02625 rad/s, w_gc = 0.0131317 rad/s   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_pi_loop_margins', Kc=0.5, tauI=20.0, K=1.5, taus=[10.0,
5.0, 2.0], t_unit='s')
# -> GM = 12.9321, PM_deg = 90.0229, w_pc = 0.365703 rad/s, w_gc = 0.0446205 rad/s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
PI loop margins: Kc = 1.0, tauI = 5.0 s, K = 2.0, taus = [1.0, 1.0, 1.0], t_unit = min	SOLVED	GM = 3.22094, PM_deg = 51.0531, w_pc = 0.02625 rad/s, w_gc = 0.0131317 rad/s
PI loop margins: Kc = 0.5, tauI = 20.0 s, K = 1.5, taus = [10.0, 5.0, 2.0], t_unit = s	SOLVED	GM = 12.9321, PM_deg = 90.0229, w_pc = 0.365703 rad/s, w_gc = 0.0446205 rad/s
PI loop margins: tauI = 5.0 s, K = 2.0, taus = [1.0, 1.0, 1.0], t_unit = min	NEEDS_INPUT	chemeng_pi_loop_margins needs Kc (controller gain). Example: solve('chemeng_pi_loop_margins', Kc=1, tauI='5 s', K=2, tau
PI loop margins: Kc = 1.0, tauI = 5.0 s, K = 2.0, taus = [1.0, 1.0, 1.0], t_unit = min, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Process gain and time constant from step-test data (least squares) —

`chemeng_first_order_step_fit`

Gain and time constant of a first-order process (dead time known) fitted by least squares to the recorded response of a step test.

$$\min_{K,\tau} \sum_k \left[y_k - KM\left(1 - e^{-(t_k-\theta)/\tau}\right)\right]^2$$

input	meaning	condition
`t`	sample times from the moment of the step, in t_unit	

input	meaning	condition
`y`	output deviations at those times	
`t_unit`	unit of the times	
`M`	size of the input step	
`theta`	dead time (known)	

output	meaning
`K`	process gain
`tau`	time constant

- **Method:** least squares of $y = K$ M (1 - exp(-(t - theta)/tau)): K solved linearly for each tau, tau from the zero of the derivative of the residual sum (bisection)
- **Assumes:** step from steady state; first-order response after the dead time; a unique interior least-squares minimum
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_first_order_step_fit', t=[0, 1, 2, 3, 4, 5, 6, 8, 10,
12, 15, 20], y=[0.0, 0.0, 1.7, 3.1, 4.3, 5.2, 6.0, 7.1, 7.9, 8.4, 8.8, 9.1], t_unit='min', M=5.0,
theta=60.0)
# -> K = 1.86707, tau = 292.778 s   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_first_order_step_fit', t=[0, 10, 20, 30, 40, 60, 80,
100, 150, 200], y=[0.0, -0.9, -1.75, -2.4, -2.95, -3.7, -4.2, -4.5, -4.85, -4.95], t_unit='s',
M=2.0, theta=0.0)
# -> K = -2.52645, tau = 46.0905 s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
First-order step test fit: t = [0.0, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 8.0, 10.0, 12.0, 15.0, 20.0], y = [0.0, 0.0, 1.7, 3.1, 4.3, 5.2, 6.0, 7.1, 7.9, 8.4, 8.8, 9.1], t_unit = min, M = 5.0, theta = 60.0 s	SOLVED	K = 1.86707, tau = 292.778 s
First-order step test fit: t = [0.0, 10.0, 20.0, 30.0, 40.0, 60.0, 80.0, 100.0, 150.0, 200.0], y = [0.0, -0.9, -1.75, -2.4, -2.95, -3.7, -4.2, -4.5, -4.85, -4.95], t_unit = s, M = 2.0, theta = 0.0 s	SOLVED	K = -2.52645, tau = 46.0905 s
First-order step test fit: y = [0.0, 0.0, 1.7, 3.1, 4.3, 5.2, 6.0, 7.1, 7.9, 8.4, 8.8, 9.1], t_unit = min, M = 5.0, theta = 60.0 s	NEEDS_INPUT	chemeng_first_order_step_fit needs t (sample times from the moment of the step, in t_unit). Example: solve('chemeng_firs
First-order step test fit: t = [0.0, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 8.0, 10.0, 12.0, 15.0, 20.0], y = [0.0, 0.0, 1.7, 3.1, 4.3, 5.2, 6.0, 7.1, 7.9, 8.4, 8.8, 9.1], t_unit = min, M = 5.0, theta = 60.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Steady radial flow of oil to a well (Darcy) and productivity index — `chemeng_radial_darcy_flow`

Steady-state reservoir flow rate into a vertical well from a circular drainage area, with an optional skin factor, and the productivity index.

$$q = \frac{2\pi k h\,(p_e - p_w)}{\mu\left(\ln\frac{r_e}{r_w} + S\right)}$$

input	meaning	condition
`k`	permeability (md is converted here)	
`h`	net pay thickness	
`mu`	fluid viscosity (cP is converted here)	
`pe`	pressure at the drainage radius	
`pw`	flowing bottom-hole pressure	
`re`	drainage radius	
`rw`	wellbore radius	
`S`	skin factor (default 0: undamaged well)	

output	meaning
`q`	flow rate at reservoir conditions
`PI`	productivity index q/(pe - pw)

- **Method:** $q = 2$ pi k h (pe - pw)/(mu (ln(re/rw) + S))
- **Assumes:** steady state, single-phase incompressible flow; $re > rw$, $pe > pw$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_radial_darcy_flow', k=9.869233e-14, h=9.144, mu=0.0015,
pe=20684271.879505083, pw=17236893.232920904, re=304.8, rw=0.09144000000000001)
# -> q = 0.00160651 m^3/s, PI = 4.6601e-10 m^3/(s*Pa)   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_radial_darcy_flow', k=5e-14, h=15.0, mu=0.002,
pe=25000000.0, pw=18000000.0, re=300.0, rw=0.1, S=3.0)
# -> q = 0.00149853 m^3/s, PI = 2.14076e-10 m^3/(s*Pa)   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Radial Darcy flow: k = 9.869233e-14 m^2, h = 9.144 m, mu = 0.0015 Pa*s, pe = 20684271.879505083 Pa, pw = 17236893.232920904 Pa, re = 304.8 m, rw = 0.09144000000000001 m	SOLVED	q = 0.00160651 m^3/s, PI = 4.6601e-10 m^3/(s*Pa)
Radial Darcy flow: k = 5e-14 m^2, h = 15.0 m, mu = 0.002 Pa*s, pe = 25000000.0 Pa, pw = 18000000.0 Pa, re = 300.0 m, rw = 0.1 m, S = 3.0	SOLVED	q = 0.00149853 m^3/s, PI = 2.14076e-10 m^3/(s*Pa)

question	outcome	result
Radial Darcy flow: h = 9.144 m, mu = 0.0015 Pa*s, pe = 20684271.879505083 Pa, pw = 17236893.232920904 Pa, re = 304.8 m, rw = 0.09144000000000001 m	NEEDS_INPUT	chemeng_radial_darcy_flow needs k (permeability (md is converted here)). Example: solve('chemeng_radial_darcy_flow', k='
Radial Darcy flow: k = 9.869233e-14 m^2, h = 9.144 m, mu = 0.0015 Pa*s, pe = 20684271.879505083 Pa, pw = 17236893.232920904 Pa, re = 304.8 m, rw = 0.09144000000000001 m, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Radial Darcy flow: k = 100 md, h = 30 ft, mu = 1.5 cP, pe = 3000 psi, pw = 2500 psi, re = 1000 ft, rw = 0.3 ft	SOLVED	q = 0.00160651 m^3/s, PI = 4.6601e-10 m^3/(s*Pa)
Radial Darcy flow: k = 100 md, h = 30 ft, mu = 1.5 cP, pe = 2000 psi, pw = 2500 psi, re = 1000 ft, rw = 0.3 ft	INVALID_INPUT	Steady radial flow of oil to a well (Darcy) and productivity index: the boundary pressure must exceed the flowing well p

Exponential decline curve: rate and cumulative production — `chemeng_exponential_decline`

Production rate and cumulative production at time t of a well in exponential (constant-percentage) decline (Arps b = 0).

$$q = q_i e^{-D_i t}, \quad N_p = \frac{q_i - q}{D_i}$$

input	meaning	condition
`qi`	initial rate (bbl/day and STB/day are converted here)	
`Di`	initial nominal decline rate, e.g. 0.3 $1/yr$	
`t`	time since the start of the decline	

output	meaning
`q`	rate at time t
`Np`	cumulative production to time t

- **Method:** $q = qi$ exp(-Di t); $Np = (qi - q)/Di$
- **Assumes:** boundary-dominated flow, constant operating conditions
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_exponential_decline', qi=0.0018401307283333335,
Di=9.506426344208685e-09, t=63115200.0)
# -> q = 0.00100989 m^3/s, Np = 87335.2 m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_exponential_decline', qi=0.01, Di=3.1688087814028953e-
09, t=157788000.0)
# -> q = 0.00606531 m^3/s, Np = 1.24169e+06 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Exponential decline: qi = 0.0018401307283333335 m^3/s, Di = 9.506426344208685e-09 1/s, t = 63115200.0 s	SOLVED	q = 0.00100989 m^3/s, Np = 87335.2 m^3
Exponential decline: qi = 0.01 m^3/s, Di = 3.1688087814028953e-09 1/s, t = 157788000.0 s	SOLVED	q = 0.00606531 m^3/s, Np = 1.24169e+06 m^3
Exponential decline: Di = 9.506426344208685e-09 1/s, t = 63115200.0 s	NEEDS_INPUT	chemeng_exponential_decline needs qi (initial rate (bbl/day and STB/day are converted here)). Example: solve('chemeng_ex
Exponential decline: qi = 0.0018401307283333335 m^3/s, Di = 9.506426344208685e-09 1/s, t = 63115200.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Exponential decline: qi = 1000 bbl/day, Di = 0.3 1/yr, t = 2 yr	SOLVED	q = 0.00100989 m^3/s, Np = 87335.2 m^3
Exponential decline: qi = 1000 bbl/day, Di = 0.3, t = 2 yr	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Hyperbolic decline curve: rate and cumulative production — `chemeng_hyperbolic_decline`

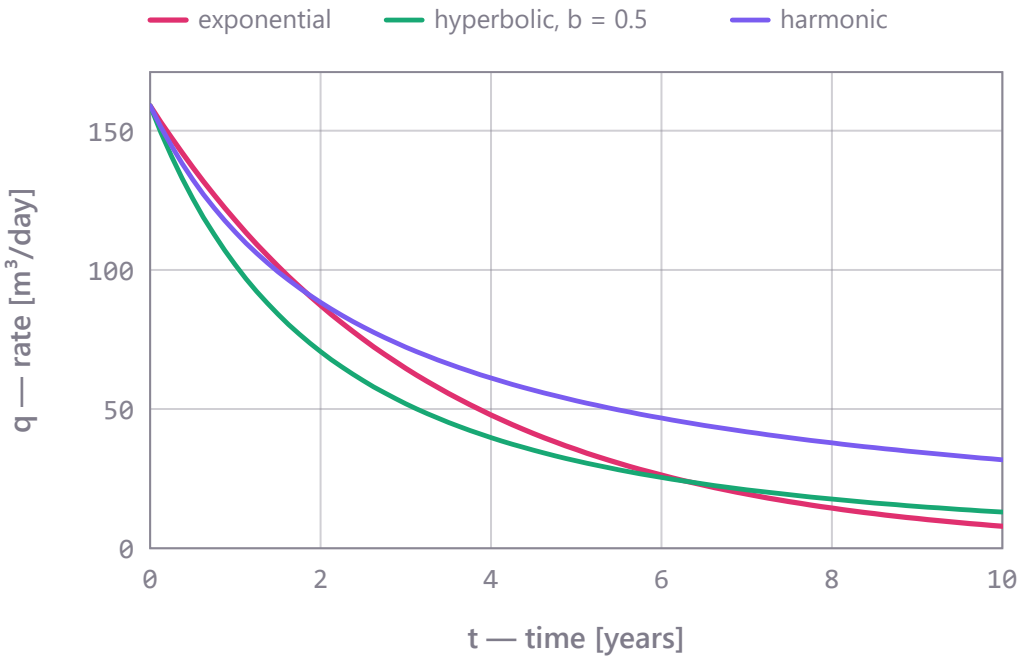
Production rate and cumulative production at time t of a well in hyperbolic decline with Arps exponent $0 < b < 1$.

$$q = \frac{q_i}{(1 + b D_i t)^{1/b}}, \quad N_p = \frac{q_i^b}{(1-b) D_i} \left(q_i^{1-b} - q^{1-b} \right)$$

input	meaning	condition
`qi`	initial rate (bbl/day and STB/day are converted here)	
`Di`	initial nominal decline rate, e.g. 0.3 $1/yr$	
`b`	Arps decline exponent $(0 < b < 1)$	
`t`	time since the start of the decline	

output	meaning
`q`	rate at time t
`Np`	cumulative production to time t

- **Method:** $q = qi$ (1 + b Di t)^(-1/b); $Np = qi^b$/($(1 - b)$ Di) (qi^(1-b) - q^(1-b))
- **Assumes:** $0 < b < 1$; boundary-dominated flow
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Well production rate against time for three decline laws from the same initial rate qi = 158.987 m³/day, each computed by the package at 81 times over 10 years. After 10 years: exponential (Di = 0.3 per year) 7.91551 m³/day; hyperbolic, b = 0.5 (Di = 0.5 per year) 12.9786 m³/day; harmonic (Di = 0.4 per year) 31.7975 m³/day.

```
chemical_engineering_ai999x.solve('chemeng_hyperbolic_decline', qi=0.0018401307283333335,
Di=1.5844043907014474e-08, b=0.5, t=94672800.0)
# -> q = 0.000600859 m^3/s, Np = 99548.8 m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_hyperbolic_decline', qi=0.01, Di=3.168808781402895e-08,
b=0.8, t=315576000.0)
# -> q = 0.0006415 m^3/s, Np = 666891 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Hyperbolic decline: qi = 0.0018401307283333335 m^3/s, Di = 1.5844043907014474e-08 1/s, b = 0.5, t = 94672800.0 s	SOLVED	q = 0.000600859 m^3/s, Np = 99548.8 m^3
Hyperbolic decline: qi = 0.01 m^3/s, Di = 3.168808781402895e-08 1/s, b = 0.8, t = 315576000.0 s	SOLVED	q = 0.0006415 m^3/s, Np = 666891 m^3
Hyperbolic decline: Di = 1.5844043907014474e-08 1/s, b = 0.5, t = 94672800.0 s	NEEDS_INPUT	chemeng_hyperbolic_decline needs qi (initial rate (bbl/day and STB/day are converted here)). Example: solve('chemeng_hyp
Hyperbolic decline: qi = 0.0018401307283333335 m^3/s, Di = 1.5844043907014474e-08 1/s, b = 0.5, t = 94672800.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Hyperbolic decline: qi = 1000 bbl/day, Di = 0.5 1/yr, b = 1, t = 3 yr	INVALID_INPUT	Hyperbolic decline curve: rate and cumulative production: the hyperbolic exponent b must lie strictly between 0 and 1 (b

Harmonic decline curve: rate and cumulative production — `chemeng_harmonic_decline`

Production rate and cumulative production at time t of a well in harmonic decline (Arps b = 1).

$$q = \frac{q_i}{1 + D_i t}, \quad N_p = \frac{q_i}{D_i} \ln \frac{q_i}{q}$$

input	meaning	condition
`qi`	initial rate (bbl/day and STB/day are converted here)	
`Di`	initial nominal decline rate, e.g. 0.3 $1/yr$	
`t`	time since the start of the decline	

output	meaning
`q`	rate at time t
`Np`	cumulative production to time t

- **Method:** q = qi/(1 + Di t); $Np = (qi/Di)$ ln(qi/q)
- **Assumes:** boundary-dominated flow
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_harmonic_decline', qi=0.0018401307283333335,
Di=1.2675235125611581e-08, t=94672800.0)
# -> q = 0.000836423 m^3/s, Np = 114465 m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_harmonic_decline', qi=0.01, Di=3.168808781402895e-08,
t=315576000.0)
# -> q = 0.000909091 m^3/s, Np = 756718 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Harmonic decline: qi = 0.0018401307283333335 m^3/s, Di = 1.2675235125611581e-08 1/s, t = 94672800.0 s	SOLVED	q = 0.000836423 m^3/s, Np = 114465 m^3
Harmonic decline: qi = 0.01 m^3/s, Di = 3.168808781402895e-08 1/s, t = 315576000.0 s	SOLVED	q = 0.000909091 m^3/s, Np = 756718 m^3
Harmonic decline: Di = 1.2675235125611581e-08 1/s, t = 94672800.0 s	NEEDS_INPUT	chemeng_harmonic_decline needs qi (initial rate (bbl/day and STB/day are converted here)). Example: solve('chemeng_harmo
Harmonic decline: qi = 0.0018401307283333335 m^3/s, Di = 1.2675235125611581e-08 1/s, t = 94672800.0 s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Oil initially in place from the material balance (Havlena-Odeh straight line) — `chemeng_havlena_odeh_oil_in_place`

Stock-tank oil initially in place from the underground withdrawal F and the total expansion term Et of successive pressure surveys (no water influx), by the straight line $F = N$ Et through the origin.

$$F = N\,E_t, \quad N = \frac{\sum F E_t}{\sum E_t^2}$$

input	meaning	condition
`F`	underground withdrawals at each survey, in v_unit at reservoir conditions	
`Et`	total expansion per unit of oil in place at each survey (reservoir volume per stock-tank volume)	
`v_unit`	unit of F: m3 or bbl	

output	meaning
`N`	oil initially in place (stock-tank volume)

- **Method:** least squares through the origin: N = sum F Et / sum Et^2
- **Assumes:** volumetric reservoir (no water influx); F and Et on one consistent basis
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_havlena_odeh_oil_in_place', F=[580700.0, 1600000.0,
2800000.0, 4100000.0], Et=[0.0101, 0.0278, 0.0485, 0.0712], v_unit='bbl')
# -> N = 9.1612e+06 m^3   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_havlena_odeh_oil_in_place', F=[120000.0, 260000.0,
410000.0], Et=[0.004, 0.0085, 0.0137], v_unit='m3')
# -> N = 3.01044e+07 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Havlena-Odeh oil in place: F = [580700.0, 1600000.0, 2800000.0, 4100000.0], Et = [0.0101, 0.0278, 0.0485, 0.0712], v_unit = bbl	SOLVED	N = 9.1612e+06 m^3
Havlena-Odeh oil in place: F = [120000.0, 260000.0, 410000.0], Et = [0.004, 0.0085, 0.0137], v_unit = m3	SOLVED	N = 3.01044e+07 m^3
Havlena-Odeh oil in place: Et = [0.0101, 0.0278, 0.0485, 0.0712], v_unit = bbl	NEEDS_INPUT	chemeng_havlena_odeh_oil_in_place needs F (underground withdrawals at each survey, in v_unit at reservoir conditions). E
Havlena-Odeh oil in place: F = [580700.0, 1600000.0, 2800000.0, 4100000.0], Et = [0.0101, 0.0278, 0.0485, 0.0712], v_unit = bbl, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Oil in place and gas-cap size from the material balance (Havlena-Odeh, gas cap) — `chemeng_havlena_odeh_gas_cap`

Oil initially in place and the gas-cap ratio m from the straight line $F/Eo = N + N$ m Eg/Eo of successive surveys (no water influx).

$$\frac{F}{E_o} = N + Nm\,\frac{E_g}{E_o}$$

input	meaning	condition
`F`	underground withdrawals, in v_unit at reservoir conditions	
`Eo`	oil and dissolved-gas expansion per unit of oil in place	
`Eg`	gas-cap expansion term per unit of oil in place	
`v_unit`	unit of F: m3 or bbl	

output	meaning
`N`	oil initially in place (stock-tank volume)
`m`	ratio of initial gas-cap to oil-zone reservoir volume

- **Method:** least squares of F/Eo against Eg/Eo: intercept N, slope N m
- **Assumes:** no water influx; consistent units
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_havlena_odeh_gas_cap', F=[5000000.0, 9300000.0,
14300000.0, 19500000.0], Eo=[0.01, 0.018, 0.027, 0.036], Eg=[0.025, 0.05, 0.08, 0.11],
v_unit='bbl')
# -> N = 5.05983e+07 m^3, m = 0.226683   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_havlena_odeh_gas_cap', F=[200000.0, 390000.0, 620000.0],
Eo=[0.005, 0.0095, 0.015], Eg=[0.008, 0.016, 0.026], v_unit='m3')
# -> N = 2.36173e+07 m^3, m = 0.434882   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Havlena-Odeh gas cap: F = [5000000.0, 9300000.0, 14300000.0, 19500000.0], Eo = [0.01, 0.018, 0.027, 0.036], Eg = [0.025, 0.05, 0.08, 0.11], v_unit = bbl	SOLVED	N = 5.05983e+07 m^3, m = 0.226683
Havlena-Odeh gas cap: F = [200000.0, 390000.0, 620000.0], Eo = [0.005, 0.0095, 0.015], Eg = [0.008, 0.016, 0.026], v_unit = m3	SOLVED	N = 2.36173e+07 m^3, m = 0.434882
Havlena-Odeh gas cap: Eo = [0.01, 0.018, 0.027, 0.036], Eg = [0.025, 0.05, 0.08, 0.11], v_unit = bbl	NEEDS_INPUT	chemeng_havlena_odeh_gas_cap needs F (underground withdrawals, in v_unit at reservoir conditions). Example: solve('cheme

question	outcome	result
Havlena-Odeh gas cap: F = [5000000.0, 9300000.0, 14300000.0, 19500000.0], Eo = [0.01, 0.018, 0.027, 0.036], Eg = [0.025, 0.05, 0.08, 0.11], v_unit = bbl, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Gas initially in place from the p/z plot — `chemeng_gas_pz_material_balance`

Original gas in place of a volumetric gas reservoir from the straight line of p/z against cumulative gas production.

$$\frac{p}{z} = \frac{p_i}{z_i}\left(1 - \frac{G_p}{G}\right)$$

input	meaning	condition
`p`	average reservoir pressures, in p_unit	
`z`	gas deviation factors at those pressures	
`Gp`	cumulative gas production at standard conditions, in gp_unit	
`p_unit`	unit of the pressures	
`gp_unit`	unit of Gp: m3, MMscf or Bscf	

output	meaning
`G`	original gas in place at standard conditions
`pz_initial`	p/z extrapolated to no production

- **Method:** least squares of p/z = (p/z)_i $(1 - Gp/G)$: G = -intercept/slope
- **Assumes:** volumetric (no water drive) gas reservoir; p/z falls with production
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_gas_pz_material_balance', p=[3200.0, 2925.0, 2525.0, 2125.0], z=[0.85, 0.84, 0.83, 0.82], Gp=[0.0, 15.0, 35.0, 55.0], p_unit='psi', gp_unit='Bscf')
# -> G = 4.99351e+09 m^3, pz_initial = 2.60849e+07 Pa   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_gas_pz_material_balance', p=[25.0, 22.1, 18.9], z=[0.9, 0.89, 0.88], Gp=[100000000.0, 300000000.0, 500000000.0], p_unit='MPa', gp_unit='m3')
# -> G = 1.86784e+09 m^3, pz_initial = 2.94209e+07 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Gas p/z material balance: p = [3200.0, 2925.0, 2525.0, 2125.0], z = [0.85, 0.84, 0.83, 0.82], Gp = [0.0, 15.0, 35.0, 55.0], p_unit = psi, gp_unit = Bscf	SOLVED	G = 4.99351e+09 m^3, pz_initial = 2.60849e+07 Pa
Gas p/z material balance: p = [25.0, 22.1, 18.9], z = [0.9, 0.89, 0.88], Gp = [100000000.0, 300000000.0,	SOLVED	G = 1.86784e+09 m^3, pz_initial = 2.94209e+07 Pa

question	outcome	result
500000000.0], p_unit = MPa, gp_unit = m3		
Gas p/z material balance: z = [0.85, 0.84, 0.83, 0.82], Gp = [0.0, 15.0, 35.0, 55.0], p_unit = psi, gp_unit = Bscf	NEEDS_INPUT	chemeng_gas_pz_material_balance needs p (average reservoir pressures, in p_unit). Example: solve('chemeng_gas_pz_materia
Gas p/z material balance: p = [3200.0, 2925.0, 2525.0, 2125.0], z = [0.85, 0.84, 0.83, 0.82], Gp = [0.0, 15.0, 35.0, 55.0], p_unit = psi, gp_unit = Bscf, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Pressure build-up test: Horner analysis for permeability and extrapolated pressure — `chemeng_horner_buildup`

Permeability and the extrapolated pressure p* from the semilog straight line of shut-in pressure against the Horner time ratio (tp + dt)/dt.

$$p_{ws} = p^* - m \log_{10} \frac{t_p + \Delta t}{\Delta t}, \quad k = \frac{q \mu \ln 10}{4 \pi m h}$$

input	meaning	condition
`dt`	shut-in times, in t_unit	
`pws`	shut-in bottom-hole pressures, in p_unit	
`t_unit`	unit of the shut-in times	
`p_unit`	unit of the pressures	
`tp`	producing time before shut-in	
`q`	flow rate before shut-in at reservoir conditions	
`mu`	fluid viscosity	
`h`	net pay thickness	

output	meaning
`m`	semilog slope, pressure change per log cycle
`k`	permeability
`p_star`	extrapolated pressure p at $(tp + dt)/dt = 1$

- **Method:** least squares of pws against log10((tp + dt)/dt): slope -m; $k = q$ mu ln 10/(4 pi m h)
- **Assumes:** infinite-acting radial flow on the straight line; the points given lie on the semilog straight line
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_horner_buildup', dt=[2.0, 4.0, 8.0, 16.0, 24.0], pws=
[2960.0, 3015.0, 3069.0, 3122.0, 3150.0], t_unit='h', p_unit='psi', tp=1800000.0,
q=0.0005520392185, mu=0.0008, h=6.096)
# -> m = 1.23929e+06 Pa, k = 1.07115e-14 m^2, p_star = 2.3388e+07 Pa   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_horner_buildup', dt=[1.0, 3.0, 10.0, 30.0], pws=[20.1,
20.6, 21.05, 21.4], t_unit='h', p_unit='MPa', tp=720000.0, q=0.002, mu=0.001, h=10.0)
# -> m = 913809 Pa, k = 4.01033e-14 m^2, p_star = 2.2236e+07 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Horner build-up analysis: dt = [2.0, 4.0, 8.0, 16.0, 24.0], pws = [2960.0, 3015.0, 3069.0, 3122.0, 3150.0], t_unit = h, p_unit = psi, tp = 1800000.0 s, q = 0.0005520392185 m^3/s, mu = 0.0008 Pa*s, h = 6.096 m	SOLVED	m = 1.23929e+06 Pa, k = 1.07115e-14 m^2, p_star = 2.3388e+07 Pa
Horner build-up analysis: dt = [1.0, 3.0, 10.0, 30.0], pws = [20.1, 20.6, 21.05, 21.4], t_unit = h, p_unit = MPa, tp = 720000.0 s, q = 0.002 m^3/s, mu = 0.001 Pa*s, h = 10.0 m	SOLVED	m = 913809 Pa, k = 4.01033e-14 m^2, p_star = 2.2236e+07 Pa
Horner build-up analysis: pws = [2960.0, 3015.0, 3069.0, 3122.0, 3150.0], t_unit = h, p_unit = psi, tp = 1800000.0 s, q = 0.0005520392185 m^3/s, mu = 0.0008 Pa*s, h = 6.096 m	NEEDS_INPUT	chemeng_horner_buildup needs dt (shut-in times, in t_unit). Example: solve('chemeng_horner_buildup', dt=[2.0, 4.0, 8.0,
Horner build-up analysis: dt = [2.0, 4.0, 8.0, 16.0, 24.0], pws = [2960.0, 3015.0, 3069.0, 3122.0, 3150.0], t_unit = h, p_unit = psi, tp = 1800000.0 s, q = 0.0005520392185 m^3/s, mu = 0.0008 Pa*s, h = 6.096 m, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Transient drawdown around a well (line-source solution) — `chemeng_line_source_drawdown`

Pressure drop and pressure at radius r and time t after a well starts producing at a constant rate in an infinite reservoir (line-source, exponential-integral solution).

$$\Delta p = \frac{q\mu}{4\pi kh} E_1\left(\frac{\phi\mu c_t r^2}{4kt}\right)$$

input	meaning	condition
`q`	flow rate at reservoir conditions	
`mu`	fluid viscosity	
`k`	permeability	
`h`	net pay thickness	
`phi`	porosity (fraction)	
`ct`	total compressibility (1/psi is converted here)	
`r`	radius from the well	
`t`	time since production started	
`pi`	initial reservoir pressure	

output	meaning
`dp`	pressure drop at r and t

output	meaning
p	pressure at r and t

- **Method:** dp = (q mu/(4 pi k h)) E1(phi mu ct r^2/(4 k t)); E1 by series or continued fraction
- **Assumes:** infinite-acting, homogeneous, slightly compressible fluid; constant rate from $t = 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_line_source_drawdown', q=0.001, mu=0.001, k=1e-13,
h=10.0, phi=0.2, ct=1e-09, r=0.1, t=3600.0, pi=30000000.0)
# -> dp = 1.02733e+06 Pa, p = 2.89727e+07 Pa   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_line_source_drawdown', q=0.002, mu=0.0015, k=5e-14,
h=20.0, phi=0.15, ct=2e-09, r=100.0, t=86400.0, pi=25000000.0)
# -> dp = 241753 Pa, p = 2.47582e+07 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Line-source drawdown: q = 0.001 m^3/s, mu = 0.001 Pa*s, k = 1e-13 m^2, h = 10.0 m, phi = 0.2, ct = 1e-09 1/Pa, r = 0.1 m, t = 3600.0 s, pi = 30000000.0 Pa	SOLVED	dp = 1.02733e+06 Pa, p = 2.89727e+07 Pa
Line-source drawdown: q = 0.002 m^3/s, mu = 0.0015 Pa*s, k = 5e-14 m^2, h = 20.0 m, phi = 0.15, ct = 2e-09 1/Pa, r = 100.0 m, t = 86400.0 s, pi = 25000000.0 Pa	SOLVED	dp = 241753 Pa, p = 2.47582e+07 Pa
Line-source drawdown: mu = 0.001 Pa*s, k = 1e-13 m^2, h = 10.0 m, phi = 0.2, ct = 1e-09 1/Pa, r = 0.1 m, t = 3600.0 s, pi = 30000000.0 Pa	NEEDS_INPUT	chemeng_line_source_drawdown needs q (flow rate at reservoir conditions). Example: solve('chemeng_line_source_drawdown',
Line-source drawdown: q = 0.001 m^3/s, mu = 0.001 Pa*s, k = 1e-13 m^2, h = 10.0 m, phi = 0.2, ct = 1e-09 1/Pa, r = 0.1 m, t = 3600.0 s, pi = 30000000.0 Pa, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Line-source drawdown: q = 3.6 m^3/h, mu = 1 cP, k = 1e-13 m^2, h = 10 m, phi = 20%, ct = 1e-9 1/Pa, r = 10 cm, t = 1 h, pi = 300 bar	SOLVED	dp = 1.02733e+06 Pa, p = 2.89727e+07 Pa

Gas-well deliverability from a back-pressure test (Rawlins-Schellhardt) —

`chemeng_gas_well_backpressure`

Exponent n, coefficient C and absolute open flow of a gas well from multi-rate test data fitted to $q = C$ (pr^2 - pwf^2)^n.

$$q = C\left(\bar{p}_r^2 - p_{wf}^2\right)^n, \quad AOF = C\,\bar{p}_r^{2n}$$

input	meaning	condition
`q`	test gas rates, in q_unit	
`pwf`	flowing bottom-hole pressures at those rates, in p_unit	
`q_unit`	unit of the rates: m3_per_day or Mscf_per_day	
`p_unit`	unit of the flowing pressures	
`pr`	average reservoir (shut-in) pressure	

output	meaning
`n`	back-pressure exponent
`log10_C`	log10 of the coefficient C, with q in m^3/s and pressures in Pa
`AOF`	absolute open flow (rate at pwf = 0)

- **Method:** least squares of log q against log(pr^2 - pwf^2): slope n, intercept log C; AOF = C pr^{2n}
- **Assumes:** stabilised flow at every rate; $0.3 < n < 1.5$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_gas_well_backpressure', q=[4288.0, 9265.0, 15552.0,
20177.0], pwf=[403.1, 394.0, 378.5, 362.6], q_unit='Mscf_per_day', p_unit='psi',
pr=2814439.927071325)
# -> n = 0.727757, log10_C = -8.06553, AOF = 20.9729 m^3/s   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_gas_well_backpressure', q=[100000.0, 200000.0,
320000.0], pwf=[18.0, 15.5, 11.0], q_unit='m3_per_day', p_unit='MPa', pr=20000000.0)
# -> n = 0.896485, log10_C = -12.3769, AOF = 5.17147 m^3/s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Gas well back-pressure test: q = [4288.0, 9265.0, 15552.0, 20177.0], pwf = [403.1, 394.0, 378.5, 362.6], q_unit = Mscf_per_day, p_unit = psi, pr = 2814439.927071325 Pa	SOLVED	n = 0.727757, log10_C = -8.06553, AOF = 20.9729 m^3/s
Gas well back-pressure test: q = [100000.0, 200000.0, 320000.0], pwf = [18.0, 15.5, 11.0], q_unit = m3_per_day, p_unit = MPa, pr = 20000000.0 Pa	SOLVED	n = 0.896485, log10_C = -12.3769, AOF = 5.17147 m^3/s
Gas well back-pressure test: pwf = [403.1, 394.0, 378.5, 362.6], q_unit = Mscf_per_day, p_unit = psi, pr = 2814439.927071325 Pa	NEEDS_INPUT	chemeng_gas_well_backpressure needs q (test gas rates, in q_unit). Example: solve('chemeng_gas_well_backpressure', q=[42
Gas well back-pressure test: q = [4288.0, 9265.0, 15552.0, 20177.0], pwf = [403.1, 394.0, 378.5, 362.6], q_unit = Mscf_per_day, p_unit = psi, pr = 2814439.927071325 Pa, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

Minimum number of stages at total reflux (Fenske) — `chemeng_fenske_minimum_stages`

The minimum number of equilibrium stages of a binary distillation at total reflux by the Fenske equation (constant relative volatility), counting the partial reboiler as a stage.

$$N_{min} = \frac{\ln\left[\frac{x_D}{1-x_D}\,\frac{1-x_B}{x_B}\right]}{\ln \alpha}$$

input	meaning	condition
`alpha`	relative volatility of the light component (constant)	
`xD`	mole fraction of the light component in the distillate	
`xB`	mole fraction of the light component in the bottoms	

output	meaning
`Nmin`	minimum number of equilibrium stages (partial reboiler included)
`Nmin_trays`	minimum number of trays (Nmin - 1, the reboiler excluded)

- **Method:** N_min = ln[(xD/(1 - xD)) ((1 - xB)/xB)]/ln(alpha)
- **Assumes:** binary, constant alpha; $0 < xB < xD < 1$; total reflux, total condenser
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_fenske_minimum_stages', alpha=2.5, xD=0.95, xB=0.05)
# -> Nmin = 6.42687, Nmin_trays = 5.42687   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_fenske_minimum_stages', alpha=1.5, xD=0.99, xB=0.01)
# -> Nmin = 22.6659, Nmin_trays = 21.6659   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Fenske minimum stages: alpha = 2.5, xD = 0.95, xB = 0.05	SOLVED	Nmin = 6.42687, Nmin_trays = 5.42687
Fenske minimum stages: alpha = 1.5, xD = 0.99, xB = 0.01	SOLVED	Nmin = 22.6659, Nmin_trays = 21.6659
Fenske minimum stages: xD = 0.95, xB = 0.05	NEEDS_INPUT	chemeng_fenske_minimum_stages needs alpha (relative volatility of the light component (constant)). Example: solve('cheme
Fenske minimum stages: alpha = 2.5, xD = 0.95, xB = 0.05, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Fenske minimum stages: alpha = 0.9, xD = 0.95, xB = 0.05	INVALID_INPUT	Minimum number of stages at total reflux (Fenske): the relative volatility alpha must exceed 1.

Pump total head and power — `chemeng_pump_head`

The total head a pump adds between its suction and discharge, H = (p2 - p1)/(rho g) + (z2 - z1) + (v2^2 - v1^2)/(2g), with the hydraulic power rho g Q H and the shaft power for an efficiency.

$$H = \frac{p_2 - p_1}{\rho g} + (z_2 - z_1) + \frac{v_2^2 - v_1^2}{2g}, \quad P = \rho g Q H$$

input	meaning	condition
`p1`	suction pressure (absolute)	
`p2`	discharge pressure (absolute)	
`rho`	liquid density	
`dz`	elevation of the discharge gauge above the suction gauge (default 0)	
`v1`	suction velocity (default 0)	
`v2`	discharge velocity (default 0)	
`Q`	volumetric flow (optional, for the power)	
`eta`	pump efficiency (optional, with Q)	

output	meaning
`H`	total head
`dp_equivalent`	head as a pressure rho g H
`P_hydraulic`	hydraulic power rho g Q H
`P_shaft`	shaft power rho g Q H/eta

- **Method:** H = (p2 - p1)/(rho g) + dz + $(v_2^2 - v_1^2)/(2$ g); $P = \rho$ g Q H; P_shaft = P/eta
- **Assumes:** absolute pressures; incompressible liquid; $H > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_pump_head', p1=100000.0, p2=400000.0, rho=1000.0)
# -> H = 30.5915 m, dp_equivalent = 300000 Pa   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_pump_head', p1=150000.0, p2=650000.0, rho=870.0, dz=2.0,
v1=1.5, v2=3.0, Q=0.02, eta=0.7)
# -> H = 60.9485 m, dp_equivalent = 520000 Pa, P_hydraulic = 10400 W, P_shaft = 14857.1 W
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Pump total head: p1 = 100000.0 Pa, p2 = 400000.0 Pa, rho = 1000.0 kg/m^3	SOLVED	H = 30.5915 m, dp_equivalent = 300000 Pa

question	outcome	result
Pump total head: p1 = 150000.0 Pa, p2 = 650000.0 Pa, rho = 870.0 kg/m^3, dz = 2.0 m, v1 = 1.5 m/s, v2 = 3.0 m/s, Q = 0.02 m^3/s, eta = 0.7	SOLVED	H = 60.9485 m, dp_equivalent = 520000 Pa, P_hydraulic = 10400 W, P_shaft = 14857.1 W
Pump total head: p2 = 400000.0 Pa, rho = 1000.0 kg/m^3	NEEDS_INPUT	chemeng_pump_head needs p1 (suction pressure (absolute)). Example: solve('chemeng_pump_head', p1='100000 Pa', p2='400000
Pump total head: p1 = 100000.0 Pa, p2 = 400000.0 Pa, rho = 1000.0 kg/m^3, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Pump total head: p1 = 400000 Pa, p2 = 100000 Pa, rho = 1000 kg/m^3	INVALID_INPUT	Pump total head and power: the pump must add head (H > 0).

Net positive suction head available — `chemeng_npsh_available`

The net positive suction head available at a pump inlet, NPSHa = (p_s - p_v)/(rho g) + z_s - h_f, from the pressure on the liquid surface, the vapour pressure, the static head and the suction-line loss; with the required NPSH, the margin.

$$NPSH_a = \frac{p_s - p_v}{\rho g} + z_s - h_f$$

input	meaning	condition
`p_s`	absolute pressure on the suction liquid surface	
`p_v`	vapour pressure of the liquid at the pumping temperature	
`rho`	liquid density	
`z_s`	liquid surface above the pump inlet (negative for a suction lift)	
`h_f`	friction head loss in the suction line	
`NPSHr`	required NPSH of the pump (optional)	

output	meaning
`NPSHa`	NPSH available
`pressure_head`	p_s/(rho g)
`vapour_head`	p_v/(rho g)
`margin`	NPSHa - NPSHr (with NPSHr)

- **Method:** NPSHa = (p_s - p_v)/(rho g) + z_s - h_f
- **Assumes:** absolute pressures; $h_f \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_npsh_available', p_s=101325.0, p_v=2339.0, rho=998.0,
z_s=2.0, h_f=0.5)
# -> NPSHa = 11.614 m, pressure_head = 10.353 m, vapour_head = 0.23899 m   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_npsh_available', p_s=101325.0, p_v=47400.0, rho=971.8,
z_s=-3.0, h_f=0.8, NPSHr=2.0)
# -> NPSHa = 1.85839 m, pressure_head = 10.6321 m, vapour_head = 4.97371 m, margin = -0.141614 m
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
NPSH available: p_s = 101325.0 Pa, p_v = 2339.0 Pa, rho = 998.0 kg/m^3, z_s = 2.0 m, h_f = 0.5 m	SOLVED	NPSHa = 11.614 m, pressure_head = 10.353 m, vapour_head = 0.23899 m
NPSH available: p_s = 101325.0 Pa, p_v = 47400.0 Pa, rho = 971.8 kg/m^3, z_s = -3.0 m, h_f = 0.8 m, NPSHr = 2.0 m	SOLVED	NPSHa = 1.85839 m, pressure_head = 10.6321 m, vapour_head = 4.97371 m, margin = -0.141614 m
NPSH available: p_v = 2339.0 Pa, rho = 998.0 kg/m^3, z_s = 2.0 m, h_f = 0.5 m	NEEDS_INPUT	chemeng_npsh_available needs p_s (absolute pressure on the suction liquid surface). Example: solve('chemeng_npsh_availab
NPSH available: p_s = 101325.0 Pa, p_v = 2339.0 Pa, rho = 998.0 kg/m^3, z_s = 2.0 m, h_f = 0.5 m, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
NPSH available: p_s = 101325 Pa, p_v = 2339 Pa, rho = 998 kg/m^3, z_s = 2 m, h_f = -0.5 m	INVALID_INPUT	Net positive suction head available: the suction friction head loss must be zero or positive.

Mass-transfer coefficient from a stated Sherwood correlation —

`chemeng_mass_transfer_coefficient`

The convective mass-transfer coefficient $k_c = Sh$ D/L from the Reynolds and Schmidt numbers by a stated correlation: a sphere (Ranz-Marshall), a flat plate in laminar flow (average) or turbulent flow in a pipe (Linton-Sherwood).

$$Sh = \frac{k_c L}{D}, \quad Sh = 2 + 0.6\, Re^{1/2} Sc^{1/3} \ \text{(sphere)}$$

input	meaning	condition
`geometry`	correlation: sphere (Ranz-Marshall), flat_plate (laminar, average) or pipe (Linton-Sherwood, turbulent)	
`u`	fluid velocity	
`L`	characteristic length: sphere or pipe diameter, plate length	
`D`	diffusivity of the solute in the fluid	
`rho`	fluid density	
`mu`	fluid viscosity	

output	meaning
`Re`	Reynolds number rho u L/mu
`Sc`	Schmidt number mu/(rho D)

output	meaning
`Sh`	Sherwood number
`kc`	mass-transfer coefficient Sh D/L

- **Method:** $Re = \rho$ u L/mu, Sc = mu/(rho D); sphere $Sh = 2 + 0.6\ Re^1/2\ Sc^1/3$; plate $Sh = 0.664\ Re^1/2\ Sc^1/3$; pipe $Sh = 0.023\ Re^{0.83}\ Sc^1/3$; $k_c = Sh$ D/L
- **Assumes:** within each correlation's stated range of Re and Sc; dilute solute
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_mass_transfer_coefficient', geometry='sphere', u=0.1,
L=0.001, D=5e-09, rho=1000.0, mu=0.001)
# -> Re = 100, Sc = 200, Sh = 37.0882, kc = 0.000185441 m/s   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_mass_transfer_coefficient', geometry='pipe', u=1.0,
L=0.025, D=1.5e-09, rho=998.0, mu=0.001)
# -> Re = 24950, Sc = 668.003, Sh = 897.2, kc = 5.3832e-05 m/s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Mass-transfer coefficient: geometry = sphere, u = 0.1 m/s, L = 0.001 m, D = 5e-09 m^2/s, rho = 1000.0 kg/m^3, mu = 0.001 Pa*s	SOLVED	Re = 100, Sc = 200, Sh = 37.0882, kc = 0.000185441 m/s
Mass-transfer coefficient: geometry = pipe, u = 1.0 m/s, L = 0.025 m, D = 1.5e-09 m^2/s, rho = 998.0 kg/m^3, mu = 0.001 Pa*s	SOLVED	Re = 24950, Sc = 668.003, Sh = 897.2, kc = 5.3832e-05 m/s
Mass-transfer coefficient: u = 0.1 m/s, L = 0.001 m, D = 5e-09 m^2/s, rho = 1000.0 kg/m^3, mu = 0.001 Pa*s	NEEDS_INPUT	chemeng_mass_transfer_coefficient needs geometry (correlation: sphere (Ranz-Marshall), flat_plate (laminar, average) or
Mass-transfer coefficient: geometry = sphere, u = 0.1 m/s, L = 0.001 m, D = 5e-09 m^2/s, rho = 1000.0 kg/m^3, mu = 0.001 Pa*s, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Mass-transfer coefficient: geometry = pipe, u = 0.01 m/s, L = 0.025 m, D = 1.5e-9 m^2/s, rho = 998 kg/m^3, mu = 0.001 Pa*s	INVALID_INPUT	Mass-transfer coefficient from a stated Sherwood correlation: Linton-Sherwood is stated for 2000 <= Re <= 70000 and 0.6

Minimum reflux of a multicomponent column (Underwood) —

`chemeng_multicomponent_underwood`

The minimum reflux ratio of a multicomponent distillation with constant relative volatilities by Underwood's equations: the root theta between the key volatilities of sum alpha_i $z_i/(alpha_i - \theta) = 1 - q$, then R_min + 1 = sum alpha_i x_D,i/(alpha_i - theta).

$$\sum_i \frac{\alpha_i z_i}{\alpha_i - \theta} = 1 - q, \quad R_{min} + 1 = \sum_i \frac{\alpha_i x_{D,i}}{\alpha_i - \theta}$$

input	meaning	condition
`alpha`	relative volatility of each component (any common reference, constant)	
`z`	feed mole fractions (sum to 1)	
`xD`	distillate mole fractions at minimum reflux (sum to 1)	
`q`	feed quality q (1 saturated liquid, 0 saturated vapour)	
`lk`	light key: its position in the lists (1 = first)	
`hk`	heavy key: its position in the lists	

output	meaning
`theta`	Underwood root between the key volatilities
`Rmin`	minimum reflux ratio L/D

- **Method:** sum alpha_i $z_i/(alpha_i - \theta) = 1 - q$, theta between alpha_HK and alpha_LK (bisection); R_min = sum alpha_i $x_D i/(alpha_i - \theta) - 1$
- **Assumes:** constant relative volatilities and molar overflow; adjacent keys (no component between them in volatility); the distillate composition at minimum reflux is stated
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_multicomponent_underwood', alpha=[2.5, 1.0, 0.5], z=
[0.4, 0.3, 0.3], xD=[0.9, 0.1, 0.0], q=1.0, lk=1, hk=2)
# -> theta = 1.40263, Rmin = 0.802   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_multicomponent_underwood', alpha=[4.0, 2.0, 1.0, 0.6],
z=[0.1, 0.3, 0.3, 0.3], xD=[0.24, 0.7, 0.06, 0.0], q=0.5, lk=2, hk=3)
# -> theta = 1.48444, Rmin = 1.97326   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Multicomponent minimum reflux: alpha = [2.5, 1.0, 0.5], z = [0.4, 0.3, 0.3], xD = [0.9, 0.1, 0.0], q = 1.0, lk = 1, hk = 2	SOLVED	theta = 1.40263, Rmin = 0.802
Multicomponent minimum reflux: alpha = [4.0, 2.0, 1.0, 0.6], z = [0.1, 0.3, 0.3, 0.3], xD = [0.24, 0.7, 0.06, 0.0], q = 0.5, lk = 2, hk = 3	SOLVED	theta = 1.48444, Rmin = 1.97326
Multicomponent minimum reflux: z = [0.4, 0.3, 0.3], xD = [0.9, 0.1, 0.0], q = 1.0, lk = 1, hk = 2	NEEDS_INPUT	chemeng_multicomponent_underwood needs alpha (relative volatility of each component (any common reference, constant)). E
Multicomponent minimum reflux: alpha = [2.5, 1.0, 0.5], z = [0.4, 0.3, 0.3], xD = [0.9, 0.1, 0.0], q = 1.0, lk = 1, hk = 2, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ

question	outcome	result
Multicomponent minimum reflux: alpha = [2.5, 1.0, 1.5], z = [0.4, 0.3, 0.3], xD = [0.9, 0.1, 0.0], q = 1, lk = 1, hk = 2	INVALID_INPUT	Minimum reflux of a multicomponent column (Underwood): the keys must be adjacent in volatility (no component between the

CSTR volume for a conversion (second order) — `chemeng_cstr_volume_second_order`

Volume and space time of a CSTR that reaches conversion X when -r_A = k C_A^2 (liquid phase, constant density).

$$V = \frac{v_0 X}{k C_{A0} (1 - X)^2}$$

input	meaning	condition
`v0`	volumetric feed flow	
`k`	second-order rate constant (-r_A = k C_A^2)	
`CA0`	feed (initial) concentration of A	
`X`	conversion of A wanted	

output	meaning
`V`	reactor volume
`tau`	space time V/v_0

- **Method:** design equation $V = F_A 0$ X/(-r_A) at the exit concentration: $V = v_0$ X/(k C_A0 (1 - X)^2)
- **Assumes:** $0 \le X < 1$; steady state, perfect mixing, constant density, $-r_A = k\, C_A^2$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_cstr_volume_second_order', v0=0.01, k=1e-05, CA0=1000.0,
X=0.8)
# -> V = 20 m^3, tau = 2000 s   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_cstr_volume_second_order', v0=0.002, k=0.0001,
CA0=200.0, X=0.5)
# -> V = 0.2 m^3, tau = 100 s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Second-order CSTR volume: v0 = 0.01 m^3/s, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3, X = 0.8	SOLVED	V = 20 m^3, tau = 2000 s
Second-order CSTR volume: v0 = 0.002 m^3/s, k = 0.0001 m^3/(mol*s), CA0 = 200.0 mol/m^3, X = 0.5	SOLVED	V = 0.2 m^3, tau = 100 s

question	outcome	result
Second-order CSTR volume: k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3, X = 0.8	NEEDS_INPUT	chemeng_cstr_volume_second_order needs v0 (volumetric feed flow). Example: solve('chemeng_cstr_volume_second_order', v0=
Second-order CSTR volume: v0 = 0.01 m^3/s, k = 1e-05 m^3/(mol*s), CA0 = 1000.0 mol/m^3, X = 0.8, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Second-order CSTR volume: v0 = 0.01 m^3/s, k = 1e-5 m^3/(mol*s), CA0 = 1000 mol/m^3, X = 100%	INVALID_INPUT	CSTR volume for a conversion (second order): the conversion X must be a fraction between 0 and 1.

Equimolar counterdiffusion flux — `chemeng_equimolar_counterdiffusion`

The molar flux of A through a film of thickness L in equimolar counterdiffusion (N_B = -N_A, ideal gas): $N_A = c$ D $(y_A1 - y_A2)/L$ with c = P/(R T).

$$N_A = \frac{P}{RT}\,\frac{D_{AB}\,(y_{A1} - y_{A2})}{L}$$

input	meaning	condition
`D`	diffusivity of A in B	
`P`	total pressure	
`T`	temperature	
`L`	film (path) thickness	
`yA1`	mole fraction of A at $z = 0$	
`yA2`	mole fraction of A at $z = L$	

output	meaning
`NA`	molar flux of A
`NB`	molar flux of B (= -N_A)

- **Method:** N_A = (P/(R T)) D $(y_A1 - y_A2)/L$
- **Assumes:** steady state, ideal gas, constant D; equimolar counterdiffusion $(N_B = -N_A)$; $yA1 > yA2$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_equimolar_counterdiffusion', D=2.3e-05, P=101325.0,
T=298.15, L=0.1, yA1=0.2, yA2=0.05)
# -> NA = 0.00141015 mol/(m^2*s), NB = -0.00141015 mol/(m^2*s)   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_equimolar_counterdiffusion', D=1e-05, P=200000.0,
T=400.0, L=0.02, yA1=0.6, yA2=0.0)
# -> NA = 0.0180409 mol/(m^2*s), NB = -0.0180409 mol/(m^2*s)   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Equimolar counterdiffusion flux: D = 2.3e-05 m^2/s, P = 101325.0 Pa, T = 298.15 K, L = 0.1 m, yA1 = 0.2, yA2 = 0.05	SOLVED	NA = 0.00141015 mol/(m^2*s), NB = -0.00141015 mol/(m^2*s)
Equimolar counterdiffusion flux: D = 1e-05 m^2/s, P = 200000.0 Pa, T = 400.0 K, L = 0.02 m, yA1 = 0.6, yA2 = 0.0	SOLVED	NA = 0.0180409 mol/(m^2*s), NB = -0.0180409 mol/(m^2*s)
Equimolar counterdiffusion flux: P = 101325.0 Pa, T = 298.15 K, L = 0.1 m, yA1 = 0.2, yA2 = 0.05	NEEDS_INPUT	chemeng_equimolar_counterdiffusion needs D (diffusivity of A in B). Example: solve('chemeng_equimolar_counterdiffusion',
Equimolar counterdiffusion flux: D = 2.3e-05 m^2/s, P = 101325.0 Pa, T = 298.15 K, L = 0.1 m, yA1 = 0.2, yA2 = 0.05, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Equimolar counterdiffusion flux: D = 2.3e-5 m^2/s, P = 101325 Pa, T = 298.15 K, L = 0.1 m, yA1 = 0.05, yA2 = 0.2	INVALID_INPUT	Equimolar counterdiffusion flux: A diffuses from yA1 to yA2: yA1 must exceed yA2.

Bubble-point pressure by the modified Raoult's law (activity coefficients) — `chemeng_modified_raoult_bubble_pressure`

The bubble-point pressure P = sum x_i gamma_i P_i^sat of a non-ideal liquid with stated activity coefficients and vapour pressures (low pressure, ideal vapour), and the vapour composition $y_i = x_i$ gamma_i P_i^sat/P.

$$P = \sum_i x_i \gamma_i P_i^{sat}, \quad y_i = \frac{x_i \gamma_i P_i^{sat}}{P}$$

input	meaning	condition
`x`	liquid mole fractions (sum to 1)	
`gamma`	activity coefficient of each component at this composition	
`Psat`	vapour pressure of each pure component at the temperature, in the unit p_unit	
`p_unit`	unit of the vapour pressures: Pa, kPa, MPa, bar, atm or psi	

output	meaning
`P`	bubble-point pressure
`y1`	vapour mole fraction of the first component
`y`	vapour mole fractions, in the order of x

- **Method:** P = sum x_i gamma_i P_i^sat, $y_i = x_i$ gamma_i P_i^sat/P
- **Assumes:** low pressure (ideal vapour, Poynting factor 1); activity coefficients stated for this liquid composition and temperature
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
chemical_engineering_ai999x.solve('chemeng_modified_raoult_bubble_pressure', x=[0.4, 0.6], gamma=
[1.8, 1.2], Psat=[55.0, 30.0], p_unit='kPa')
# -> P = 61200 Pa, y1 = 0.647059, y = [0.647058823529, 0.352941176471]   (SOLVED)
```

```
chemical_engineering_ai999x.solve('chemeng_modified_raoult_bubble_pressure', x=[0.2, 0.3, 0.5],
gamma=[2.0, 1.1, 1.05], Psat=[1.2, 0.8, 0.3], p_unit='bar')
# -> P = 90150 Pa, y1 = 0.532446, y = [0.532445923461, 0.292845257903, 0.174708818636]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Modified Raoult bubble pressure: x = [0.4, 0.6], gamma = [1.8, 1.2], Psat = [55.0, 30.0], p_unit = kPa	SOLVED	P = 61200 Pa, y1 = 0.647059, y = [0.647058823529, 0.352941176471]
Modified Raoult bubble pressure: x = [0.2, 0.3, 0.5], gamma = [2.0, 1.1, 1.05], Psat = [1.2, 0.8, 0.3], p_unit = bar	SOLVED	P = 90150 Pa, y1 = 0.532446, y = [0.532445923461, 0.292845257903, 0.174708818636]
Modified Raoult bubble pressure: gamma = [1.8, 1.2], Psat = [55.0, 30.0], p_unit = kPa	NEEDS_INPUT	chemeng_modified_raoult_bubble_pressure needs x (liquid mole fractions (sum to 1)). Example: solve('chemeng_modified_rao
Modified Raoult bubble pressure: x = [0.4, 0.6], gamma = [1.8, 1.2], Psat = [55.0, 30.0], p_unit = kPa, zz = 1	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemical_engineering_ai999x.templ
Modified Raoult bubble pressure: x = [0.4, 0.7], gamma = [1.8, 1.2], Psat = [55, 30], p_unit = kPa	INVALID_INPUT	Bubble-point pressure by the modified Raoult's law (activity coefficients): x: the fractions must sum to 1.